

Measurement of $\nu_\mu + \bar{\nu}_\mu$ Charged-Current Single Pion Production Differential Cross Sections on Argon with the MicroBooNE Detector in the NuMI Beam

Analysis Note, Full FHC, RHC, and Combined Exposure

Draft 1.4, 2026-09-04

dc0545b · 51 extractions · blind

Ishanee Pophale, Jarek Nowak

Lancaster University

Abstract

We present a measurement of $\nu_\mu + \bar{\nu}_\mu$ charged-current single charged-pion ($\text{CC}\pi^\pm$) differential cross sections on argon-40 using the full MicroBooNE NuMI exposure, in three configurations: forward horn current (FHC; Runs 1, 2, 4, 5; 8.857×10^{20} protons on target, POT), reverse horn current (RHC; Runs 1, 2, 3, 4a, 4b, 4c; 11.082×10^{20} POT), and their combination (19.939×10^{20} POT). The analysis is blind: results are extracted on Poisson-thrown central-value fake data. Events are selected requiring a reconstructed muon candidate track and exactly one contained pion candidate track emerging from a fiducial neutrino vertex, passing an electromagnetic-shower veto, and with a muon–pion opening angle less than 2.6 rad. In the measured phase space the selection achieves $\approx 59\%$ signal purity and $\approx 17\%$ efficiency (Table 10). Differential cross sections are extracted for each configuration via Wiener-SVD unfolding in five kinematic quantities: muon momentum p_μ , pion momentum p_π , muon polar angle, pion polar angle $\cos\theta_\pi$ and the muon–pion opening angle $\theta_{\mu\pi}$. The muon polar angle is reported in both its $\cos\theta_\mu$ and its θ_μ parameterisation, so six differential distributions are released per configuration. Each configuration is normalised with its own integrated $\nu_\mu + \bar{\nu}_\mu$ flux (FHC 6.81159 , RHC 6.44646 , combined $6.60865 \times 10^{-10} \text{ cm}^{-2} \text{ POT}^{-1}$; new-G4 GEANT4 values above 60 MeV, Sec. 5); the RHC $\nu_\mu + \bar{\nu}_\mu$ integral is lower because the reverse-horn focusing accepts a softer, less intense hadron spectrum at the off-axis MicroBooNE location, not because the beam is $\bar{\nu}_\mu$ -dominant, which is a composition property. The primary total is the cut-and-count flux-averaged cross section (Table 21): FHC 1.03 ± 0.42 , RHC 0.93 ± 0.35 and combined 0.98 ± 0.34 ($\times 10^{-38} \text{ cm}^2/\text{Ar}$), with a cut-and-count RHC/FHC ratio of 0.901. The Wiener-SVD integrated cross sections are of order $0.66\text{--}0.96 \times 10^{-38} \text{ cm}^2/\text{Ar}$ (Table 20); they differ between observables because the additional-smearing matrix A_C is not norm preserving, so a ratio of two of them is observable-dependent and is not quoted as a result. The RHC deficit is a property of the injected sample rather than a normalisation artefact: the exposures sum exactly and each configuration carries its own flux, and on fake data the cut-and-count total is by construction the central-value tune. Compared with the standalone generators

in the same phase space, the injected central-value tune lies $0.5\text{--}0.6\sigma$ above GENIE v3 and within 0.7σ of GiBUU, NEUT and NuWro (Table 21; Table 17 of the technical supplement). No beam-on data is unblinded in this note: every “data” point shown is Poisson-thrown from the central-value Monte Carlo (§6), so no statement here is a measurement of a real deficit or excess. The extraction is validated by a same-sample self-closure, and cross-checked against the D’Agostini iterative-Bayesian method: its flat, iteration-stable integral (20–29% above Wiener-SVD, and flat to $\leq 1.0\%$ across observables where the Wiener-SVD spread is 15–25%; re-evaluated on this release) confirms that the observable-to-observable spread of the Wiener-SVD integrated cross sections is the additional-smearing matrix A_C , not a physical or acceptance effect (Sec. 4.5 of the technical supplement). All results use the XSECANALYZER framework on the resolution-optimised binning of Sec. 6 of the technical supplement. The unfolded integral reproduces the realised fake-data truth to 1.018 on average over the eighteen inclusive extractions (second-derivative regularisation; the earlier 1.001 was measured against a truth that still carried the central-value weight once too often, §16). Earlier versions of this note carried a 13% excess whose origin was unknown; it has since been traced to a double application of the central-value weight to the externally-thrown fake data, and correcting it moved every one of the fifty-one extractions by between 3.8% and 12.9% (§16). **Scope.** The primary results are the total cross section, the angular observables, $d\sigma/dp_\mu$, and a deliberately coarse two-bin $d\sigma/dp_\pi$. The proton-tagged subsample is secondary scope because the exposure supports an integrated cross section rather than a resonance shape, so the differential $W_{\pi p}$ is withdrawn; its systematics are otherwise complete. All fifty-one extractions carry a full covariance including detector variations, the proton-tagged ones having been rebuilt and re-unfolded for that purpose on 2026-08-31 — they previously carried none, the detector variation samples having been processed but never referenced by the proton-tagged configuration. Sec. 12.1 gives a per-result status table. **Validation.** Two results bound what the extraction can be trusted to do. An ensemble of thirty-two independent Poisson realisations, each carried through the whole chain, gives per-bin pull widths of 1.04 ($d\sigma/dp_\mu$) and 0.96 ($d\sigma/d\cos\theta_\mu$) against the statistical covariance, with 68% and 95% interval coverage recovered: the statistical uncertainties are correctly sized, and the observable whose additional smearing is most severe behaves no differently from the best-conditioned one (ensemble of the 2026-08-31 release; the beam-off correction cancels in fake-data closure and does not change it). Separately, all fifty-one extractions rebuild bit-identically from the processed ntuples with every cached universe deleted, so the released numbers are regenerable from the raw inputs rather than from a preserved intermediate. The same ensemble shows a small negative bias, 2.5% on the integral, which is stated rather than corrected. **Companion documents.** This note describes the current state of the analysis. The validation and diagnostic studies behind it — the pion-momentum estimator survey, the binning and threshold optimisation, and the closure programme — are in the technical validation supplement. Every correction that changed a result, with the values before and after, is in the change log. The additional-smearing and covariance matrices for all fifty-one extractions are released as text in `report/data_release/`; a prediction cannot be compared with these results without the former, nor used quantitatively without the latter.

Contents

1	Introduction	1
2	Request to open the background-control regions	1
3	Theoretical Background and Prior $\text{CC}\pi^+$ Measurements	4
3.1	Production Mechanisms	4
3.2	Nuclear Effects and Final-State Interactions	4
3.3	Generator Predictions and Known Tensions	5
3.4	Previous Experimental Measurements	6

4	Detector	9
4.1	Operating Principle	9
4.2	Active Volume and Electric Field	9
4.3	Wire Readout and Signal Formation	9
4.4	Photon Detection System	10
4.5	Calibration and Corrections	10
4.6	Cosmic Ray Environment and Reconstruction	10
5	Beam and Neutrino Flux	11
5.1	NuMI beam and flux composition	11
5.2	Beam direction in detector coordinates	12
5.3	Data exposure	13
5.4	Integrated flux	14
6	Monte Carlo Samples	15
7	Signal Definition	18
8	Fiducial Volume	20
8.1	Which volume applies where	20
9	Event Selection	21
9.1	BDT-based particle identification	21
9.2	Cut definitions	21
9.3	Kinematic reconstruction	23
10	Cut-Flow and Selection Performance	23
10.1	Key N−1 distributions	25
10.2	Pion proton-rejection variables	26
10.3	Final-cut data/MC distributions	27
10.4	Background decomposition	28
10.5	Event displays	28
11	Unfolding	29
11.1	The extraction chain, stage by stage	29
11.2	Signal extraction	32
11.3	Response matrix	32
11.4	Regularised unfolding: the bias–variance trade-off	33
11.5	Wiener-SVD unfolding	34
11.6	Reconstructed spectra and response matrices	34
11.7	Reconstruction resolution	38
11.8	Model comparison: χ^2 against truth, tune and generators	39
12	Full-exposure results: FHC, RHC, and combined	41
12.1	Scope and status of each result	41
12.2	How far A_C is from preserving normalisation	49
12.3	Total flux-averaged cross section (cut-and-count)	50
12.4	Background-control sidebands	51
12.5	Region overlaps	54
12.6	Full-stack pseudo-data test	54
12.7	Transfer factors and control-region uncertainties	56
12.8	Frozen procedure for opening the control regions	58
12.9	Scope: inclusive regions only	59

12.10	Control regions for the proton-tagged subsample	60
13	Secondary scope: proton-tagged subsample (hadronic invariant mass and transverse kinematic imbalance)	61
13.1	Extracted cross sections	62
14	Systematic Uncertainties	67
14.1	Systematic breakdown by source	68
14.2	Anatomy of the flux term: genuine flux vs. unfolding amplification	69
15	Analysis Pipeline and Automated Guards	70
16	Limitations	71
17	Summary	72
A	Observable Binning	74
B	Selection Cut Parameter Summary	75

1 Introduction

2 Request to open the background-control regions

This section is the concise, self-contained case for the one decision requested of the collaboration at this stage: permission to inspect beam-on data in four reconstruction-level control regions while the signal region stays blind. Everything here is expanded in §12.4; the numbers are those of the current release.

Request. We request approval to inspect beam-on data in four control regions of the inclusive $CC1\mu1\pi Xp$ selection: $CC0\pi$, multi- π , π^0 and large muon-pion opening angle (“cosmic”). Each is anchored by the inversion of one nominal signal requirement; requirements that become undefined once that inversion is made, or that would sculpt the targeted control sample, are explicitly dropped as listed in Table 1. Each region is disjoint from the signal region at reconstruction level (demonstrated event by event, §12.5); their expected signal contaminations are 8–23%. The signal region remains inaccessible. Before the regions are opened, the selections, the run-dependent normalisation, the beam-off scaling, the displayed observables and binning, the covariance construction, the goodness-of-fit tests and the permitted responses to a discrepancy are frozen as written in §12.8. Control-region data will be compared with the complete prediction (neutrino MC + beam-off + dirt) separately in FHC and RHC, with the combined configuration as a cross-check. Any discrepancy beyond the pre-declared criteria requires a documented treatment and renewed approval before the signal region is opened. The proton-tagged control regions are *not* part of this request (§12.9).

What the present pseudo-data studies do and do not show. The sideband machinery has been exercised on Poisson-thrown central-value pseudo-data. Those studies validate the selection logic, the per-run exposure scaling, the central-value weighting and, in the full-stack variant of §12.6, the exact bookkeeping path that beam-on data will follow. They do *not* validate the background model: the pseudo-data are drawn from the same simulation that supplies the prediction, so agreement is guaranteed up to statistics. The background model is the largest untested assumption in the analysis, and testing it is what the control regions are for.

Table 1: Control-region definitions at branch level (`CC1mulpiXp.cxx`). The common preselection `pre` is software trigger, reconstructed vertex in the fiducial volume, topological score > 0.67 and a qualifying muon candidate (Cuts 1–4 of §9); `shower` is the shower veto (Cut 8), `final` the non-proton multiplicity cut (Cut 10), `contained/gap` the pion containment and three-plane requirements (Cuts 5–7), `opening` the $\theta_{\mu\pi} < 2.6$ rad cut (Cut 9), and n_π the number of qualifying contained pion candidates. The signal region is `pre && contained && gap && shower && opening && final && n $_\pi$  = 1`. “Dropped” lists nominal requirements not applied in the region; “target” the signal-region background the region is meant to constrain. The regions are stored as the branches `CC1mulpiXp_sb_*`.

Region	Definition (branch level)	Dropped vs. signal	Target background
CC0 π	<code>pre && shower && final && n$_\pi$ = 0</code>	contained, gap, opening (no pion to define them)	CC0 π events with a proton or other track taken as the pion; pion absorption feed-in
multi- π	<code>pre && final && n$_\pi$ $\geq$ 2</code>	shower, contained/-gap of the extra pion, opening	multi-pion events with one pion lost
π^0	<code>pre && !shower && n$_{\text{shower}}$ $\geq$ 1</code>	contained, gap, opening, final	CC/NC π^0 with both photons missed; shower-veto efficiency
cosmic	<code>pre && contained && gap && shower && final && !opening</code>	opening only	residual beam-off (cosmic) in the signal region; its normalisation

The π^0 region is the furthest from a single inversion: besides the pion requirements, which have no meaning once no pion candidate is demanded, it also drops the final multiplicity cut. That cut counts primary tracks and non-proton-like tracks, and in a π^0 event the photon showers are frequently reconstructed as additional track-like primaries, so retaining it would preferentially remove exactly the multi-shower π^0 topologies the region is meant to sample and sculpt its composition towards the single-shower events that already resemble the signal region. The size of that effect on the region’s yield, purity, transfer factor and overlap with multi- π cannot be quantified after the fact on the present processed files, which do not store the multiplicities; it is listed as a diagnostic to be produced with the next reprocessing, and the frozen definition is the one in the table.

Two points the collaboration is asked to ratify explicitly. (i) The π^0 definition without the final multiplicity cut (Table 1 and the paragraph above): the comparison with the version retaining that cut is not available in the current processed files and will *not* be used to modify the region after beam-on inspection. (ii) The roles the transfer factors of §12.7 assign to the regions: CC0 π is a quantitative test of the 0 π background whose extrapolation to the signal region carries a 31–34% uncertainty, dominated by the detector variations; π^0 is the strongest quantitative transfer handle (14–17%); multi- π is diagnostic only ($T \simeq 2.5$); and cosmic validates the beam-off normalisation and shape but, at 34–41% beam-off purity, is not eligible for a fitted correction under the purity rule of §12.8.

Expected yields and precision. Table 23 gives the expected composition per region and beam. The raw statistical precision of the data samples on the dominant component is $\sim 1\%$ (CC0 π), $\sim 1.5\%$ (π^0), $\sim 6\%$ (multi- π) and $\sim 17\%$ on the beam-off normalisation (cosmic) for the combined exposure. The regions *test* the background components they target and provide sensitivity to them; whether a region can be used to *constrain* its target in the signal region is decided by the transfer factors and their uncertainties (§12.7), and for multi- π the answer is already no. Although the combined CC0 π sample has $\sim 1\%$ raw statistical precision, its target-class transfer to the signal region has a 31–34% uncertainty, dominated by the detector

variations, so the region is primarily a high-statistics validation test rather than a per-cent-level signal-region constraint. The neutrino-MC prediction in each region carries the flux, cross-section and re-interaction uncertainties of Table 25, including the detector variations.

Decision checklist. The prerequisites are split by the decision they gate.

Required before the control regions are opened (this request):

- ✓ Result tables regenerated from one release and checked automatically (`report/check_tables.py`, §15); cut-flow with the full central-value weight (Table 10).
- ✓ Control-region selections frozen at branch level (Table 1); signal-region overlap demonstrated to be zero (§12.5); displayed distributions and binning frozen (`configs/sideband_plots.txt`, Figs. 25–26).
- ✓ Beam-off gates and scale factors released per run (`data.release/ext_gates.tsv`, Fig. 2), each (file, run) pair entering once, asserted by the normalisation manifest (§15).
- ✓ Real beam-on files are refused by the framework unless `XSEC_UNBLIND=1` is set (§15).
- ✓ Full-stack pseudo-data closure (§12.6).
- ✓ Control-region uncertainties including detector variations, and transfer-factor uncertainties computed as ratios per universe (§12.7).
- Ratification of the comparison procedure and the decision rule of §12.8 by the conveners.

Required before the signal region is opened (a later request):

- Confirmation that the flux constants are the PPFX-corrected central values (§5.4); the control-region comparisons are event-yield comparisons and do not depend on it.
- Documented treatment of any control-region discrepancy under §12.8, with renewed approval where item 4(d) applies.
- A final clean-cache reproduction of the release to be unblinded.
- For any proton-tagged result: the inverted-proton-PID control region (§12.9).

Charged-current single charged-pion ($\text{CC}\pi^\pm$) production,

$$\nu_\mu(\bar{\nu}_\mu) + \text{Ar} \longrightarrow \mu^\mp + \pi^\pm + X, \quad (1)$$

is among the most frequent neutrino interaction channels at energies of 1–5 GeV and constitutes a significant background for oscillation analyses at the same energy scale, most notably for ν_e appearance measurements at experiments such as NOvA [3] and T2K [4], as well as the future DUNE [5]. Despite its importance, the cross section on nuclear targets remains poorly constrained: major neutrino interaction generators (GENIE v3, NuWro, NEUT) disagree by 20–30% in absolute rate and show significant shape differences for individual kinematic variables.

MicroBooNE is a single-phase liquid argon time-projection chamber (LArTPC) [33] at Fermilab that operated from 2015–2021, providing the world’s highest-statistics ν_μ -on-argon dataset below 5 GeV during its NuMI running period. This note presents a simultaneous measurement of differential cross sections in five kinematic quantities (six released distributions, the muon polar angle being given in both $\cos\theta_\mu$ and θ_μ) in the $\text{CC}\pi^\pm$ channel using the full NuMI exposure in FHC, RHC, and combined configurations, employing Pandora pattern recognition [6, 12] and Wiener-SVD unfolding [7].

3 Theoretical Background and Prior $\text{CC}\pi^+$ Measurements

3.1 Production Mechanisms

At neutrino energies of 0.5–5 GeV, charged-current single charged-pion production proceeds through three mechanisms.

Baryon resonance production dominates at $E_\nu \lesssim 2$ GeV. The neutrino excites the struck nucleon to a baryon resonance, most prominently the $\Delta(1232)$ isobar, which then decays to a nucleon and pion:

$$\nu_\mu n \rightarrow \mu^- \Delta^{++} \rightarrow \mu^- \pi^+ p, \quad \nu_\mu p \rightarrow \mu^- \Delta^+ \rightarrow \mu^- \pi^0 p \text{ (or } \pi^+ n). \quad (2)$$

The standard model is due to Rein and Sehgal [35], which uses the Feynman–Kislinger–Ravndal (FKR) quark model to compute form factors for all baryon resonances up to $W \lesssim 2$ GeV. The modern implementation used in GENIE for resonance (RES) production is the Kuzmin–Lyubushkin–Naumov Berger–Sehgal (KLN-BS) model [36, 37, 38, 39], which combines the lepton-mass-corrected vector and axial form factors of Kuzmin *et al.* [36] with the updated $\Delta(1232)$ electromagnetic form factors of Graczyk and Sobczyk [37] and the improved treatment of the Δ line shape and the lepton-mass effects of Berger and Sehgal [38]. (Berger and Sehgal also discuss the interference between the resonant and non-resonant amplitudes; that interference term is not implemented in the GENIE v3 configuration underlying the MicroBooNE tune [16], where the non-resonant contribution is added incoherently.) Higher resonances ($N(1440)P_{11}$, $N(1520)D_{13}$, etc.) are included up to $W \lesssim 2$ GeV and contribute at $E_\nu \gtrsim 1.5$ GeV.

Non-resonant and DIS backgrounds. A non-resonant continuum contributes at all W ; in a full amplitude-level treatment it interferes (constructively or destructively by isospin) with the resonant amplitude, although the generators used here add it incoherently. Deep-inelastic scattering (DIS) becomes significant above $W \approx 1.6$ GeV. Together these processes represent $\sim 20\%$ of the $\text{CC}\pi^\pm$ rate in the energy range of this analysis and are modelled in GENIE via the Bodek–Yang formalism.

Coherent pion production. The neutrino can scatter coherently off the entire nucleus [30, 31], $\nu_\mu A \rightarrow \mu^- \pi^+ A^*$, leaving the nucleus in its ground state. The cross section is small ($\lesssim 10\%$ of the total) below 5 GeV. The signal definition of Sec. 7 is a final-state topology (one muon, one charged pion, no other mesons) and carries no truth-level veto on the interaction mode, so coherent $\text{CC}\pi^\pm$ events that satisfy it are *included in the signal*; they are a small signal component, not a background.

3.2 Nuclear Effects and Final-State Interactions

On a heavy nucleus such as argon ($A = 40$), two classes of nuclear correction modify the $\text{CC}\pi^\pm$ rate and kinematics significantly.

Initial-state effects. Fermi motion of the struck nucleon broadens kinematic distributions, and Pauli blocking suppresses transitions to low-momentum nucleon final states near threshold. Nuclear binding also shifts the effective nucleon mass relative to free-nucleon predictions.

Final-state interactions (FSI). The produced pion must traverse the nuclear medium before being detected. Intranuclear rescattering occurs via:

- *Pion absorption* ($\pi A \rightarrow NN \dots$): probability $\sim 25\text{--}30\%$ on Ar; converts $\text{CC}\pi^\pm$ events into $\text{CC}0\pi$ topology.
- *Charge exchange* ($\pi^+ n \rightarrow \pi^0 p$): produces a $\text{CC}\pi^0$ event.
- *Quasi-elastic scattering*: pion survives but changes direction; modifies p_π and θ_π distributions.
- *Inelastic production* ($\pi N \rightarrow \pi\pi N$): rare below $p_\pi \approx 400$ MeV/ c .

FSI are modelled in GENIE via the INTRANUKE (*hA*) cascade. Because absorption and charge exchange mix $CC\pi^\pm$, $CC\pi^0$, and $CC0\pi$ topologies at the $\sim 30\%$ level, the signal in any $CC\pi$ analysis is necessarily defined by the final-state particle content observed in the detector, not by the underlying production process (Section 7).

3.3 Generator Predictions and Known Tensions

Major neutrino event generators reproduce the inclusive CC cross section to 10–15% but show systematic spread in the $CC\pi^\pm$ rate, owing to differences in the resonance model, the treatment of the transition region between resonance and DIS, and the FSI cascade.

GENIE v3 [9, 20]. For resonance production GENIE implements the KLN-BS model [36, 37, 38, 39] for the full tower of 18 baryon resonances up to $W \lesssim 2$ GeV, including the $\Delta(1232)P_{33}$, $N(1440)P_{11}$ (Roper), $N(1520)D_{13}$, $N(1535)S_{11}$, $\Delta(1620)S_{31}$, $\Delta(1700)D_{33}$, $N(1680)F_{15}$, $\Delta(1950)F_{37}$, and their isospin partners. The hadronic invariant mass range $1.0 < W < 1.7$ GeV is bridged by a resonance–DIS transition using the AGKY (Andreopoulos–Gallagher–Kehayias–Yang) model, which applies KNO scaling in the transition region and the Bodek–Yang DIS model above $W \approx 1.7$ GeV. Non-resonant single-pion production is included through the low- W extension of the Bodek–Yang DIS model and is added incoherently to the resonant contribution; no resonant/non-resonant interference is implemented. Coherent π^+ production uses the Rein–Sehgal coherent model. FSI are handled by the INTRANUKE *hA* empirical model, in which each hadron undergoes at most one effective interaction whose probability and outcome are tuned to hadron–nucleus data; it is the alternative *hN* microscopic cascade, also available in GENIE, that propagates hadrons step-by-step using free-nucleon πN and NN cross sections. GENIE has been reported to over-predict the $CC\pi^\pm$ rate by 20–35% relative to bubble-chamber and MiniBooNE data in several comparisons when normalised to the BNL dataset; the tension is reduced but not eliminated in the AR23 tune.

NuWro [21]. NuWro models resonance production via the $\Delta(1232)$ only; no higher baryon resonances are included. The Δ is treated with the Paschos–Yu formalism for the Δ excitation amplitude, retaining the full angular structure of the pion decay. A smooth analytic interpolation connects the single- Δ resonance region to the DIS regime at higher hadronic invariant mass ($W \gtrsim 1.4$ GeV), using Bodek–Yang structure functions for the deep inelastic contribution; the transition avoids a hard boundary between resonance and continuum. Non-resonant single-pion production is included as a separate amplitude. FSI are implemented using the microscopic Salcedo–Oset model, which treats pion propagation in nuclear matter through the pion self-energy in the random-phase approximation, naturally accounting for medium modifications such as Δ broadening and pion absorption via two-body processes ($\pi NN \rightarrow NN$) that are absent in purely cascade-based approaches. NuWro has historically shown better agreement with MiniBooNE and bubble-chamber data than GENIE, partly attributable to the absence of overestimated higher-resonance contributions.

NEUT [40, 23]. NEUT, the generator used by T2K and Super-Kamiokande, implements resonance production via a modified Rein–Sehgal model including 18 resonances up to $W \lesssim 2$ GeV, with Graczyk–Sobczyk [37] electromagnetic form factors for the $\Delta(1232)$. The axial mass M_A^{RES} is a tunable parameter; T2K applies a data-driven tune that reduces the overall $CC\pi^\pm$ normalisation by ~ 10 –20% relative to the default. The transition to DIS is handled similarly to GENIE, using GRV98 LO parton distribution functions with Bodek–Yang corrections above $W \approx 2$ GeV. Coherent pion production follows the Rein–Sehgal coherent model with a Berger–Sehgal-style correction for finite lepton mass. FSI are modelled by a custom NEUT intranuclear cascade that tracks each produced hadron through the nucleus using step-by-step interactions sampled from

πN cross sections, with nuclear density and mean-free-path corrections derived from the optical potential.

GiBUU [22]. The Giessen Boltzmann–Uehling–Uhlenbeck (GiBUU) generator takes a fundamentally different approach: instead of a probabilistic step-by-step cascade, it solves a coupled set of kinetic transport equations for the full hadronic system inside the nuclear medium. Resonance production includes the $\Delta(1232)$ and higher resonances whose in-medium spectral functions are broadened by collisional damping; the Δ width in the nuclear medium is substantially larger than its free-space value of $\Gamma_0 \approx 118$ MeV, suppressing threshold production. Pion propagation after production is governed by a mean-field potential (the pion optical potential), which shifts pion momenta as they traverse the nucleus – an effect absent in cascade models and particularly important for low-momentum pions ($p_\pi \lesssim 200$ MeV/ c) near the pion production threshold. The transition to DIS uses PYTHIA for string fragmentation above $W \approx 1.6$ GeV. GiBUU predictions for pion angular distributions and for the backward-hemisphere $d\sigma/d\cos\theta_\pi$ can differ substantially from cascade-based generators, making it a valuable cross-check of FSI systematics.

Because the present results are extracted on central-value fake data, no data/MC deficit is measured here. What the analysis does provide is the spread of the standalone generators in the measured phase space, a factor of ~ 1.6 between GENIE and NEUT (Table 21), which brackets the injected MicroBooNE tune; the over-prediction observed for several generator configurations in parts of the relevant phase space in the measurements surveyed below is the context for that spread, not a result of this note.

3.4 Previous Experimental Measurements

Table 2 summarises published $\text{CC}\pi^+$ measurements on nuclear targets ordered chronologically.

Table 2: Selected $\text{CC}\pi^+$ cross-section measurements on nuclear targets. $\langle E_\nu \rangle$ is the approximate mean or peak neutrino energy. “Type” indicates whether a total integrated (T) or differential (D) cross section was reported. “Target” gives the nucleus primarily used for cross-section extraction.

Experiment	Target	$\langle E_\nu \rangle$ (GeV)	Type	Key result / reference
ANL 12-ft [41]	D	1.0	T	Bubble chamber; foundational σ vs E_ν
BNL 7-ft [42]	D	1.5	T	Higher statistics; 15% normalisation tension with ANL
K2K [43]	C	1.3	T	First accelerator $\text{CC}\pi^+$ on carbon near detector
MiniBooNE [44]	CH ₂	0.7	D	Q^2 , W , p_μ , θ_μ ; 20% excess over NUANCE
MINERvA [46]	CH	3.6	D	p_π , θ_π , Adler angles; NuMI medium-energy
T2K ND280 [49]	CH	0.85	D	p_μ , $\cos\theta_\mu$; off-axis beam similar to this work
ArgoNEUT [50]	Ar	9.6	T	First $\text{CC}\pi^+$ on Ar; 115 ν + 158 $\bar{\nu}$ events (1.25×10^{20} POT)
MicroBooNE BNB [52]	Ar	0.7	D	Full BNB dataset (1.11×10^{21} POT); $\sigma =$ (3.75 ± 0.80) $\times 10^{-38}$ cm ² /Ar
MicroBooNE NuMI ν_e [51]	Ar	0.73	D	First $\nu_e + \bar{\nu}_e$ $\text{CC}\pi^+$ on Ar; $\sigma = (0.93 \pm 0.28) \times$ 10^{-39} cm ² /nucleon

Deuterium bubble chambers: ANL and BNL (1979–1986)

The ANL 12-foot bubble chamber [41] and BNL 7-foot bubble chamber [42] provided the first high-statistics measurements of $\nu_\mu d \rightarrow \mu^- \pi^+ pp$ in the 1–3 GeV range. These measurements are important because the deuterium target is close to a free nucleon, minimising nuclear corrections relative to heavier targets. The ANL and BNL total cross sections disagree by $\sim 15\%$ in absolute normalisation, a discrepancy long attributed to flux uncertainties; the BNL flux is now believed to be better constrained. This ANL–BNL tension directly propagated into the spread of generator predictions: models tuned to ANL tend to predict lower rates than those tuned to BNL. The MINERvA experiment [48] subsequently resolved much of this tension by simultaneously constraining the flux and the cross section.

K2K near detector (2008)

The K2K experiment [43] made the first accelerator-based total $\text{CC}\pi^+$ cross-section measurement on a carbon target at $\langle E_\nu \rangle = 1.3$ GeV using its fine-grained near detector. The result was consistent with NEUT predictions and established the nuclear-target correction from deuterium to carbon at intermediate energies.

MiniBooNE (ν_μ on CH_2 , BNB beam, 2011)

The MiniBooNE detector (818-tonne mineral-oil Cherenkov detector) published differential $\text{CC}\pi^+$ cross sections on CH_2 in the Booster Neutrino Beam ($E_\nu^{\text{peak}} \approx 0.7$ GeV) [44]. MiniBooNE was the first experiment to provide high-statistics differential distributions as a function of Q^2 , the invariant hadronic mass W , and the outgoing muon and pion kinematics. The flux-integrated total cross section was measured to be $\sigma \approx 3.4 \times 10^{-38}$ cm² per CH_2 molecule, or equivalently $\sim 2.0 \times 10^{-38}$ cm² per carbon nucleus at $\langle E_\nu \rangle \approx 1$ GeV [52]. The measured rate exceeded the NUANCE prediction by $\sim 20\%$, i.e. that model *under*-predicted the data; the “generator over-prediction” discussed in the later literature refers to the opposite tension, in which GENIE-based predictions lie above subsequent measurements in parts of the phase space, and is generator- and phase-space-dependent rather than universal. The hydrogen content of CH_2 complicates the extraction of the carbon-only cross section, and the spherical Cherenkov geometry makes it difficult to measure low-energy pions or reconstruct proton recoils. These limitations drove subsequent adoption of finer-grained tracking detectors and liquid-argon targets.

MINERvA (ν_μ on CH, NuMI medium energy, 2015–)

The MINERvA experiment [46] uses a finely-segmented scintillator-strip tracking calorimeter in the NuMI medium-energy beam ($\langle E_\nu \rangle \approx 3.6$ GeV). It has published the most kinematically complete $\text{CC}\pi^+$ measurements to date, reporting differential cross sections as a function of: pion momentum p_π , pion angle θ_π , muon momentum p_μ , muon angle θ_μ , and the full set of Adler angles describing the pion in the hadronic centre-of-mass frame. MINERvA also measured $\text{CC}\pi^+$ on iron and lead targets [47], providing direct constraints on nuclear A -dependence at the same beam energy. GENIE consistently over-predicts MINERvA by 20–35% across all kinematic variables in the $\Delta(1232)$ peak, the same qualitative tension seen at lower energies. The higher beam energy at MINERvA means that the $p_\pi > 0.5$ GeV/ c region is well-populated, making it complementary to the MicroBooNE NuMI result which is statistics-limited at high pion momentum.

T2K ND280 (ν_μ on CH, off-axis beam, 2020)

T2K measured $\text{CC}\pi^+$ at its near detector ND280 in the off-axis T2K beam (OAB), which peaks at $\langle E_\nu \rangle \approx 0.85$ GeV [49]. The T2K OAB spectrum closely overlaps with the MicroBooNE NuMI

off-axis spectrum in the kinematically-sensitive 0.3–1.5 GeV range, making this the most directly comparable prior measurement to the present work. T2K reported differential cross sections in p_μ and $\cos\theta_\mu$, finding a data/MC deficit of $\sim 30\%$ relative to NEUT. The carbon target ($A = 12$) differs from argon ($A = 40$), but the overlapping beam spectrum enables direct comparison of nuclear-mass dependence at fixed neutrino energy.

ArgoNEUT ($\nu_\mu + \bar{\nu}_\mu$ on Ar, NuMI, 2017)

ArgoNEUT [50] was a 175-litre LArTPC placed upstream of the MINOS near detector in the NuMI low-energy beam ($\langle E_\nu \rangle \approx 9.6$ GeV). Using 1.25×10^{20} POT, ArgoNEUT selected 115 ν_μ and 158 $\bar{\nu}_\mu$ CC π^+ candidate events after background subtraction [52], publishing the first CC π^+ total cross section on argon as a function of neutrino energy. The result was consistent with GENIE predictions at these high energies, where sensitivity to the $\Delta(1232)$ resonance is suppressed. The LArTPC technology and argon target make ArgoNEUT the direct predecessor to MicroBooNE, but the factor ~ 10 higher beam energy means that extrapolation to the $E_\nu < 2$ GeV regime relevant for oscillation experiments requires generator assumptions and limits the constraining power on FSI models in the resonance region.

MicroBooNE BNB channel: inclusive precedent (2022)

The same MicroBooNE detector used in this analysis published the first measurement of energy-dependent inclusive ν_μ charged-current cross sections on argon in the Booster Neutrino Beam [32] ($E_\nu^{\text{peak}} \approx 0.7$ GeV), establishing the detector’s absolute cross-section capability on argon together with the BNB flux modelling and reconstruction performance relevant to the present work. The dedicated exclusive CC π^+ measurement in the same beam followed on the full BNB dataset (below), and it is that exclusive result to which the present NuMI CC π^+ measurement is most directly compared.

MicroBooNE BNB channel: full dataset (2026)

Reference [52] reports CC π^+ differential cross sections using the complete MicroBooNE BNB dataset of 1.11×10^{21} POT [52], selecting 6816 events after background subtraction and measuring differential cross sections in muon momentum, muon angle, pion momentum, and pion angle – the same four observables as the present analysis. The flux-integrated total cross section is

$$\sigma = (3.75 \pm 0.07_{\text{stat}} \pm 0.80_{\text{syst}}) \times 10^{-38} \text{ cm}^2/\text{Ar}$$

at mean neutrino energy $\langle E_\nu \rangle \approx 0.8$ GeV. Most generators reproduce the data within uncertainties; a notable exception is the muon angle with respect to the beam, where all tested generators overestimate the cross section at very forward angles ($\cos\theta_\mu \rightarrow 1$) [52]. The pion momentum differential cross section reported in this analysis represents the first such measurement on an argon target.

The BNB and NuMI analyses probe complementary phase space: BNB is on-axis with a narrow ν_μ spectrum peaked at ~ 700 MeV and negligible $\bar{\nu}_\mu$ content, while NuMI FHC at 135 mrad off-axis provides a softer, broader spectrum with a $\sim 38\%$ $\bar{\nu}_\mu$ admixture and extends sensitivity to $E_\nu \sim 0.1$ – 0.5 GeV.

MicroBooNE NuMI $\nu_e + \bar{\nu}_e$ CC π^+ (2025)

Reference [51] presents the first measurement of $\nu_e + \bar{\nu}_e$ CC single charged-pion differential cross sections on argon, using the full MicroBooNE NuMI dataset (2.0×10^{21} POT, FHC and RHC combined). The observable set – electron energy E_e , electron angle θ_e , pion angle θ_π , and

electron–pion opening angle $\theta_{e\pi}$ – mirrors the muonic observables of the present analysis. The flux-integrated total cross section is measured to be

$$\sigma = (0.93 \pm 0.13_{\text{stat}} \pm 0.27_{\text{syst}}) \times 10^{-39} \text{ cm}^2/\text{nucleon}$$

at a mean ν_e energy of 730 MeV, in good agreement with all tested generator predictions (GENIE, NuWro, NEUT, GiBUU) within the combined uncertainties. Prior to this measurement, $\text{CC}\pi^+$ production by electron neutrinos on argon had never been measured; the result is a key input to the DUNE ν_e -appearance analysis, where $\text{CC}\pi^+$ events are a background to the ν_e quasi-elastic signal.

4 Detector

4.1 Operating Principle

A liquid argon time projection chamber records three-dimensional ionisation images of charged-particle tracks by exploiting two properties of ultra-pure liquid argon:

1. Ionisation electrons produced by a traversing charged particle are not captured by argon atoms and can drift centimetres or metres under an applied electric field without significant recombination, provided the purity is sufficient (O_2 -equivalent $\lesssim 100$ ppt).
2. Argon scintillates at 127 nm (vacuum ultraviolet), producing $\sim 24\,000$ photons per MeV of deposited energy, enabling nanosecond-scale timing.

An applied electric field drifts the ionisation electrons to a set of sense-wire planes at the anode, which record three two-dimensional projections of each track. The third coordinate (along the drift direction) is reconstructed from the known drift velocity and the timing of the signals relative to the beam-gate start, combining the two-dimensional wire images into a full 3D reconstruction.

MicroBooNE operated from October 2015 to July 2021 at the Fermilab Accelerator Laboratory, collecting data in both the Booster Neutrino Beam (BNB) and, as a parasitic experiment, the NuMI beam [19]. It was the world’s largest operating LArTPC until the start of SBND and ICARUS data-taking.

4.2 Active Volume and Electric Field

The MicroBooNE active volume has rectangular geometry: 256 cm (drift, x) $\times$ 233 cm (height, y) $\times$ 1037 cm (beam, z), containing ≈ 87 tonnes of liquid argon at ≈ 87 K. The cathode is a stainless-steel plane at $x = 256$ cm held at -70 kV; the anode wire planes are at $x = 0$. The resulting 273 V/cm drift field gives an electron drift velocity of ≈ 1.11 mm/ μs , so the maximum drift time across the full gap is approximately 2.3 ms.

Liquid argon purity is maintained by continuous recirculation through oxygen and water absorbers, keeping the electron lifetime well above 5 ms (corresponding to O_2 -equivalent contamination below ~ 100 ppt). At this purity level, charge attenuation over the full 2.56 m drift is less than 10%.

4.3 Wire Readout and Signal Formation

Three wire planes at the anode sample the drifting charge:

- **U plane** (first induction): 2400 wires at $+60^\circ$ from the vertical, 3 mm pitch.
- **V plane** (second induction): 2400 wires at -60° from the vertical, 3 mm pitch.
- **Y plane** (collection): 3456 vertical wires, 3 mm pitch.

Induction-plane wires are biased to be transparent to the drifting electrons, producing bipolar (derivative-like) induced signals; collection-plane wires absorb the charge and produce unipolar signals. The combination of two stereo induction views at $\pm 60^\circ$ and a vertical collection view provides sufficient 2D information in each plane to resolve ambiguities during 3D reconstruction.

Front-end electronics (application-specific integrated circuits, ASICs) are mounted inside the cryostat on the wire planes and cold-multiply each channel at ≈ 87 K, reducing thermal noise. The analogue signals are digitised at 16 MHz inside the cryostat by 12-bit ADCs and transmitted over ~ 110 m of cold cables to warm readout electronics. The data stream is sampled into a circular buffer; beam triggers and cosmic triggers select 4.8 ms readout windows (corresponding to $\approx$ two full drift windows) for permanent storage.

A dead-wire region at $z \in [675.1, 775.1]$ cm interrupts the collection plane over a 100 cm span due to a shorted wire bundle. Tracks crossing this region have degraded hit-counting and range-based momentum estimation; the fiducial volume (Section 8) excludes this region.

4.4 Photon Detection System

Thirty-two 8-inch Hamamatsu R5912-MOD photomultiplier tubes (PMTs) are installed behind the collection-plane wire frame facing the active volume. The PMTs detect the 127 nm argon scintillation light via a tetraphenyl-butadiene (TPB) wavelength-shifting coating that converts VUV photons to visible (~ 430 nm) light. The system serves two purposes in this analysis:

1. **Beam-gate synchronisation:** the PMT flash time identifies which 1.6 μ s NuMI spill window produced the neutrino interaction.
2. **Flash-matching for cosmic rejection:** the Pandora reconstruction associates PFParticle hypotheses with the PMT flash closest in time and position, suppressing out-of-time cosmic-ray muons that overlap with beam-induced events in the long 2.3 ms drift window.

4.5 Calibration and Corrections

Three independent calibration inputs are used:

- **UV laser system** [11]: two pulsed laser beams produce straight ionisation tracks at precisely known locations. Deviations of the reconstructed track positions from the known trajectories are used to build a three-dimensional space-charge distortion map [10], which is applied to all reconstructed hits as a position correction. Distortions reach ~ 10 cm near the anode and cathode edges.
- **Stopping cosmic-ray muons:** the Bragg peak of cosmic muons stopping inside the active volume calibrates the absolute energy scale and the effective recombination factor as a function of local electric field and drift distance. Through-going muons calibrate relative wire-by-wire gains and the electron lifetime.
- **Michel electrons:** $\mu^- \rightarrow e^- \bar{\nu}_e \nu_\mu$ decay products from stopped muons provide a check of the calorimetric reconstruction at $E_e \lesssim 55$ MeV.

4.6 Cosmic Ray Environment and Reconstruction

Operating at the surface with minimal overburden, MicroBooNE is exposed to a cosmic-ray muon rate of ≈ 5 kHz through the active volume. Each 4.8 ms readout window contains on average ~ 10 cosmic-ray tracks unrelated to the beam interaction. The 1.6 μ s NuMI spill itself contains $\lesssim 0.1$ expected cosmics on average, but the drift-window pile-up is the dominant background at early selection stages (Section 9).

Cosmic rejection proceeds in stages:

1. Flash-matching (PMT timing): candidate neutrino slices must have a consistent optical flash within the beam window.

2. Topological score: the Pandora-based topological neutrino score discriminates between neutrino-like vertex structures and through-going/stopping cosmic topologies.
3. Dedicated BDT scores: the `bdt_mip` and `bdt_cosmic` scores trained on track-level and event-level features further suppress residual cosmics that passed the topological cut.

The CRT (Cosmic Ray Tagger) sub-detector [34], consisting of scintillator panels surrounding the cryostat, provides an additional cosmic veto but is not used in the present analysis because the relevant branch variables are zero-filled in the NuMI data ntuples.

Reconstruction uses the Pandora [6, 12] multi-algorithm pattern recognition framework, which groups hits into PFParticles organised in a hierarchy with the neutrino interaction as the top-level particle. Track candidates are fitted with a Kalman-filter-based track fitter to obtain momentum from track curvature (negligible for <300 MeV/ c muons at 273 V/cm) and from range for contained tracks. Log-likelihood ratio particle identification (LLR PID) [13] using the dE/dx profile along the track is the primary tool for muon/pion/proton separation (Section 9).

5 Beam and Neutrino Flux

5.1 NuMI beam and flux composition

The NuMI beam is produced by 120 GeV protons from the Fermilab Main Injector striking a graphite target. Charged secondary hadrons are focused by two magnetic horns whose polarity selects the sign of the focused mesons. In Forward Horn Current (FHC) mode the horns focus positive pions and kaons, giving a beam with predominantly ν_μ content; in Reverse Horn Current (RHC) mode the polarity is reversed to focus negatives, giving a predominantly $\bar{\nu}_\mu$ beam. MicroBooNE is located 135 mrad (7.71°) off-axis from the NuMI beam axis¹, reducing the peak neutrino energy to the 0.5–2.5 GeV range and narrowing the energy spectrum. Because MicroBooNE sits off-axis and at comparatively low energy, the horn sign-selection is imperfect and both modes carry a sizeable wrong-sign component, most notably RHC, whose wrong-sign ν_μ fraction is large.

The flux composition at the MicroBooNE site is taken from the new-G4 central-value flux histograms (`flux_newg4.root`), integrated over $E_\nu > 60$ MeV to exclude an unphysical sub-threshold pile-up near $E_\nu = 0$ (removed by the flux group’s `RemoveFluxPeak` step; below the reaction threshold and irrelevant to the measurement). The muon-flavour split of these central-value histograms matches the author’s authoritative integrated fluxes exactly (Sec. 5.4):

Table 3: NuMI flux composition at the MicroBooNE site by neutrino type, for the FHC and RHC horn modes (new-G4 central-value flux `flux_newg4.root`, integrated over $E_\nu > 60$ MeV, the same threshold as the authoritative integrals of Table 5; above 0.1 GeV the RHC split would read 42.0/55.2%). FHC is ν_μ -dominant; RHC is $\bar{\nu}_\mu$ -dominant with a large wrong-sign ν_μ fraction. The combined row weights each mode by its data exposure (FHC 8.857, RHC 11.082×10^{20} POT; Table 4), giving a near-balanced $\nu_\mu/\bar{\nu}_\mu$ mix. The signal is charge-blind ($|\nu_{\text{pdg}}| = 14$), so both ν_μ and $\bar{\nu}_\mu$ contribute in every mode.

Horn mode	ν_μ	$\bar{\nu}_\mu$	ν_e	$\bar{\nu}_e$
FHC	63.4%	34.0%	1.7%	0.9%
RHC	44.2%	53.3%	1.4%	1.2%
Combined	53.0%	44.4%	1.5%	1.1%

Flux corrections from the Package to Predict the FluX (PPFX) [8] are applied per event via

¹From the detector position in beam coordinates, (5502, 7259, 67270) cm: 91.1 m transverse at 672.7 m downstream. Earlier versions of this note quoted 110 mrad. See §5.2.

the central-value weight `ppfx_cv` stored in the analysis ntuples. The absolute flux normalisation of Sec. 5.4 uses the same new-G4 histograms.

5.2 Beam direction in detector coordinates

Two different angles describe the MicroBooNE–NuMI geometry, and they must not be confused (Fig. 1). The *off-axis angle* is a property of the beamline: the detector sits 91.1 m off the NuMI axis at 672.7 m downstream of the target, a baseline of 678.8 m and an off-axis angle of 7.71° (134.6 mrad).² That angle says nothing about how the TPC is oriented.

The angle that enters the measurement is a different one: the direction in which the neutrinos travel *relative to the detector z axis*. Averaged over interactions in the active volume, the true neutrino direction in detector coordinates is $(0.4624, 0.0489, 0.8853)$, which lies 28.6° away from z . The detector z axis is therefore a poor stand-in for the beam: it is not the direction the neutrinos came from, nor the NuMI beamline axis (itself 23.2° from z).

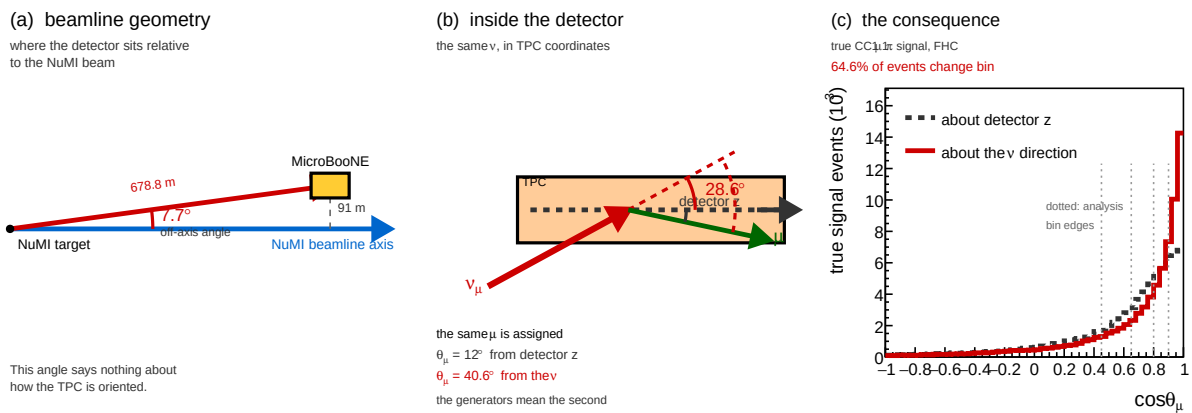


Figure 1: The NuMI geometry seen by MicroBooNE. (a) Beamline geometry, to scale: the detector sits 91.1 m off the NuMI axis at a baseline of 678.8 m, an off-axis angle of 7.7° . (b) The same neutrinos in TPC coordinates, to scale (the x – z projection; the quoted 28.6° is the full three-dimensional angle): the beam arrives 28.6° away from the detector z axis, so one and the same muon is assigned $\theta_\mu = 12^\circ$ about z but $\theta_\mu = 40.6^\circ$ about the neutrino. The generators mean the second. (c) The resulting shift in the true $\cos \theta_\mu$ distribution; dotted lines mark the analysis bin edges. Under the two conventions 64.6% of signal events fall in a different bin (44.6% for $\cos \theta_\pi$).

This matters because every polar angle and every transverse-kinematic-imbalance variable is defined with respect to the neutrino direction. In the generator predictions the neutrino defines $+z$ by construction — the GENIE event records used here have $p_x^\nu = p_y^\nu = 0$ and $p_z^\nu = E_\nu$ exactly — so a measurement that refers its angles to the detector z axis is not being compared with the prediction it appears to be compared with. The effect is not small: 64.6% of signal events change $\cos \theta_\mu$ bin between the two conventions, and 44.6% change $\cos \theta_\pi$ bin. For the TKI variables the axis error exceeds the observable itself, the mean δp_T being 0.742 GeV/ c about z against 0.264 GeV/ c about the beam.

Accordingly, all angular and TKI observables in this analysis are referred to the neutrino direction, following the β convention of the NuMI ν_e CC1 π internal note [1] and of the earlier NuMI ν_e cross-section measurement it derives from [2]. At truth level the exact per-event neutrino direction is used, which reproduces the generator convention exactly. At reconstruction level the neutrino direction is approximated by the vector from the NuMI target — at

²Computed from the detector position in beam coordinates, (5502, 7259, 67270) cm, taken from the same rotation matrix used for the NuMI beamline geometry weights.

(−31388, −3316, −60100) cm in detector coordinates, a 684 m baseline — to the reconstructed interaction vertex. Adopting the same β as the ν_e analysis keeps the two NuMI measurements directly comparable.

The choice of reconstruction-level axis is in any case not a sensitive one. A single fixed axis and the per-event β differ by a median of 2.4 mrad (maximum 7.3 mrad), which moves only 0.42% of muons across a $\cos\theta_\mu$ bin edge — to be compared with the 64.6% moved by the detector- z convention. Against the true per-event direction the two perform almost identically, with a mean opening angle of 1.95° for β and 2.00° for a fixed axis. What neither can recover is the physical divergence of the beam: neutrinos reaching the detector come from decays distributed along the decay pipe, not from the target, and over a 684 m baseline a 2.5 m detector subtends only ~ 4 mrad, so the vertex position carries almost no information about the direction. That residual is a resolution effect, absorbed by the response matrix like any other.

Quantities that are invariant under this choice of axis are unaffected: p_μ , p_π , the muon–pion opening angle $\theta_{\mu\pi}$, the invariant mass $W_{\pi p}$, and all integrated cross sections. The event selection is likewise unaffected, as no selection cut and no BDT input variable uses a polar angle.

5.3 Data exposure

The analysis is *blind*: the real beam-on data are not used. Each configuration is extracted on Poisson-thrown central-value fake data, generated and scaled *per run* to that run’s data POT (Table 4). The FHC measurement uses the full FHC exposure (Runs 1, 2, 4 and 5); RHC uses Runs 1, 2, 3, 4a, 4b and 4c; the combined measurement uses all fourteen overlay files. Every run is normalised independently to its own data exposure, so the prediction carries the data-POT mixture of run periods and the framework is ready for real per-run beam-on data (each fake-data sample is a drop-in replacement for its beam-on counterpart).

Table 4: Run-period exposure accounting (final, full FHC+RHC). “MC POT” is each period’s native simulated exposure; “Data POT” is the per-run exposure to which the (blind, per-run) fake data is scaled. *Each run is normalised independently* by its own factor, $\text{Scale} = (\text{run Data POT})/(\text{run MC POT})$, so the prediction is the data-POT-weighted mixture of run periods; the normalisation a real per-run beam-on dataset produces. This replaces an earlier single per-mode pooling factor: because the per-run MC/data POT ratios vary by up to a factor of three (Scale ranges 0.051–0.141 within FHC alone), a single factor would weight the runs by their MC statistics rather than by the data exposure. The RHC Run 4 row sums the 4a/4b/4c sub-files and Run 3 sums aa–ae; sub-files sharing a run’s data POT are scaled by the group MC total so they pool within that run. The combined measurement applies each run’s own factor. Run 5 uses the reweighted-PPFX ν_μ overlay. The scheme is validated by the FHC closure (unfolded vs. smeared truth $\chi^2 = 1.6/7$, $p = 0.98$).

Run period	MC POT ($\times 10^{20}$)	Data POT ($\times 10^{20}$)	Scale
Run 1 (FHC)	23.282	3.283	0.1410
Run 2 (FHC)	24.934	1.268	0.0509
Run 4 (FHC)	28.335	2.075	0.0732
Run 5 (FHC)	19.300	2.231	0.1156
FHC total	95.85	8.857	<i>per run</i>
Run 1 (RHC)	8.997	0.605	0.0673
Run 2 (RHC)	57.865	2.591	0.0448
Run 3 (RHC, aa–ae)	55.185	5.003	0.0907
Run 4 (RHC, 4a–4c)	32.586	2.883	0.0885
RHC total	154.63	11.082	<i>per run</i>
Combined (FHC+RHC)	250.49	19.939	<i>per run</i>

The full exposure substantially reduces the data-statistical term (Table 18 of the technical supplement–Table 20 of the technical supplement), leaving the measurement flux- and detector-systematic limited. When the analysis is unblinded, the real beam-on POT (and gate counts) replace the data-POT values above.

5.4 Integrated flux

The absolute flux normalisation uses the new-G4 GEANT4 NuMI flux (`flux_newg4.root`), integrated over $E_\nu > 60$ MeV (the flux group’s `RemoveFluxPeak` cut, which removes an unphysical ~ 25 – 30 MeV artifact spike). The signal is charge-blind ($|\nu_{\text{pdg}}| = 14$, covering both $\nu_\mu \rightarrow \mu^- \pi^+$ and $\bar{\nu}_\mu \rightarrow \mu^+ \pi^-$), so the normalisation uses the combined $\nu_\mu + \bar{\nu}_\mu$ flux $\Phi = \Phi_{\nu_\mu} + \Phi_{\bar{\nu}_\mu}$. Because RHC is $\bar{\nu}_\mu$ -dominant with a different absolute flux than FHC, each configuration uses its *own* integrated flux:

Table 5: Authoritative integrated ν_μ and $\bar{\nu}_\mu$ fluxes per POT (new-G4 flux, $E_\nu > 60$ MeV), and the combined $\nu_\mu + \bar{\nu}_\mu$ value used as the per-configuration normalisation. The combined row is the data-POT-weighted average of the two modes (weights 8.857 : 11.082; Table 4). The FHC value 6.81159×10^{-10} is the previously confirmed normalisation; RHC and combined were previously normalised with the FHC flux and are corrected here (raising the RHC cross section by 5.7% and the combined by 3.1%).

Config	Φ_{ν_μ} ($10^{-10} \text{ cm}^{-2} \text{ POT}^{-1}$)	$\Phi_{\bar{\nu}_\mu}$ ($10^{-10} \text{ cm}^{-2} \text{ POT}^{-1}$)	$\Phi_{\nu_\mu + \bar{\nu}_\mu}$ ($10^{-10} \text{ cm}^{-2} \text{ POT}^{-1}$)	$\bar{\nu}_\mu$ frac.
FHC	4.43515	2.37644	6.81159	34.9%
RHC	2.92348	3.52298	6.44646	54.6%
Combined	3.59497	3.01368	6.60865	45.6%

The mean $\bar{\nu}_\mu$ energy is 0.500 GeV (FHC) and 0.417 GeV (RHC), the lower RHC value reflecting the softer wrong-sign-enhanced spectrum. These fluxes are set per configuration through the `Flux` line of each cross-section configuration (`CrossSectionExtractor.hh`). Because every simulated event carries the PPFX central-value flux correction `ppfx_cv` (§6; mean 0.944 on selected signal), the integrated flux that divides the event rate must be the PPFX-corrected central value, not the raw GEANT4 flux; the `*_cv_*` histograms of `flux_newg4.root` are taken to be that central value, and this should be confirmed with the flux author before unblinding, since a raw-G4 integral would put every absolute cross section $\sim 5\text{--}6\%$ low. The correction relative to the superseded flux (`uboone_numi_flux_histograms.root`, whose combined value 2.3725×10^{-9} was $\sim 3.48\times$ too large and made every extracted cross section $\sim 3.5\times$ too small) was validated four independent ways, standalone NuWro and GENIE on the old vs. corrected flux, the flux author’s numbers, and a GENIE closure, documented in `generator_predictions/`.

The number of argon target nuclei in the fiducial volume is computed from first principles as

$$N_{\text{Ar}} = \frac{\rho_{\text{LAr}} \cdot V_{\text{FV}} \cdot N_A}{A_{\text{Ar}}} = 8.710 \times 10^{29}, \quad (3)$$

with $\rho_{\text{LAr}} = 1.3836 \text{ g/cm}^3$, $A_{\text{Ar}} = 39.948 \text{ g/mol}$ and $N_A = 6.02214076 \times 10^{23} \text{ mol}^{-1}$. The fiducial box is $10 < x < 246$, $-101 < y < 101$, $10 < z < 986 \text{ cm}$ ($4.6528 \times 10^7 \text{ cm}^3$), from which the $675.1 < z < 775.1 \text{ cm}$ dead region is subtracted, leaving $V_{\text{FV}} = 4.1761 \times 10^7 \text{ cm}^3$. Without that subtraction the same expression would give 9.705×10^{29} . This is the value the code uses, computed by `num_Ar_targets_in_FV(FV, 675.1, 775.1)` in `CrossSectionExtractor.hh`, and it is the value entering the conversion factor of §11 and the cut-and-count normalisation of §12.3. and the per-configuration normalisation factor is $N_{\text{norm}} = \Phi_{\text{mode}} \cdot N_{\text{POT,mode}} \cdot N_{\text{Ar}}$, evaluated with each mode’s flux and data POT (Table 4).

6 Monte Carlo Samples

Monte Carlo events are generated with GENIE v3 [9, 20] (the MicroBooNE tune [16]) on argon-40, propagated through the detector with Geant4 [24], and overlaid with real cosmic-ray data (the “overlay” technique) so that simulated events experience in-situ detector noise and cosmogenic activity. Table 6 summarises the input samples.

Table 6: Input samples (final, full FHC+RHC). “ n ” is the number of overlay files; “MC POT” is the summed native simulated exposure ($\times 10^{20}$). The blind Poisson-thrown fake data stands in for the (unused) beam-on data; the detector variations use the native Run-4 FHC and RHC samples. The per-file lists and POT/trigger values are in the `configs/file_properties_numi_*` tables.

Sample	Composition	n	MC POT
ν_μ overlay (FHC)	Runs 1, 2, 4 (pandora) + Run 5 (reweighted-PPFX)	4	95.85
ν_μ overlay (RHC)	Runs 1, 2, 3, 4a, 4b, 4c (pandora)	10	154.63
Dirt MC	GENIE NuMI dirt overlay (Run 1)	1	16.74
EXT (beam-off)	cosmic data, per run, horn-mode-independent	7	n/a
Detector variations	Run-4 FHC + Run-4 RHC, CV + 8 knobs each	18	n/a
Fake data (blind)	Poisson-thrown central value, per configuration	n/a	n/a
Beam-on data	blinded, not used until unblinding	n/a	n/a

Event weights. Each MC event is weighted by

$$w = \text{Scale}[\text{run}] \times w_{\text{PPFX}} \times w_{\text{tune}}, \quad (4)$$

where $\text{Scale}[\text{run}] = (\text{run Data POT})/(\text{run MC POT})$ is the *per-run* POT factor of Table 4: each run period is scaled independently to its own data exposure, so the prediction reproduces the data-POT mixture of runs rather than the MC-statistics mixture (see the caption of Table 4). Operationally this is exactly what the framework does natively each overlay file carries its run’s data POT via a matching per-run beam-on (here, per-run fake-data) sample, and is scaled by `run_bnb_pot/summed_pot`. w_{PPFX} is the PPFX central-value flux correction (`ppfx_cv`) and w_{tune} is the GENIE/GEANT4 reweighting correction (`weightTune`). One weight rule is used everywhere (`safe_weight()` in `UniverseMaker.hh`):

$$w_{\text{safe}} = \begin{cases} 1, & w \text{ non-finite, } w < 0, \text{ or } w > 30, \\ w, & \text{otherwise (a weight of exactly zero is kept),} \end{cases} \quad (5)$$

applied to the product $w = \text{Scale} \times w_{\text{PPFX}} \times w_{\text{tune}}$ (the 0.65 dirt normalisation weight included). The universe construction, the covariance, the unfolding inputs, the cut-flow histograms (since the 2026-09-03 reprocessing), the count-level tables, the control-region yields, the transfer factors and the beam-on comparison at unblinding all use Eq. (5); the count-level macros carried a variant that zeroed invalid weights and did not clamp until 2026-09-03, which is why earlier drafts’ sideband yields differed from the systematics tables by a few per cent. At unblinding the per-run fake-data samples are replaced by the corresponding per-run beam-on files (same runs, same POT), so the normalisation is already correct.

EXT scaling. Beam-off data are cosmic-only, so a beam-off sample is matched to a beam-on run by *run period* alone, irrespective of the horn polarity it was recorded under. Each run period uses the beam-off sample(s) taken in that period, scaled by that run’s beam-on/beam-off gate ratio (times the 0.98 NuMI beam-occupancy factor); where a run has several beam-off files the framework sums their gate counts and pools them:

$$\text{EXT}_{\text{run}} = 0.98 \frac{\text{bnb_gates}[\text{run}]}{\sum_{\text{files} \in \text{run}} \text{ext_gates}} \sum_{\text{files} \in \text{run}} N_{\text{EXT}}^{\text{sel}}. \quad (6)$$

The gate counts are the analyser’s trigger table (2026-09-02): Run 1 4 582 248.27, Run 3b 18 512 842.15 + 14 136 286.5 = 32 649 128.65, Run 4 4a+4b 9 338 778.225, 4c 8 060 024.70, 4d 17 432 345.70 (all four pooled under Run 4), Run 5 19 256 341.475; the beam-on gates per run

are those of Table 4's runs (9 846 635, 3 535 129, 4 131 149, 5 154 196 for FHC Runs 1, 2, 4, 5 and 1 458 253, 5 422 907, 10 349 610, 6 304 167 for RHC Runs 1–4). No Run-2 beam-off sample exists and none will be produced, so the Run-1 sample, the nearest pre-CRT period, stands in for Run 2 permanently, scaled by its own gate count. The same beam-off files serve the FHC and RHC beam-on runs of a given period, and the combined configuration lists each (file, run) pair once. The selected beam-off rate per 10^6 gates is 1.75 (Run 1), 0.43 (Run 3b), 1.15 (Run 4) and 0.52 (Run 5): the pre-CRT Run 1 sample is roughly three times the CRT-era ones, which is why period matching rather than a pooled sample is used. At the FHC final cut the beam-off background is 30 events (23 RHC, 53 combined), 1.9% of the selected sample. Figure 2 shows the per-period rates, scale factors and resulting yields; the machine-readable version is `data_release/ext_gates.tsv`, and the normalisation manifest (§15) asserts that every (beam-off file, run) pair enters each configuration exactly once, that every symlink resolves, and that every gate count is a gate count.

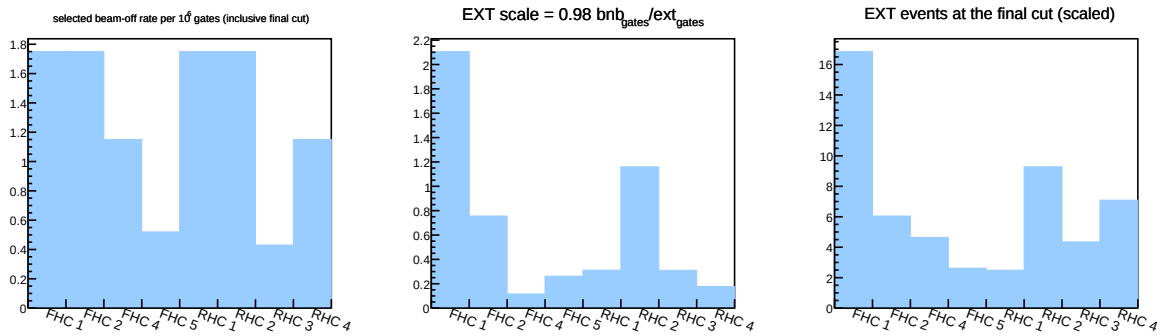


Figure 2: Beam-off validation by run period, from `data_release/ext_gates.tsv`: selected beam-off rate per 10^6 gates at the inclusive final cut (left), the per-run scale factor $0.98 \text{ bnb_gates}/\text{ext_gates}$ (centre) and the resulting beam-off events at the final cut (right), for each beam-on run period of the FHC and RHC configurations. The Run-1 sample serves Run 2 (both modes) and the RHC Run 1 as a stand-in. The pre-CRT Run-1 rate is three to four times the CRT-era ones.

The beam-off gate count was wrong until 2026-09-02. Earlier drafts scaled a single pooled Run-1+Run-3b beam-off sample by 3 821 593, a number inherited from the earlier custom selection where it was the sum $610\,496 + 3\,211\,097$ of the two samples' *event* counts; their gate count is 37 231 377, $9.7\times$ larger. The selected cosmic rate per gate exposed it (5.8 per 10^6 gates against 0.4–1.8 for the dedicated per-period files). With the pooled sample correctly scaled the FHC final-cut beam-off background was 13 events; the period-matched treatment above gives 30, because the pre-CRT Run-1 and Run-2 exposure is weighted by its own, higher, rate. The fake-data closure was never affected, because the same EXT prediction is added to the pseudo-data and subtracted again, but a real-data measurement with the old value would have over-subtracted ≈ 100 events, $\approx 10\%$ of the selected signal.

A directional cut against residual cosmics: considered and rejected. Part of the surviving beam-off sample is thought to be broken cosmic muon tracks, where the break is reconstructed as a vertex and the track direction bears no relation to it. That leaves a directional signature, and it is present in the data: among selected events the beam-off sample is enriched at backward reconstructed muon angles by nearly an order of magnitude relative to signal.

reco $\cos \theta_\mu$	EXT (beam-off)	MC signal	enrichment
< 0	31.8%	4.1%	$7.8\times$
< -0.5	9.1%	1.4%	$6.4\times$

The enrichment is real but the arithmetic does not support a cut. At the final cut the FHC sample is 943 signal, 30 EXT and 1 599 predicted in total (Tab. 10). A $\cos \theta_\mu < 0$ requirement would remove $\approx 32\%$ of the EXT, about 10 events, and $\approx 4\%$ of the signal, about 39 events. Because the beam-off sample is subtracted from the data directly, and that subtraction is unbiased by construction, removing part of it buys only a reduction in the subtraction’s *statistical* uncertainty, from $\sqrt{30} \approx 5.5$ to $\sqrt{20} \approx 4.5$ events. Spending 39 signal events to gain ≈ 1 event of uncertainty is a net loss.

The same bound applies to any cosmic cut, not just this one: EXT is 1.9% of the selected sample and is already subtracted without bias, so removing *all* of it for *zero* signal loss would gain about 5 events of statistical uncertainty on a sample of 1 599. (Earlier drafts, with the beam-off over-scaled by an order of magnitude, made the same argument with 128 EXT events; the conclusion is unchanged and the case is now stronger.) Any cut costing more than about 1% of the signal is a net loss regardless of how efficiently it tags cosmics. The residual beam-off contribution is therefore left in place and handled by subtraction.

A more surgical variant remains untested. The CC0 π analysis tags mis-directed tracks with the calorimetric flag `trk.bragg_mu_fwd_preferred_v` in coincidence with a poor forward-muon fit, $\chi_\mu^2 > 6$, which fires on tracks whose dE/dx profile disagrees with the assigned direction rather than on backward-going tracks as such. That would not cut genuine backward muons, which outnumber mis-reconstructed ones here by roughly 8:1. Both branches exist in the input ntuples but are not currently propagated by `ProcessNTuples`; the study is scoped in `report/BACKWARD_MUON_PLAN.md`. It is bounded by the same $\sqrt{30} \approx 5.5$ -event ceiling on the EXT statistical uncertainty, so it is worth measuring but cannot become a significant improvement.

Software trigger. Data and EXT files are pre-filtered to require the software trigger (`swtrig == 1`). MC events failing `swtrig = 0` are removed at the NeutrinoSlice cut stage. In practice this has no measurable effect at the current working point: all MC events passing the topological cut are verified to have `swtrig == 1` (46 810/46 810 events confirmed numerically).

7 Signal Definition

The signal is defined at generator level and is used only to construct the selection efficiency and response matrix. It is *never* applied as a reconstruction cut. The definition is *charge-blind*: both $\nu_\mu \rightarrow \mu^- \pi^+$ and $\bar{\nu}_\mu \rightarrow \mu^+ \pi^-$ are treated as signal on the same footing.

Neutrino flavour

$$|\nu \text{ PDG}| = 14 \text{ } (\nu_\mu \text{ or } \bar{\nu}_\mu).$$

Interaction type

Charged-current (`ccnc == 0`).

True vertex

Inside the fiducial volume (Section 8).

Primary multiplicity

Exactly one muon ($|\text{PDG}| = 13$), exactly one charged pion ($|\text{PDG}| = 211$), zero neutral pions ($|\text{PDG}| = 111$), zero kaons ($K^\pm, K^0, K_S, K_L$) and no other mesons (`sig_no_heavy_mesons_` in `CC1mu1piXp.cxx`). Any number of protons (and neutrons) is allowed.

Kinematic thresholds

$p_\mu > 0.15$ GeV/ c , $p_\pi > 0.175$ GeV/ c , and $\theta_{\mu\pi} < 2.6$ rad. These define the *measured phase space*, chosen from the efficiency turn-on curves (Section 10). The momentum efficiency turns on sharply at ~ 0.15 – 0.20 GeV/ c (set by the 10 cm muon and 20 cm pion track-length cuts); below that it is only ~ 1 – 3% with a strongly model-dependent correction. The opening-angle upper bound (2.6 rad) excludes the large-angle region, which is dominated by cosmic and dirt background ($\sim 7\%$ purity in the $\theta_{\mu\pi} > 2.513$ rad bin). No *lower* opening-angle cut is applied: the small-angle region is high-purity ($\sim 58\%$), so cutting it would only remove signal (it is low-efficiency but not low-purity). The same cut is applied at reco level so the signal and selection acceptances match. The first p_μ/p_π bins and the $\theta_{\mu\pi}$ upper edge start/end at these thresholds.

(Earlier iterations used $p_\mu, p_\pi > 0.1$ GeV/ c with the full 0 – π angle; that left near-dead efficiency bins at low momentum and at the large-angle extreme. The phase space above raises the efficiency from 12.1% to 15.6% at $\sim 55\%$ purity at the beam-off scaling then in use, see Section 10.)

Pion threshold and the sibling ν_e measurement. The companion MicroBooNE NuMI ν_e CC 1π analysis defines its pion signal by kinetic energy, $\text{KE}_\pi > 40$ MeV, equivalent to $p_\pi > 0.113$ GeV/ c . Our $p_\pi > 0.175$ GeV/ c ($\text{KE}_\pi > 84$ MeV) is therefore a higher threshold. Lowering it toward the ν_e value is possible, the signal definition is a truth-level choice and the unfolding maps reco to truth but the pion momentum reconstruction (track range under the muon hypothesis, mass-corrected) degrades rapidly below 0.175 GeV/ c . The table below is the study that led to *rejecting* a lower threshold; it is not part of the measurement. The measured phase space is $p_\pi > 0.175$ GeV/ c throughout this note, and no cross section is reported below it:

Table 7: Pion momentum reconstruction versus true p_π (correctly reconstructed signal pions). Below the 0.175 GeV/ c threshold the bias and resolution blow up and the reconstructed fraction halves.

True p_π [GeV/ c]	Bias $\langle (p^{\text{reco}} - p^{\text{true}})/p^{\text{true}} \rangle$	RMS resolution	Reco fraction
0.080–0.113	+131%	73%	3.4%
0.113–0.150	+66%	56%	3.4%
0.150–0.175	+30%	51%	4.3%
0.175–0.250 (threshold)	−8%	31%	10.0%
0.250–0.400	−30%	21%	10.4%

At the threshold the resolution is 31%; in the $[0.113, 0.175]$ GeV/ c region it is 51–56% with a +30–66% bias and only ~ 3 – 4% reconstructability. Extending the measurement to the ν_e threshold would thus add bins carrying large, strongly model-dependent unfolding systematics. For consistency with the ν_e measurement the threshold could be lowered to 0.113 GeV/ c with the $[0.113, 0.175]$ GeV/ c region kept in a single coarse bin to contain those systematics, or both phase spaces reported; this is a systematics-versus-cross-analysis-consistency trade rather than a reconstruction limitation.

Why the threshold was not lowered. Extending to the sibling ν_e threshold was investigated and rejected: the $[0.113, 0.175]$ GeV/ c region has 2.2% selection efficiency and a strongly model-dependent reconstruction correction. The study is recorded in the technical supplement. It is not part of this measurement, and no cross section is reported below $p_\pi > 0.175$ GeV/ c .

8 Fiducial Volume

The fiducial volume (FV) is defined in `CC1mu1piXp.cxx` (the independent cross-check selection carries the same numbers in `FV_new.h`):

$$x \in [10.0, 246.0] \text{ cm}, \quad y \in [-101.0, +101.0] \text{ cm}, \quad z \in [10.0, 986.0] \text{ cm}, \quad (7)$$

with the dead-wire region $z \in [675.1, 775.1] \text{ cm}$ excluded. The total FV volume is

$$\begin{aligned} V_{\text{FV}} &= (246 - 10) \times 202 \times (976 - 100) \text{ cm}^3 \\ &= 236 \times 202 \times 876 \text{ cm}^3 \approx 4.17 \times 10^7 \text{ cm}^3. \end{aligned} \quad (8)$$

A wider *containment* region (2 cm inset from the physical TPC walls) is used to require that the pion candidate track endpoint is contained:

$$x \in [2.0, 254.35] \text{ cm}, \quad y \in [-113.53, +107.47] \text{ cm}, \quad z \in [2.1, 1034.9] \text{ cm}. \quad (9)$$

8.1 Which volume applies where

Three distinct volumes appear above and it is worth stating plainly which is used for what, since “contained” and “in the fiducial volume” are not the same requirement.

volume	extent	used for
fiducial (FV)	$[10, 246] \times [-101, 101] \times [10, 986] \text{ cm}$ minus $z \in [675.1, 775.1] \text{ cm}$	neutrino vertex, true <i>and</i> reco
containment	$[2, 254.35] \times [-113.53, 107.47] \times [2.1, 1034.9] \text{ cm}$	track endpoints
active TPC	$[0, 256.35] \times [-115.5, 117.5] \times [0, 1036.9] \text{ cm}$	(reference only)

True and reco use identical dimensions. The framework keeps the two as separate quantities, `true.FV()` and `reco.FV()`, and other selections in the code base use that freedom. Here they are defined as the same box (`CC1mu1piXp.cxx`, `define_true_FV / define_reco_FV`). The signal definition tests the *true* neutrino vertex and the selection tests the *reconstructed* vertex, both through the same helper, so both inherit the dead-region veto as well. The truth and reconstructed boundaries are geometrically identical, so there is no *intentional* acceptance mismatch. Vertex-resolution and space-charge effects can nevertheless move a reconstructed vertex across these boundaries, and events that migrate across the FV or dead-region edge are represented in the simulated response (true-in, reco-out as inefficiency; true-out, reco-in as the out-of-FV background category of Table 10) rather than being absent.

The dead-region veto applies to the vertex only. An event whose vertex lies outside the unresponsive-wire slab is kept even if one of its tracks crosses that slab. This is the conventional choice, but it is an assumption rather than a necessity: tracks crossing dead wires have degraded calorimetry and range reconstruction, and the associated inefficiency is absorbed into the response matrix rather than removed by a cut.

The dead slab is also removed from the target count. The argon target number is computed from the FV with the slab subtracted, not from the full box. Retaining the slab would inflate the target count by $100/976 \approx 10\%$ and bias every cross section low by the same factor. Note that this subtraction is applied on the NuMI code path specifically; the generic path computes the target number without it, so the two must not be mixed.

Relation to the active volume. The FV is inset from the active TPC by approximately 10 cm in x , 15 cm in y , 10 cm at the upstream z face and 51 cm at the downstream face. The containment box, by contrast, is inset by only 2 cm: track endpoints are required to be inside the detector, not inside the fiducial region, which is the correct requirement for range-based momentum and for the pion-containment cut.

9 Event Selection

The event selection is implemented in the framework selection class `src/selections/CC1mu1piXp.cxx` (which produces the response matrices and spectra for all results in this note); the standalone `selection/ccpi_selection.C` pipeline used for the cross-pipeline validation of Sec. C of the technical supplement implements the byte-identical cut sequence. The selection comprises 11 sequential cuts.

9.1 BDT-based particle identification

Three gradient-boosted decision trees (TMVA [26]), trained on the MicroBooNE overlay MC, supply the calorimetric particle identification used in the muon- and pion-candidate cuts. All three share a common set of per-track inputs: the three Bragg-peak dE/dx likelihoods under the proton, muon and MIP hypotheses (`trk_bragg_{p,mu,mip}_v`), the log-likelihood-ratio PID score (`trk_llr_pid_score_v`), the Pandora track-vs-shower score (`trk_score_v`), and the space-charge-corrected track end-point coordinates (`trk_sce_end_{x,y,z}_v`):

- **MIP BDT** (`bdt_mip`; weights `dataset_MIP_BDT_no_len`): separates minimum-ionising tracks (muons, pions) from stopping protons. Both the muon and the pion candidate are required to have `bdt_mip > -0.1` (Cuts 4–5).
- **Pion BDT** (`bdt_pi`; weights `dataset_pion_BDT_no_len`): a dedicated pion-versus-proton discriminant applied to the pion candidate, `bdt_pi > -0.1` (Cut 5).
- **Muon BDT** (weights `dataset_muon_BDT`): the same inputs plus the track length (`trk_len_v`). It applies no threshold: the set of *qualifying* muon-candidate tracks is defined by the cuts of Table 8, and the muon BDT only *orders* that set; the qualifying track with the highest muon-BDT score is chosen as the muon (`CC1mu1piXp.cxx`, which earlier chose the longest qualifying track).

The two candidate-PID BDTs deliberately omit track length so the score does not bias the momentum spectra. The weight files are resolved from `$CC1MU1PIXP_BDT_WEIGHTS_DIR`, falling back to the in-tree `booster_decision_tree/` copy. A fourth score, `bdt_cosmic`, is available for cosmic rejection but is disabled by default (no benefit in an A/B test; the Pandora topological score already provides the cosmic rejection at Cut 3).

9.2 Cut definitions

Cut 0. EventsInTrueFV. All events. Reference denominator for truth-level studies.

Cut 1. NeutrinoSlice. ≥ 1 reconstructed track (`ntracks > 0`), `bdt_cosmic > 0.0` (currently disabled; no benefit found in A/B test), and `swtrig == 1`.

Cut 2. InFiducialVol. The SCE-corrected reconstructed neutrino vertex lies inside the FV.

Cut 3. Topological. `topological_score > 0.67`. This Pandora-produced multivariate score distinguishes cosmic-ray topologies from neutrino-like topologies and provides the primary cosmic rejection: EXT events are reduced by a factor of ~ 33 between Cut 2 and Cut 3.

- Cut 4. MuonCandidate.** At least one primary PFP track satisfies the muon PID criteria of Table 8. Among the qualifying tracks, the one with the highest muon-BDT score (§9.1) is the muon candidate; track length enters only through the 10 cm qualifying cut and as a BDT input, not as the ordering.
- Cut 5. ContainedPion.** Exactly one primary track (other than the muon candidate) satisfies the pion PID criteria of Table 9 and has its endpoint inside the containment volume. The track with the highest LLR PID score is the pion candidate.
- Cut 6. MuonIn3Planes.** The muon candidate has ≥ 1 hit on each of the U, V, Y planes.
- Cut 7. PionIn3Planes.** The pion candidate has ≥ 1 hit on each of the U, V, Y planes.
- Cut 8. ShowerCut.** Zero primary-level EM showers, unless exactly one shower is present with fewer than 50 Y-plane hits and a vertex distance > 10 cm (consistent with a delta-ray or low-energy nuclear photon).
- Cut 9. OpeningAngle.** The reconstructed $\mu\text{-}\pi$ opening angle satisfies $\theta_{\mu\pi} < 2.6$ rad ($\approx 149^\circ$), rejecting the large-angle cosmic/dirt background. No lower bound is applied (the small-angle region is high-purity). The same cut is imposed on the signal definition (Section 7).
- Cut 10.**
Nonprotons (final cut). Fewer than 4 primary tracks have LLR PID > 0.1 (non-proton-like), and the total number of primary reconstructed tracks is less than 5.

Table 8: Muon candidate PID thresholds (applied at Cut 4).

Variable	Threshold	Description
<code>trk_score_v</code>	> 0.80	Track-like PFP
Vertex distance	≤ 4.0 cm	Attached to ν vertex
Track length	> 10.0 cm	Minimum track length
LLR PID (<code>trk_llr_pid_score_v</code>)	> 0.20	Muon-like
MIP BDT (<code>bdt_mip</code>)	≥ -0.10	Minimum-ionising-particle

Table 9: Pion candidate PID thresholds (applied at Cut 5). Endpoint containment requires the track endpoint inside the containment volume. `trk_bragg_pion_v`: pion-hypothesis Bragg-peak likelihood score (rejects stopping protons).

Variable	Threshold	Description
<code>trk_score_v</code>	≥ 0.50	Track-like PFP
LLR PID	> 0.10	Pion / non-proton-like
Vertex distance	≤ 4.0 cm	Attached to ν vertex
Track length	> 20.0 cm	Minimum track length
MIP BDT (<code>bdt_mip</code>)	> -0.10	MIP-like
Pion BDT (<code>bdt_pi</code>)	> -0.10	Pion-like (vs. proton)
Bragg-pion score	≥ 0.08	Proton rejection
Endpoint contained	yes	Inside containment volume

9.3 Kinematic reconstruction

Muon momentum. A *hybrid* estimator, switched on containment of the muon-candidate track: the range-based momentum `trk_range_muon_mom_v` is used when the track ends inside the detector, and the multiple-Coulomb-scattering momentum `trk_mcs_muon_mom_v` when it exits (range is meaningless for an exiting track). The reported p_μ is therefore `candidate_muon_mom_reco`, not either branch alone. In the selected FHC sample the split is 37% contained (range) and 63% exiting (MCS [14]), i.e. the measurement is *MCS-dominated*: the NuMI muons are energetic and the MicroBooNE TPC is small, so most of them leave it. This matters for the p_μ resolution and for the detector systematic, since the range and MCS estimators carry different uncertainties.

Pion momentum. Taken from the *muon-hypothesis* range momentum (`trk_range_muon_mom_v`). The muon mass (105.7 MeV) is close to the pion mass (139.6 MeV), so the range-to-momentum conversion is a good proxy, and, crucially it is a *momentum*, directly comparable to the true p_π it is binned against. An earlier version of this analysis used the proton-hypothesis range *kinetic energy* (`trk_energy_proton_v`) instead. That is a different physical quantity from the momentum used for the true axis with the same bin edges, so the migration matrix was systematically shifted rather than near-diagonal: a true 0.2 GeV/c pion gives $KE \approx 0.104$ GeV, two bins low, and a true 0.175 GeV/c pion falls below the first reco bin entirely. Range momentum is meaningful only for contained tracks, which the containment requirement in the selection guarantees.

Angles. Polar angles are defined with respect to the *neutrino direction* $\hat{u}_\nu$, not the detector z -axis: $\cos \theta_\mu = \hat{u}_\mu \cdot \hat{u}_\nu$, $\cos \theta_\pi = \hat{u}_\pi \cdot \hat{u}_\nu$. The two are far from equivalent — the NuMI beam reaches MicroBooNE 28.6° away from z , and referring the angles to z would place 64.6% of signal events in a different $\cos \theta_\mu$ bin than the generator predictions they are compared with (Sec. 5.2). For reconstructed quantities $\hat{u}_\nu$ is the β direction of Refs. [1, 2], from the NuMI target to the reconstructed vertex; at truth level it is the exact per-event neutrino direction. The opening angle $\theta_{\mu\pi} = \arccos(\hat{u}_\mu \cdot \hat{u}_\pi)$ is computed from reconstructed track directions and, being an angle between two measured tracks, is independent of this choice.

10 Cut-Flow and Selection Performance

Figure 3 shows the full-exposure cut-flow yields for all three configurations as stacked plots, and Table 10 gives the detailed per-category, per-cut breakdown for each.

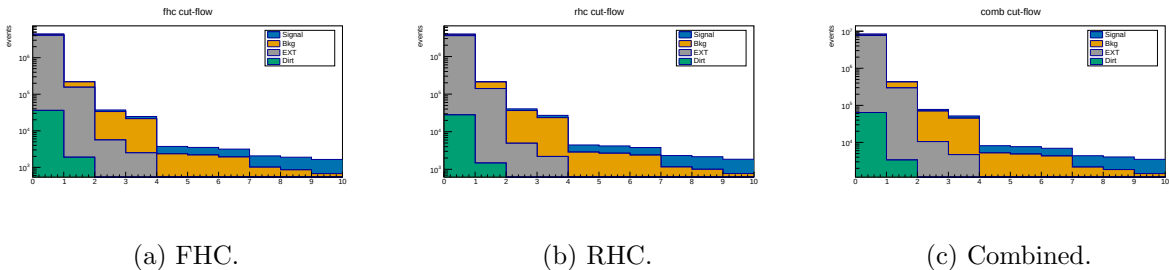


Figure 3: Cut-flow yields at *full exposure* for FHC, RHC and combined (log scale; MC-prediction stack: signal blue, background orange, EXT grey, dirt green; colourblind-safe Okabe-Ito), signal and background POT-scaled per mode. Data is blind, so this shows the MC-prediction stack; the detailed per-category breakdown is in Table 10.

Table 10: Cut-flow yields at *full exposure* for FHC, RHC and combined (MC prediction; data is blind: the fake-data *expectation* equals the central-value prediction by construction, while the realised pseudo-data fluctuate according to the Poisson throw). The histograms behind this table carry the framework’s full central-value weight, $\text{tune} \times \text{PPFX} \times \text{normalisation}$ (§6; reprocessed 2026-09-03 with the safe-weight rule of Eq. (5)), the same weight as the fake-data throw and the unfolding, so the final-cut yields agree with the cut-and-count Table 21 to $< 1\%$. Signal and “Bkg” (all non-signal MC: OOFV, NC, $\text{CC}1\pi^0$, $\text{CC}0\pi$, $\text{CCN}\pi$, CCK, CCothers) are from the ν_μ overlays, POT-scaled *per run* (each run by its own $D_{\text{run}}/\text{MC}_{\text{run}}$, Table 4); EXT (beam-off cosmic) is scaled by the (per-run summed) per-run beam-on/beam-off *gate* ratio times the 2% NuMI beam-occupancy factor, each run period using the beam-off sample taken in that period (§6); dirt from the GENIE NuMI dirt overlay. Pred. = Signal + Bkg + EXT + Dirt. Eff. is the cumulative signal efficiency relative to the generated signal (the None stage, where all true signal is counted); Pur. = Signal/Pred. [release 2026-09-04, `current_results.tsv`, per-run EXT, re-thrown fake data]

Cut	Signal	Bkg MC	EXT	Dirt	Pred.	Eff. [%]	Pur. [%]
<i>FHC (Runs 1,2,4,5; 8.857×10^{20} POT)</i>							
None	5 417	268 316	4 049 292	33 638	4 356 662	100.0	0.1
InFiducialVol	4 504	60 235	153 533	1 763	220 036	83.1	2.0
Topological	3 407	27 640	5 409	192	36 647	62.9	9.3
MuonCandidate	3 056	18 786	2 363	129	24 333	56.4	12.6
ContainedPion	1 358	2 080	223	24	3 685	25.1	36.8
MuonIn3Planes	1 299	1 964	185	16	3 464	24.0	37.5
PionIn3Planes	1 204	1 759	144	6.8	3 114	22.2	38.7
ShowerCut	1 016	891	108	5.8	2 021	18.8	50.3
OpeningAngle	1 009	807	41	1.9	1 859	18.6	54.3
Nonprotons	943	637	30	1.3	1 612	17.4	58.5
<i>RHC (Runs 1,2,3,4a,4b,4c; 1.108×10^{21} POT)</i>							
None	5 778	286 540	3 560 975	26 089	3 879 381	100.0	0.1
InFiducialVol	4 828	65 474	139 122	1 367	210 791	83.6	2.3
Topological	3 743	30 446	4 767	149	39 104	64.8	9.6
MuonCandidate	3 363	20 709	2 090	100	26 262	58.2	12.8
ContainedPion	1 488	2 576	191	19	4 274	25.7	34.8
MuonIn3Planes	1 426	2 430	158	13	4 028	24.7	35.4
PionIn3Planes	1 329	2 194	121	5.3	3 650	23.0	36.4
ShowerCut	1 101	1 026	86	4.5	2 218	19.1	49.7
OpeningAngle	1 093	933	33	1.5	2 060	18.9	53.1
Nonprotons	1 015	720	23	1.0	1 759	17.6	57.7
<i>Combined FHC+RHC (1.994×10^{21} POT)</i>							
None	11 195	554 856	7 610 266	59 727	8 236 044	100.0	0.1
InFiducialVol	9 332	125 709	292 655	3 130	430 826	83.4	2.2
Topological	7 149	58 086	10 176	340	75 752	63.9	9.4
MuonCandidate	6 419	39 495	4 453	228	50 595	57.3	12.7
ContainedPion	2 845	4 656	414	43	7 958	25.4	35.8
MuonIn3Planes	2 725	4 394	343	29	7 492	24.3	36.4
PionIn3Planes	2 533	3 954	266	12	6 764	22.6	37.4
ShowerCut	2 117	1 917	194	10	4 238	18.9	50.0
OpeningAngle	2 102	1 740	74	3.3	3 919	18.8	53.6
Nonprotons	1 959	1 357	53	2.3	3 372	17.5	58.1

Historical optimisation sample (Run-1-era reference phase space, $p_\mu, p_\pi > 0.1 \text{ GeV}/c$; *not the full-exposure result*). The numbers in this paragraph and in Table 11 come from the smaller sample on which the phase space and the opening-angle cut were optimised. They are kept because they document why the thresholds were chosen; the full-exposure yields, efficiency

and purity of the released measurement are those of Table 10 above and must be quoted from there.

- Signal MC (POT-weighted): 324.7 events
- Total background: $242.6 = 198.5 (\nu\text{-bkg}) + 40.1 (\text{EXT}) + 3.8 (\text{dirt})$
- True signal generated (efficiency denominator): 2577–2684 events (observable-dependent)
- **Signal purity: 57.2%**
- **Signal efficiency: 12.1–12.6%**
- **Measured phase space (reported result): efficiency 15.6%, purity 54.9%** (reference phase-space study: selected signal 311.9, generated 1999, background 253.7; Table 11)

Per-run scaling. The yields above and in Table 10 use the per-run POT factors of Table 4 (each run scaled by its own $D_{\text{run}}/\text{MC}_{\text{run}}$), and the EXT by the per-run trigger ratios (per-run gate ratios, §6). Relative to the earlier single per-mode factor this shifts the FHC final-cut signal by -1.9% ($993 \rightarrow 974$ events) and rescales the beam-off by the updated ratio; because the shift is a small, nearly uniform re-weighting of the run mixture, the efficiency and purity move by $< 0.5\%$ and the quoted percentages are unchanged at this precision. The per-observable figures are finalised with the full per-run re-extraction; the FHC p_μ closure (unfolded vs. smeared truth $\chi^2 = 1.6/7$, $p = 0.98$) confirms the scheme is unbiased.

Measured phase space and efficiency turn-on. The reference selection above has near-dead efficiency at low momentum and at the large opening-angle extreme: from the fine truth distributions, the per-bin efficiency is only $\sim 2\text{--}5\%$ for $p_\mu, p_\pi < 0.15 \text{ GeV}/c$ (the 10/20 cm track-length cuts) and the region above the reco $\theta_{\mu\pi} = 2.6$ rad cut is dominated by cosmics ($\sim 7\%$ purity). The measured phase space $p_\mu > 0.15$, $p_\pi > 0.175 \text{ GeV}/c$, $\theta_{\mu\pi} < 2.6$ rad (Section 7) raises and flattens the efficiency to $\sim 13\text{--}20\%$ (Fig. 20). Table 11 compares the selection across the iterations. Two lessons: (i) the 2.6 rad *upper* cut is essential, dropping it admits $3.4\times$ more cosmic background and cuts the purity to 45%; (ii) a *lower* opening-angle cut is counterproductive, the small-angle region is high-purity ($\sim 58\%$), so cutting it lowers the selected signal without improving purity, which is why no lower cut is used.

Table 11: Historical selection-optimisation study across the phase-space iterations, on the Run-1 FHC sample of the era in which NuWro served as the fake data. Every entry is Monte-Carlo derived (GENIE overlay signal and background, EXT, dirt) and does not depend on the fake data, so the choice of fake-data generator did not enter the optimisation; the thresholds chosen here were carried unchanged into the full-exposure selection of Table 10. “no upper cut” drops the $\theta_{\mu\pi} < 2.6$ cut; “[0.314, 2.6]” adds a lower OA cut. The final choice keeps the 2.6 rad upper cut with no lower cut.

	Reference ($p > 0.1$)	no upper cut ($p > .15/.175$)	[0.314, 2.6] (lower cut)	Final ($\theta < 2.6$)
Generated signal	2684	2055	1916	1999
Selected signal	324.7	320.3	301.8	311.9
Efficiency	12.1%	15.6%	15.8%	15.6%
EXT (cosmic)	40.1	136.2	38.0	40.1
Total bkg	243.6	385.4	252.3	256.4
Purity	57.1%	45.4%	54.5%	54.9%
sig $\times$ purity	185.6	145.4	164.4	171.2

10.1 Key N–1 distributions

Figure 4 shows the N–1 distributions (all cuts applied except the one under study) for the two cuts instrumented per event; the topological score (Cut 3) and the $\mu\text{--}\pi$ opening angle (Cut 9),

for all three configurations. The shapes are consistent across horn modes, as expected since the selection is identical; RHC and combined simply carry the larger exposure. Signal (blue) and background (orange) are stacked and the dashed line marks the applied threshold. The remaining cut variables are shown at the final cut in Sec. 10.3.

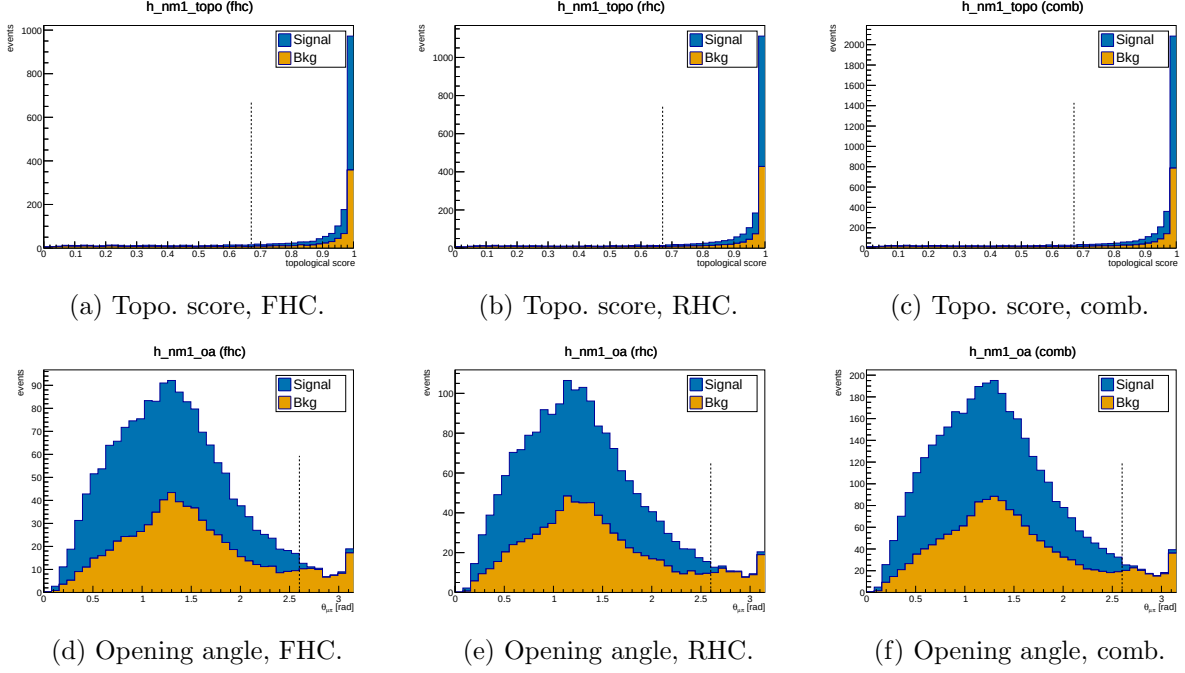
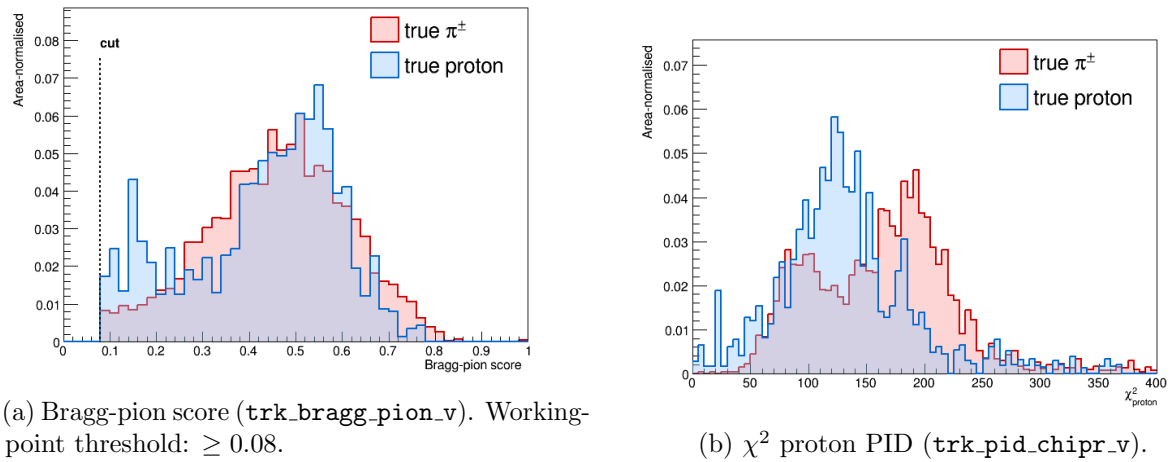


Figure 4: N-1 distributions for the topological score (Cut 3, top) and μ - π opening angle (Cut 9, bottom), for FHC, RHC and combined. Signal (blue) and background (orange) stacked; dashed line = applied threshold. Colourblind-safe Okabe-Ito palette.

10.2 Pion proton-rejection variables

Figure 5 shows the two main proton-rejection variables on the pion candidate, evaluated on MC truth-labelled pions ($\pi^\pm$) and protons at the final cut stage, used to tune the working-point thresholds.



(a) Bragg-pion score (`trk_bragg_pion_v`). Working-point threshold: ≥ 0.08 .

(b) χ^2 proton PID (`trk_pid_chi2_v`).

Figure 5: Pion-candidate PID distributions for true $\pi^\pm$ (red) and true protons (blue) at the final cut stage. The Bragg-pion score provides the primary proton rejection; χ^2_p is studied as cross-check.

10.3 Final-cut data/MC distributions

Figures 6 and 7 show the candidate-level distributions at the final selection cut for the four instrumented PID and track-length variables, for all three configurations. Signal (blue) and background (orange) are stacked (colourblind-safe Okabe-Ito palette).

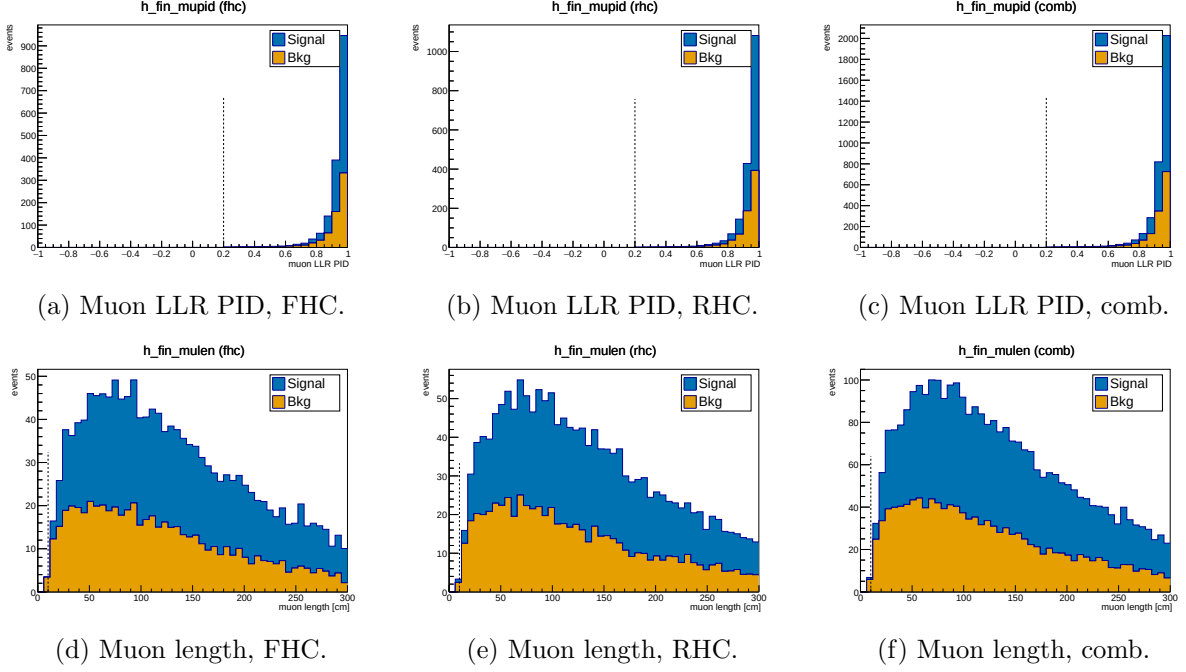


Figure 6: Muon-candidate LLR PID (top) and track length (bottom) at the final cut, for FHC, RHC and combined. Signal (blue) / background (orange) stacked.

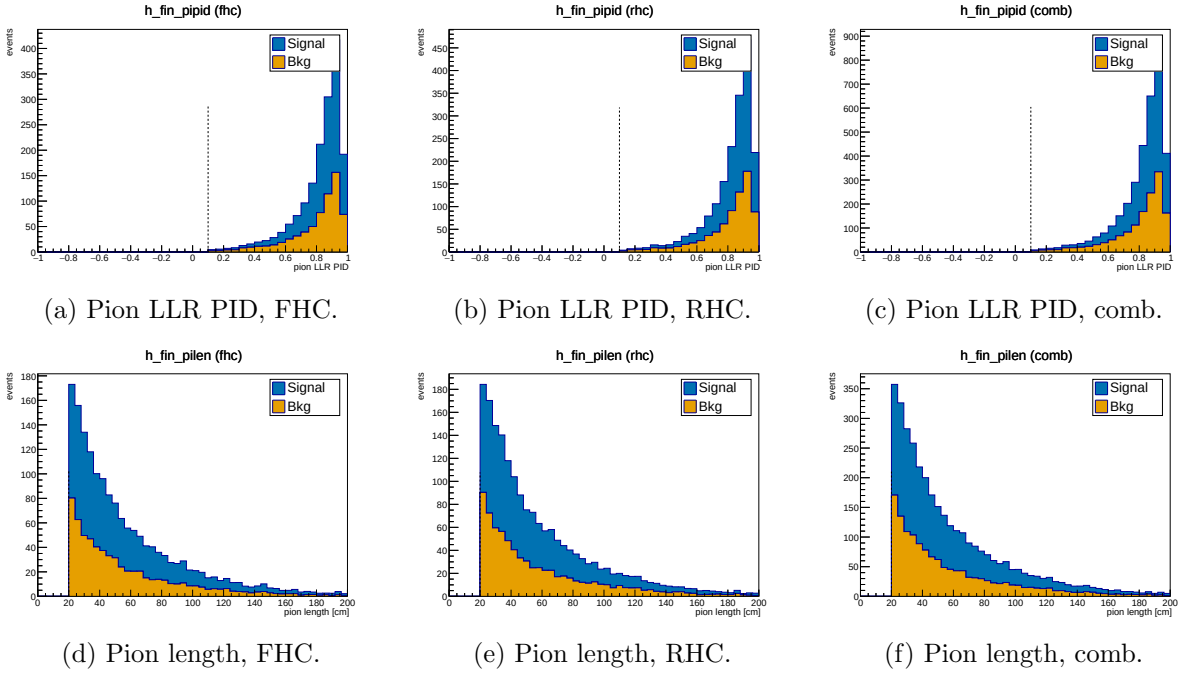


Figure 7: Pion-candidate LLR PID (top) and track length (bottom) at the final cut, for FHC, RHC and combined. Signal (blue) / background (orange) stacked.

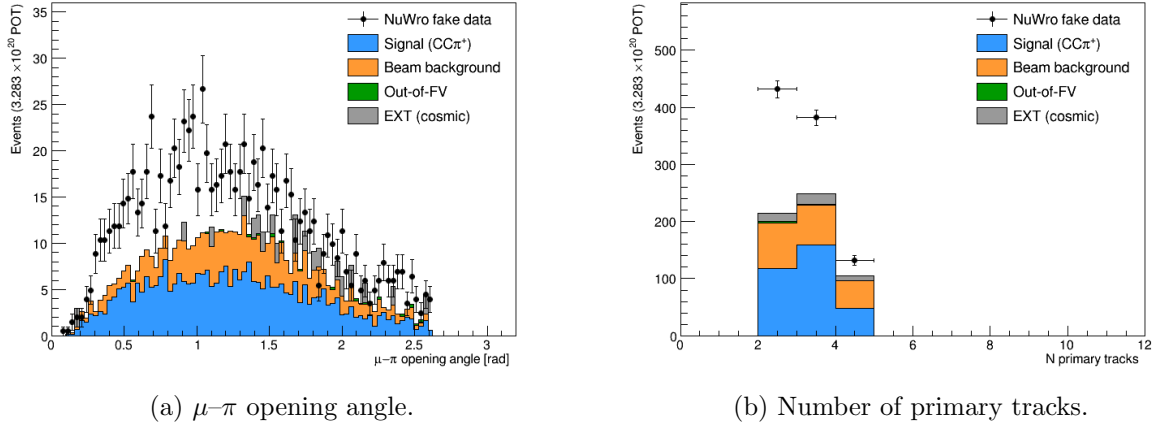


Figure 8: Data/MC comparison for topological variables at the final cut.

10.4 Background decomposition

Table 12 shows the truth-level background breakdown at the final cut.

Table 12: Background composition at the final cut (truth-level MC categories, POT-weighted).

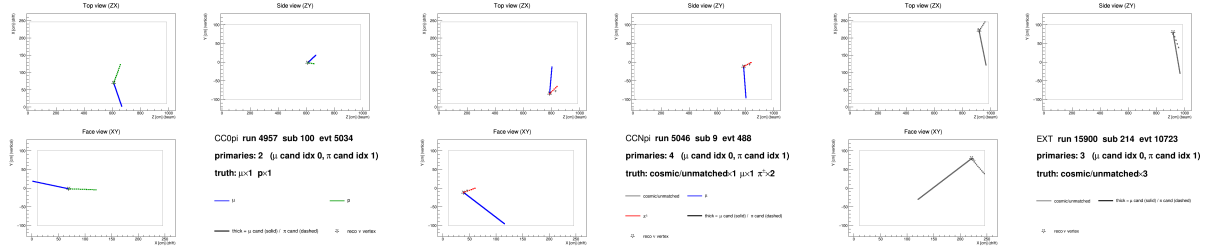
Category	Events	Fraction
$CC0\pi$ (proton mis-ID as pion)	~ 82	$\sim 41\%$
$CCN\pi$ (≥ 2 pion events)	~ 67	$\sim 34\%$
OOFV (vertex outside FV)	~ 18	$\sim 9\%$
NC (neutral current)	~ 14	$\sim 7\%$
$CC1\pi^0$	~ 8	$\sim 4\%$
CC other / CC kaon	~ 6	$\sim 3\%$
ν_e / other flavour	~ 4	$\sim 2\%$
EXT (cosmic)	40.1	n/a
Dirt	3.8	n/a

The dominant background ($\sim 41\%$) is $CC0\pi$ interactions in which a proton is reconstructed as the pion candidate. This mis-identification was verified directly by 2D event displays (truth-labelled by backtracked PDG): the pion candidate track shows a green (proton) truth colour with a stopping Bragg peak, while the genuine muon track is correctly identified.

At the current working point ($\text{trk_bragg_pion_v} \geq 0.08$), the pion candidate is truth-matched to a proton in $\approx 18\%$ of cases. Tighter Bragg-score thresholds (0.20–0.30) were studied but shown not to improve the figure of merit: the pion-to-proton ratio in $CC\pi^\pm$ signal is $\approx 4 : 1$, so rejecting protons removes genuine pions at nearly the same rate.

10.5 Event displays

Figure 9 shows representative 2D truth-labelled event displays for the two dominant backgrounds and the EXT cosmic sample. Tracks are coloured by backtracked truth PDG: μ (blue), $\pi^\pm$ (red), p (green), cosmic/unmatched (grey). The muon candidate is drawn solid-thick; the pion candidate dashed-thick; the reconstructed ν vertex is shown as a star.



(a) CC0 π background: a proton (green, dashed) is reconstructed as the pion candidate. (b) CCN π background: two genuine charged pions present; one is selected as the pion candidate. (c) EXT cosmic event: all tracks are grey (no truth match), faking a neutrino vertex topology.

Figure 9: Truth-labelled 2D event displays (ZX projection) for representative background events at the final cut. Three views (ZX top, ZY side, XY face) are generated per event by `selection/event_display.C`.

11 Unfolding

11.1 The extraction chain, stage by stage

Before the formalism, it is useful to see the measurement assembled in the order it is actually performed. Each stage is shown for all three beam configurations in the flat reco-bin space in which the unfolding operates (every analysis slice concatenated).

Stage 1, selected events, POT-scaled (Fig. 10). The starting point is the selected reconstructed spectrum, scaled to the full exposure of each configuration with the per-run POT factors of Sec. B of the technical supplement. The Monte Carlo is split into signal and background. At the final cut the purity is $\simeq 59\%$ (Table 10), so some 40% of the selected events have to be removed before anything can be interpreted as a signal rate. Because the analysis is blind, the points are the Poisson-thrown fake data rather than beam-on data.

Stage 2, background subtraction (Fig. 11). The estimated background is subtracted bin by bin, leaving the background-subtracted data alongside the central-value MC signal. The neutrino background is taken from the simulation, the cosmic component from beam-off (EXT) data, and the dirt component from the dedicated overlay; the four control regions of §12.4 are what validate this step. This subtracted spectrum, not the raw selection, is the input to the unfolding.

Stage 3, unfolding. The subtracted spectrum still folds in the detector response and the selection efficiency, encoded in the response matrix of §11.3. With 50–99% bin migration a direct inversion is unusable (§11.4), so the measurement is regularised with Wiener–SVD. The price is the additional-smearing matrix A_C : what is recovered is $A_C \times (\text{true signal})$, and *every* model must therefore be multiplied by A_C before being compared with the data.

Stage 4, comparison with models. The unfolded result is then compared with the Micro-BooNE tune and the four external generators, all A_C -smeared into the same space (§12). It is worth separating two effects that are easily conflated. A_C suppresses the integral of each prediction, but it does so by a *common* factor: for FHC p_μ it is 0.64, 0.64, 0.59 and 0.67 for GENIE, GiBUU, NEUT and NuWro respectively. The truth-level totals of those same generators span 0.82 to $1.32 \times 10^{-38} \text{ cm}^2/\text{Ar}$ (Table 21), a factor 1.6. A common $\simeq 0.6$ factor cannot produce a $1.6\times$ spread, so the differences between the generators are genuine model differences and not an

artefact of the regularisation. The cut-and-count measurement of §12.3 provides the independent check, since it involves no unfolding and hence no A_C at all.

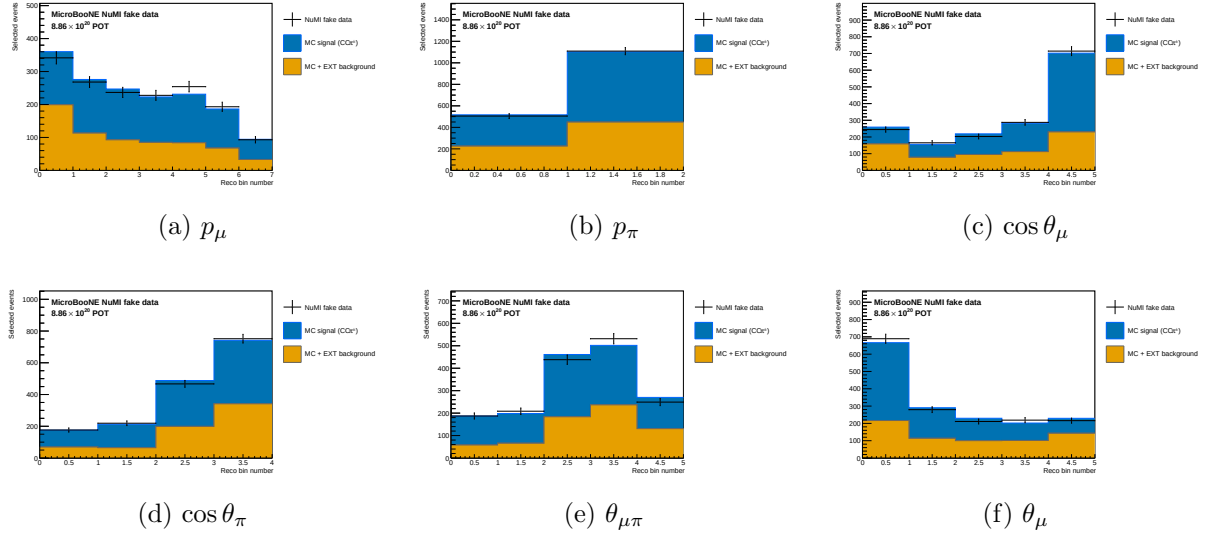


Figure 10: Stage 1, FHC: selected reconstructed spectra scaled to the full exposure, with the MC split into signal and background. Points are the Poisson-thrown fake data. The RHC and combined equivalents are Figs. 12 and 14.

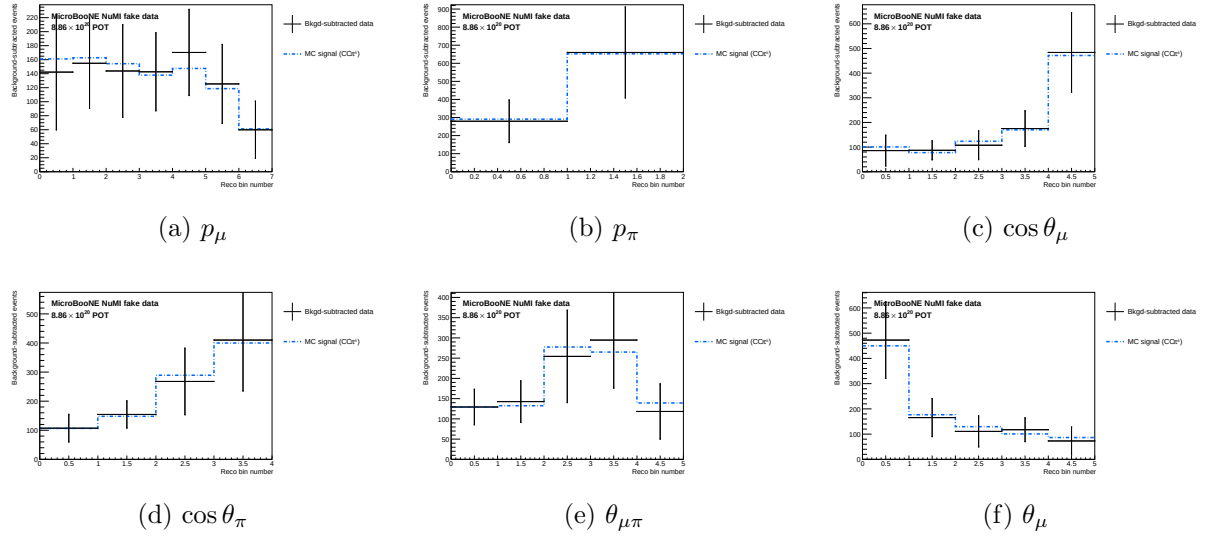


Figure 11: Stage 2, FHC: background-subtracted data compared with the central-value MC signal; the input to the unfolding. RHC and combined: Figs. 13 and 15.

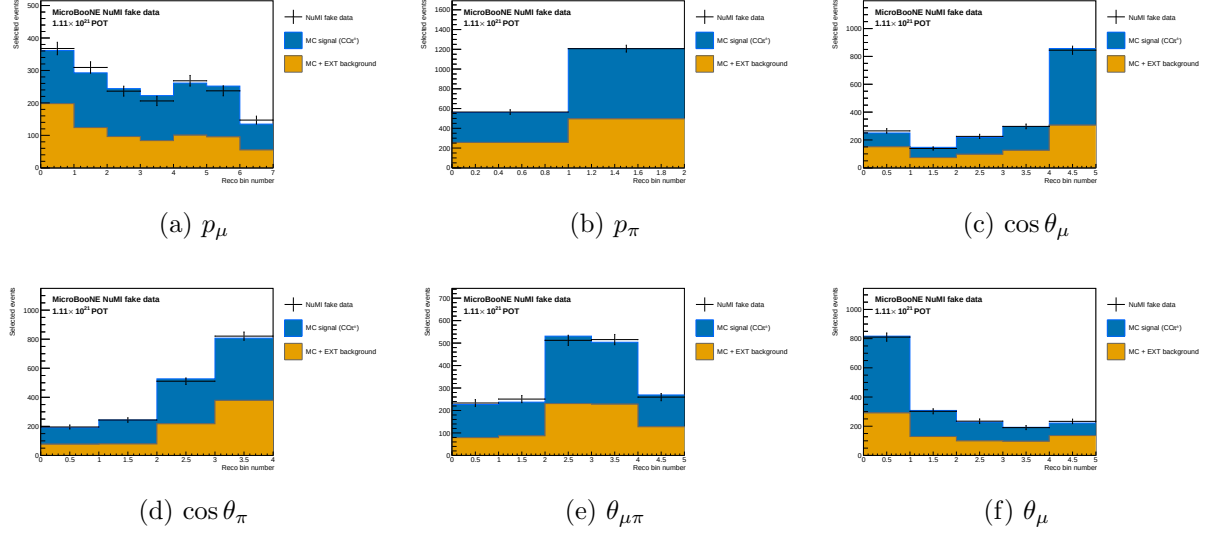


Figure 12: Stage 1, RHC: selected reconstructed spectra scaled to the full RHC exposure, MC split into signal and background.

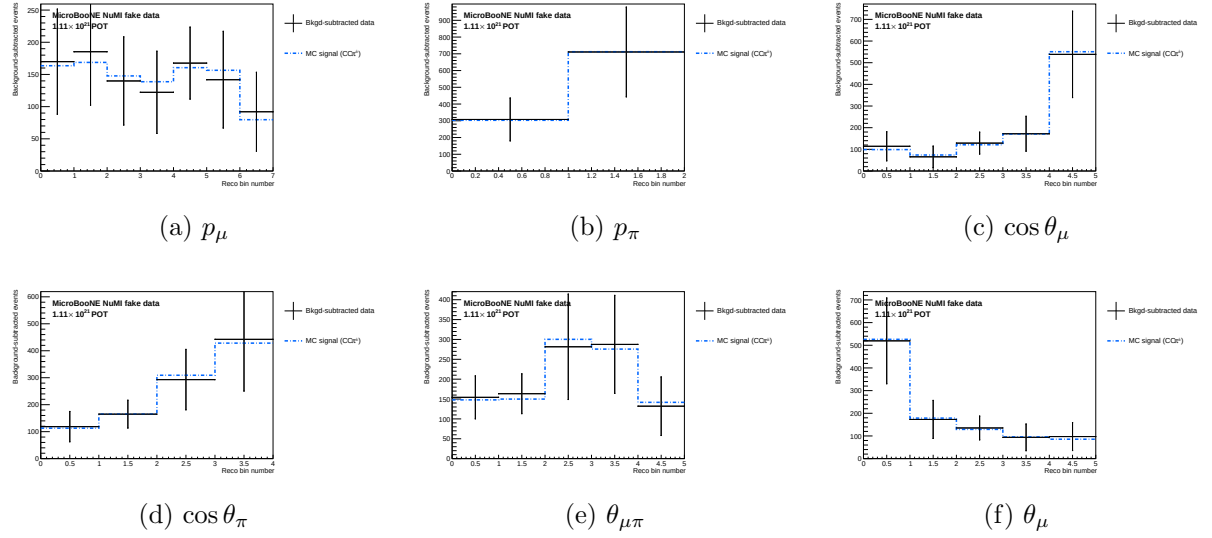


Figure 13: Stage 2, RHC: background-subtracted data compared with the central-value MC signal.

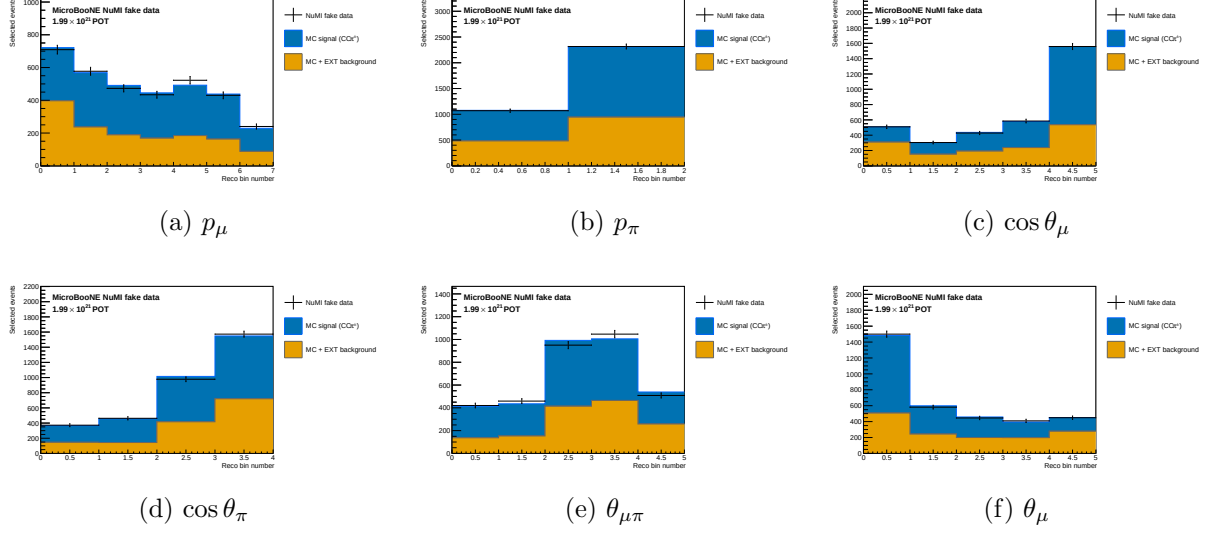


Figure 14: Stage 1, combined: selected reconstructed spectra at the combined FHC+RHC exposure.

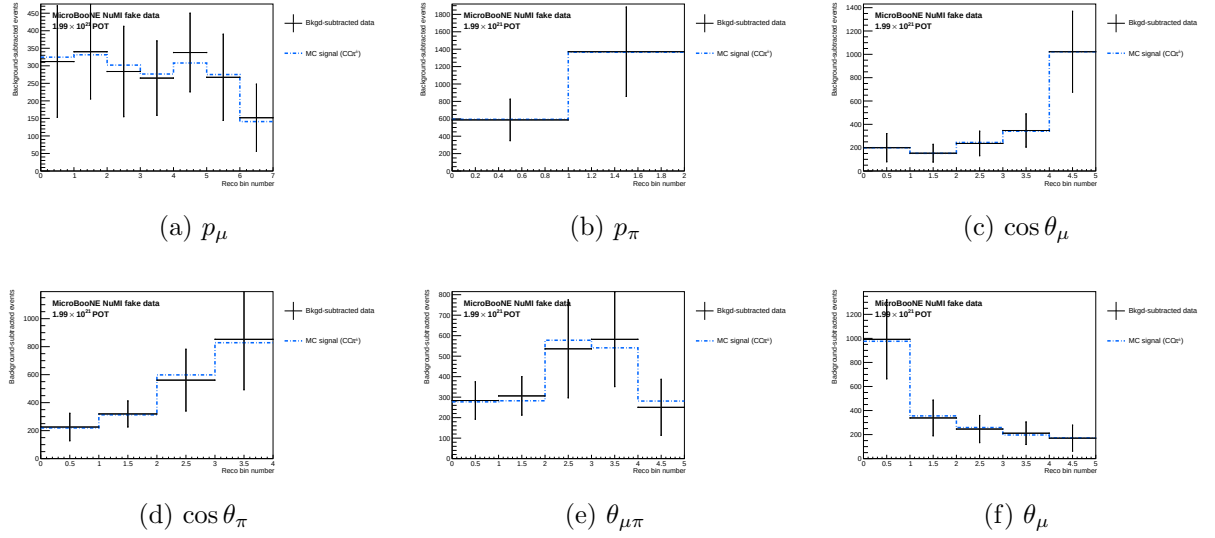


Figure 15: Stage 2, combined: background-subtracted data compared with the central-value MC signal.

11.2 Signal extraction

After the event selection, the background-subtracted reco-space signal spectrum for observable x is:

$$d_i^{\text{sig}} = d_i^{\text{data}} - b_i^{\nu\text{-bkg}} - b_i^{\text{EXT}} - b_i^{\text{dirt}}, \quad (10)$$

where i runs over reconstructed bins.

11.3 Response matrix

Two distinct matrices are in play and it is worth separating them, since they differ by exactly the efficiency. The *migration* matrix

$$M_{ij} = P(\text{reco in bin } i \mid \text{true in bin } j, \text{ selected}), \quad \sum_i M_{ij} = 1, \quad (11)$$

describes only where selected events move between bins. The *smearceptance* matrix additionally carries the probability of being selected at all,

$$S_{ij} = \varepsilon_j M_{ij}, \quad \sum_i S_{ij} = \varepsilon_j, \quad (12)$$

where ε_j is the ratio of selected (`h_smeas_sel`) to generated (`h_true_gen`) signal events in true bin j .

It is S , not M , that enters the unfolding: `UnfolderNuMI.C` passes `get_cv_smearceptance_matrix()` to the unfolders, and the efficiency of Fig. 20 is recovered as its column sums. S is built from selected signal MC using the same binning on both axes. The folding relation the unfolders invert is therefore

$$n_i^{\text{reco}} = \sum_j S_{ij} N_j^{\text{true}} + b_i, \quad S_{ij} = \varepsilon_j M_{ij}, \quad (13)$$

and the unfolded $\hat{N}_j^{\text{true}}$ is already efficiency-corrected: no separate division by ε_j is applied afterwards (§14). Figure 19 plots S itself, exactly the matrix passed to the unfolders (colour scale $P(\text{reco}|\text{true}) \times \varepsilon$, so each column sums to the efficiency of that true bin rather than to unity). The migration diagonals quoted in this note, for example 91% and 76% for the two p_π bins, are elements of the column-normalised M , obtained from the dumped S by dividing each column by its sum (`smear_matrix_dump.root, h_diagonal`); they are not read off the colour scale of Fig. 19.

11.4 Regularised unfolding: the bias–variance trade-off

Unfolding inverts the folding relation $\vec{m} = A \vec{s} + \vec{b}$ to estimate the true signal spectrum $\vec{s}$ from the measured $\vec{m}$. The naive inverse $A^{-1}(\vec{m} - \vec{b})$ is unbiased but useless: bin migration makes A nearly singular, so its inverse amplifies statistical fluctuations into large, anticorrelated bin-to-bin oscillations. Every practical method therefore *regularises*, trading a small, controlled bias for a large reduction in variance. This analysis uses two complementary methods (a fuller pedagogical treatment is given in the companion note `report/unfolding_note.tex`).

Wiener-SVD (primary). Diagonalising the response with an SVD, each mode k is weighted by the Wiener filter $W_k = \sigma_k^2 / (\sigma_k^2 + N_k)$; the minimum-mean-squared-error weighting, $\simeq 1$ for well-measured modes and $\rightarrow 0$ for noise-dominated ones, so no regularisation strength is tuned by hand. The price is a *known* linear bias: the unfolded data estimates not $\vec{s}$ but the smeared spectrum $A_C \vec{s}$, where the additional-smearing matrix A_C is published so that every theory prediction is smeared by the same A_C before comparison.

D’Agostini iterative Bayesian [27] (cross-check). Bayes’ theorem inverts the migration, $P(C_j|E_i) = R_{ij}\pi_j / \sum_l R_{il}\pi_l$, and the unfolded estimate is $\hat{s}_j = \varepsilon_j^{-1} \sum_i P(C_j|E_i)(m_i - b_i)$. The MC prior π_j is updated with the result and iterated; the *number of iterations* is the regularisation knob (few $\Rightarrow$ prior-biased, many $\Rightarrow$ noisy). It carries no A_C , so its integral is essentially unbiased and theory is compared raw.

Table 13: Trade-offs of the two unfolding methods. Neither is “more correct”; they make complementary choices about how to trade bias for variance and how to expose the residual bias.

	Wiener-SVD	D’Agostini (iter. Bayes)
Regularisation	Wiener filter (automatic, from S/N)	Iteration count n (chosen)
Residual bias	Explicit A_C , published	Implicit; set by early stopping
Result integral	Suppressed, observable-dependent	Essentially unbiased (physical)
Theory comparison	Smear theory by A_C	Raw theory
Positivity	Not guaranteed	Guaranteed
Uncertainties	Closed-form, linear	Correlated across iters.; needs care

Wiener-SVD is the primary result, for its objective regularisation and its explicit, reportable bias A_C ; D’Agostini serves as the quantitative cross-check (Sec. 4.5 of the technical supplement; re-evaluated on this release), where its flat, iteration-stable integral, 20–29% above Wiener-SVD (config-dependent), confirms that the observable-to-observable spread of the Wiener-SVD integrated cross sections (Table 20) is the A_C smearing rather than a physical or acceptance effect.

11.5 Wiener-SVD unfolding

The primary method is Wiener-SVD [7], implemented in `xsec/WienerSVD.h` (`RunWienerSVD.Fw`) as a faithful replica of the official `XSecAnalyzer` [18, 17] `WienerSVDUnfolder`: Cholesky whitening of the data covariance, regularisation inside the SVD of RC^{-1} , and the BNL filter (Eq. 3.24 of Ref. [7]). The method applies a Wiener filter in the SVD basis, providing adaptive regularisation whose strength is set by the data covariance. Because the custom pipeline does not throw systematic universes, the covariance is imported from the framework (the prediction covariance scaled to the custom prediction, plus the custom data Poisson; Sec. B of the technical supplement); a stat-only covariance under-regularises and over-corrects the integral by $\sim 26\%$.

The differential cross section is:

$$\left. \frac{d\sigma}{dx} \right|_j = \frac{\hat{N}_j^{\text{true}}}{\Delta x_j \cdot \Phi_{\text{total}} \cdot N_{\text{Ar}}}, \quad (14)$$

where $\hat{N}_j^{\text{true}}$ is the Wiener-SVD estimate of the true signal count in true bin j obtained by inverting Eq. (13), $\Phi_{\text{total}} = \Phi_{\text{per POT}} \times T_{\text{POT}}$ is the total integrated flux, and N_{Ar} is the number of argon target nuclei in the FV. Because the response supplied to the unfolders is the smearacceptance matrix $S = \varepsilon M$, the efficiency correction is part of the inversion and *no explicit* $1/\varepsilon_j$ appears here; an earlier draft of this equation carried one, which would have applied the correction twice. The implementation (`UnfolderNuMI.C`) forms each bin as $\hat{N}_j^{\text{true}}/(\text{conv} \cdot \Delta x_j)$ with the events-to-cross-section conversion $\text{conv} = N_{\text{Ar}} \Phi_{\text{total}}/10^{-38}$ and applies no further efficiency factor. The per-configuration inputs at the final exposures are:

Quantity	FHC	RHC	Combined
N_{Ar} (FV, dead-region excluded)		8.710×10^{29}	
Flux constant ϕ ($E_\nu > 60$ MeV; 10^{-10} cm $^{-2}$ /POT)	6.81159	6.44646	6.60865
Data POT (10^{20})	8.857	11.082	19.939
Integrated flux $\Phi_{\text{total}} = \phi \times \text{POT}$ (10^{11} cm $^{-2}$)	6.033	7.144	13.177
$\text{conv} = N_{\text{Ar}} \Phi_{\text{total}}/10^{-38}$ (events per 10^{-38} cm 2 /Ar)	5255	6223	11478

(An earlier draft tabulated a single FHC example at a superseded fake-data exposure of 7.63×10^{20} POT, $\text{conv} = 4527$; that exposure is not used for any released result.) An integrated cross section is computed as the width-weighted sum,

$$\sigma_{\text{int}} = \sum_j \left. \frac{d\sigma}{dx} \right|_j \Delta x_j. \quad (15)$$

It is *not* required to agree between observables: the Wiener-SVD result estimates $A_C x_{\text{true}}$ and A_C is not norm preserving, so σ_{int} is observable-dependent (§12.2). The observable-independent total is the cut-and-count value of §12.3.

11.6 Reconstructed spectra and response matrices

Figure 16 shows the background-subtracted reco-space signal spectra for all six observables, overlaid with the MC signal prediction and the individual background components.

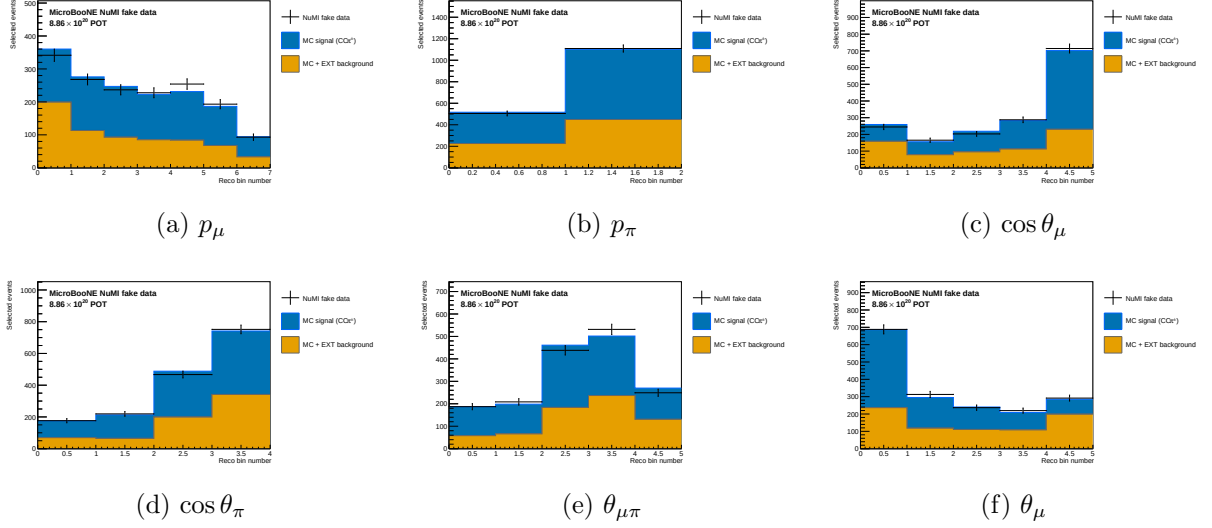


Figure 16: Reconstructed-space spectra at the final cut for all six observables, FHC full exposure (Runs 1+2+4+5): Poisson-thrown central-value fake data (black points), stacked signal (blue), beam background (orange), dirt (green), and EXT (grey). The fake data is drawn from the central-value Monte Carlo and so agrees with the stacked prediction by construction (a reco-level self-closure); the differential results and their closure against the A_C -smeared truth are shown in Fig. 22.

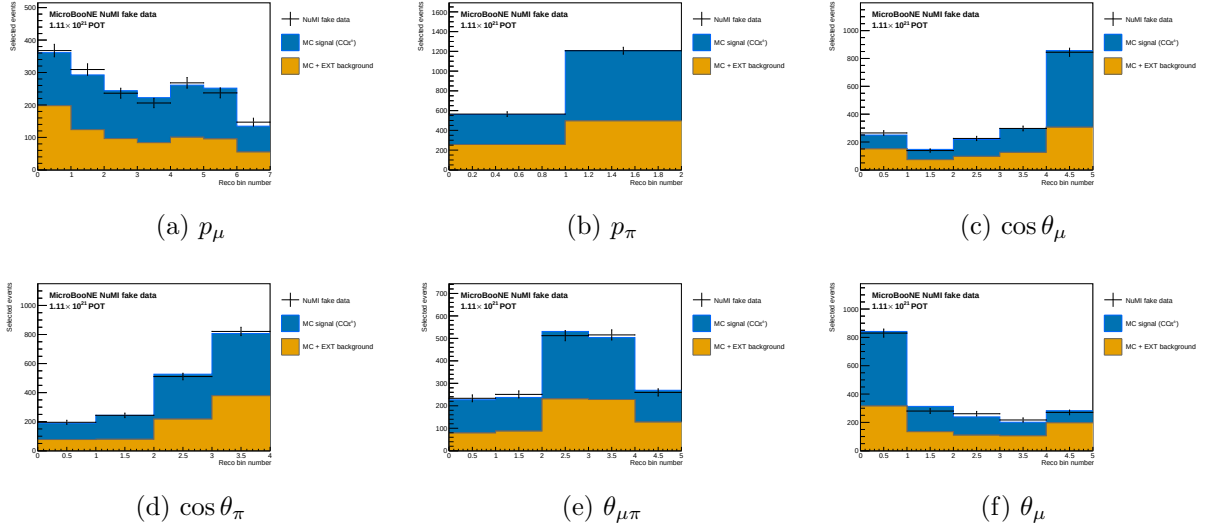


Figure 17: Reconstructed-space spectra, RHC full exposure (Runs 1+2+3+4a+4b+4c); same components as Fig. 16.

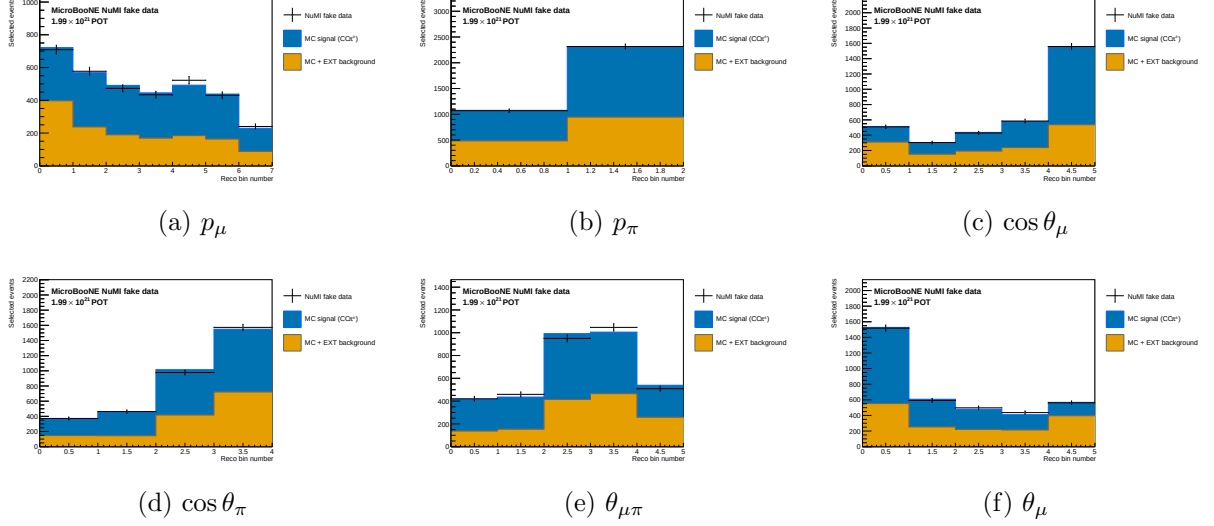


Figure 18: Reconstructed-space spectra, combined FHC+RHC (per-mode normalised); same components as Fig. 16.

Figure 19 shows the smearceptance matrices S (§11.3; colour scale includes the efficiency, so the columns sum to ε_j , not to unity) for the six observables (the p_π panel at the superseded five-bin binning, as a diagnostic; the rest at the released binning). The angular observables and p_μ are near-diagonal in the column-normalised sense; for p_μ the width is set mainly by the MCS estimator, which reconstructs 63% of the selected sample and is intrinsically less precise than range.

p_π . At any fine binning the p_π response is strongly non-diagonal, because the range estimator saturates near 0.34 GeV/ c once pions interact hadronically before stopping; that is a physical limit rather than an algorithmic one, and the fine-bin study is in the technical supplement (Sec. 1.1 of the technical supplement). The adopted two-bin response shown here is sufficiently diagonal, with column-normalised diagonals of 91% and 76%, but it carries only threshold-versus-above-threshold information: the upper bin is an inclusive region, not a resolved spectral point (§12.1).

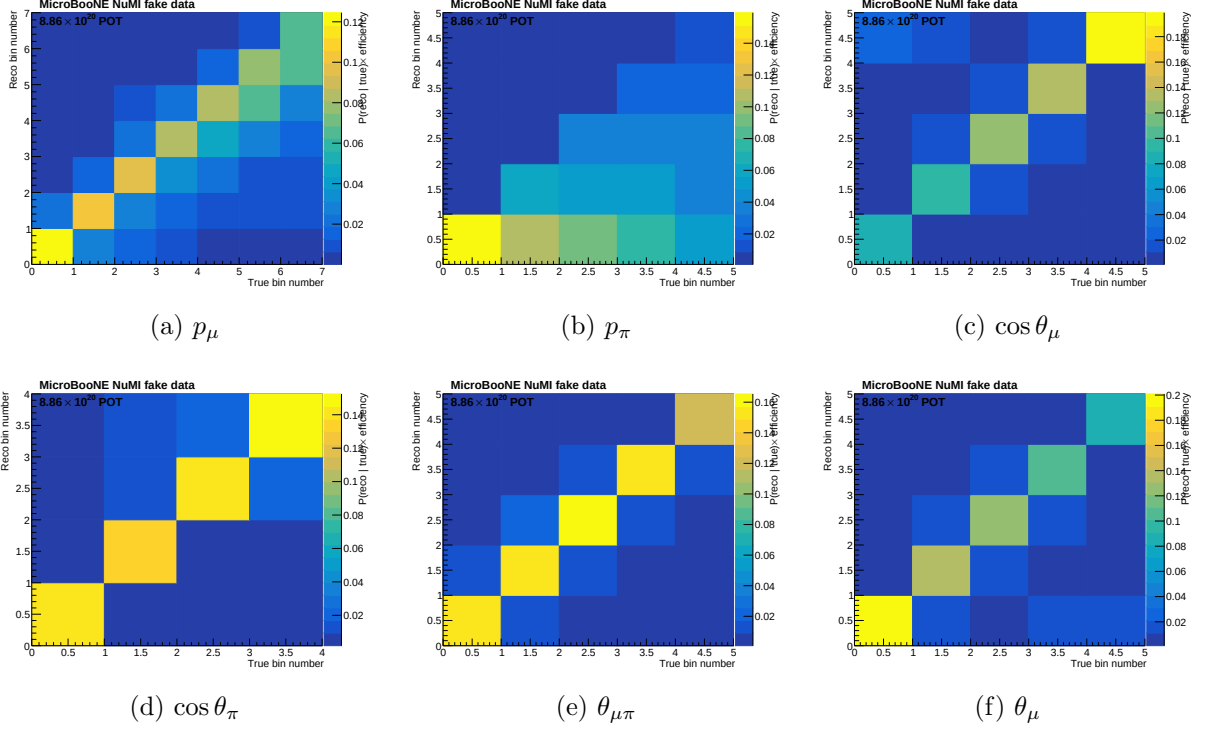


Figure 19: Smearceptance matrices $S_{ij} = \varepsilon_j M_{ij}$, FHC, exactly as passed to the unfold (`plot_step3_smearing_matrix`). Five panels are at the released binning; the p_π panel is deliberately the *superseded five-bin* response, kept as the diagnostic that shows reconstructed events piling into the lowest bin from every true bin and motivated the two-bin scheme (the released two-bin S is 2×2 with column-normalised diagonals 91% and 76%). The colour scale is $P(\text{reco} = i | \text{true} = j) \times \varepsilon_j$, so each column sums to the selection efficiency of that true bin ($\sim 0.1\text{--}0.2$), *not* to unity; the column-normalised migration matrix M is obtained by dividing each column by its sum, and the diagonals quoted in the text are of M . Axes are flat bin numbers (true on x , reco on y).

Figure 20 shows the selection efficiency per true bin for all six observables, computed as the ratio of selected to generated signal MC.

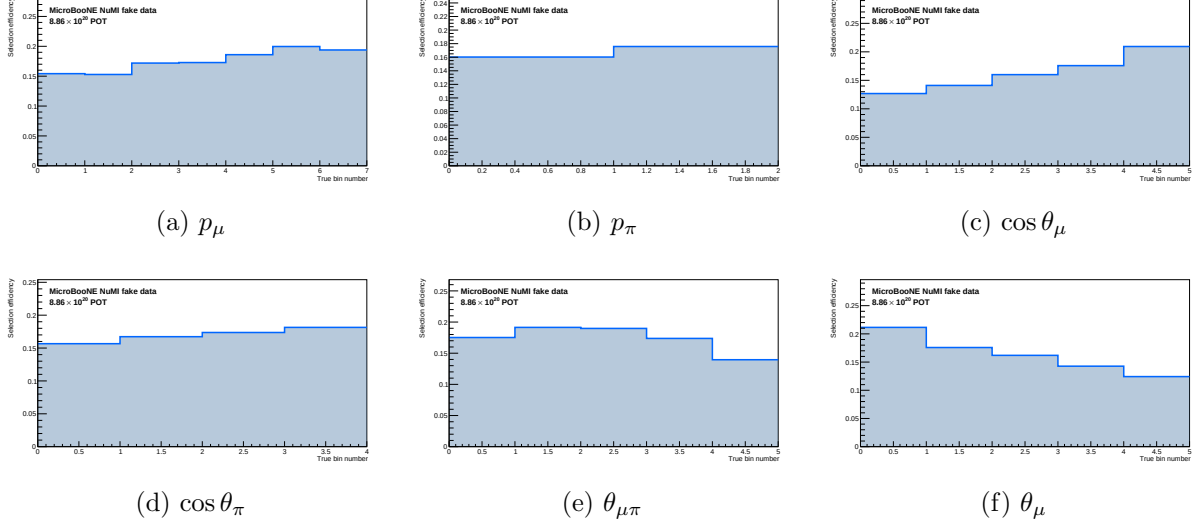


Figure 20: Selection efficiency as a function of true observable value, in the measured phase space ($p_\mu > 0.15$, $p_\pi > 0.175$ GeV/c, $\theta_{\mu\pi} < 2.6$ rad). Efficiency is the ratio of selected to generated signal MC ($\varepsilon_i = \sum_j A_{ij}$, integrated over reco bins). With the momentum thresholds and the 2.6 rad opening-angle cut the efficiency is flat at ~ 13 – 20% across all observables; the former $\sim 5\%$ low-momentum turn-on bins and the cosmic-dominated large-angle region are removed. The lowest point is the smallest opening-angle bin ($\sim 9\%$ efficiency); it is retained because it is high-purity ($\sim 58\%$), not low-purity.

11.7 Reconstruction resolution

Figure 21 shows the reconstruction resolution of each measured observable, evaluated on selected signal events (GENIE ν_μ overlay, measured phase space). For the momenta the relative resolution $(\text{reco} - \text{true})/\text{true}$ is shown; for the angular observables the absolute residual $(\text{reco} - \text{true})$ is used, since the relative metric diverges as $\cos \theta \rightarrow 0$ / $\theta_{\mu\pi} \rightarrow 0$. The red points are the mean residual per true bin; the quoted numbers are the overall mean (bias) and RMS (resolution).

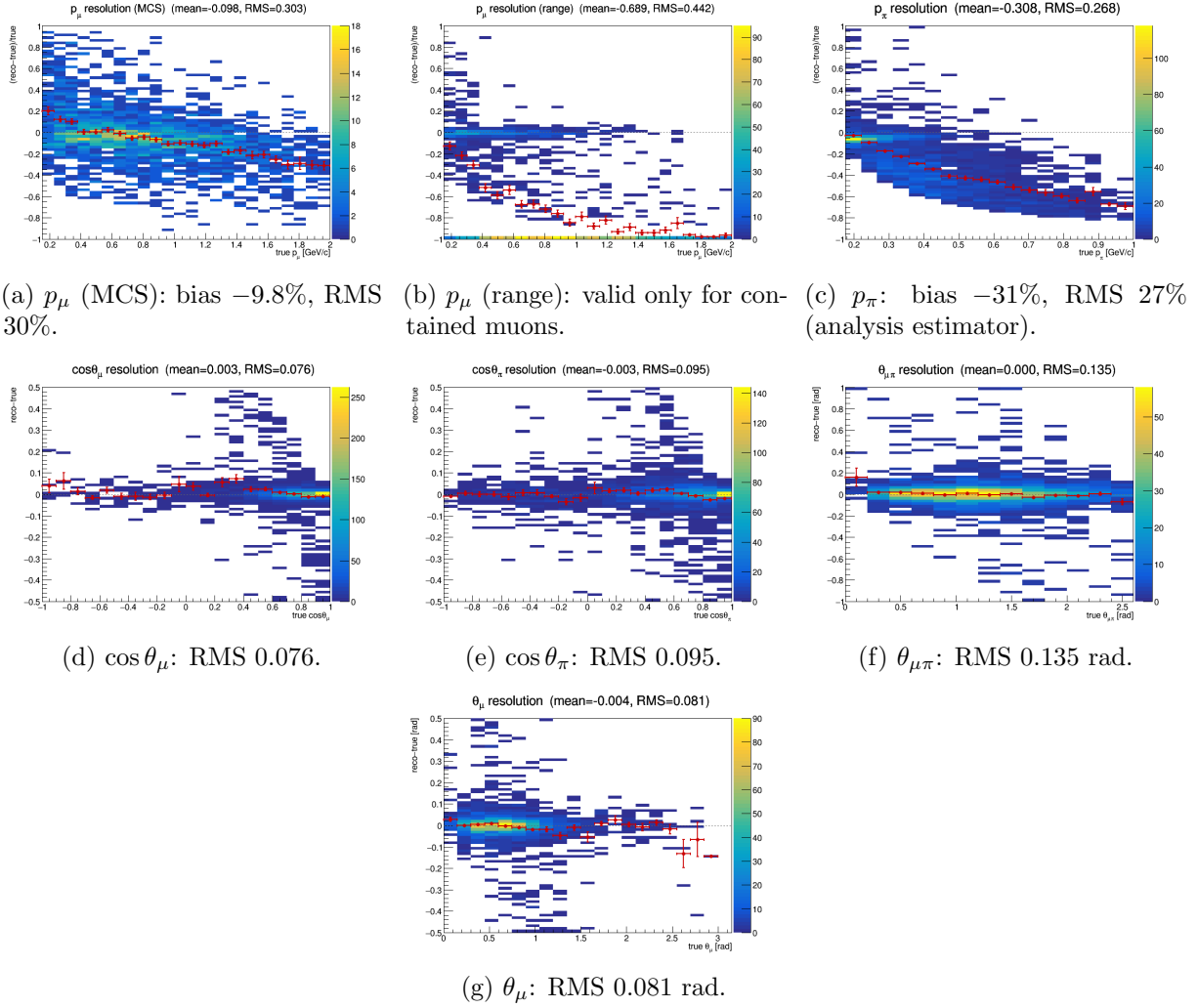


Figure 21: Reconstruction resolution vs. true value, selected signal. Momenta: relative ($\text{reco} - \text{true}$)/ true ; angles: absolute ($\text{reco} - \text{true}$). Red points: mean residual per true bin. θ_μ is derived as $\arccos(\cos \theta_\mu)$, exactly as the bin config does it, so the panel shows the quantity the analysis bins on; it is unbiased (-0.004 rad) with RMS 0.081 rad, the drift below zero above ~ 2.5 rad being a handful of backward-going muons. The muon MCS momentum slightly over-estimates at low p_μ and increasingly under-estimates above ~ 1 GeV/c; the range estimator is badly biased for *exiting* muons (range = distance to wall), which is why the analysis uses range only for contained muons (range-or-MCS). The pion panel shows the estimator the analysis actually bins on, not the raw `candidate_pion_mom_reco` branch (Sec. 1.1 of the technical supplement): mean bias -31% , RMS 27% . That average is dominated by high momenta; the bias is only -2% in the first true bin and reaches -65% above 0.8 GeV/c; because the reconstructed value saturates near 0.34 GeV/c once pions interact hadronically before stopping. The angular variables are essentially unbiased with narrow resolution.

11.8 Model comparison: χ^2 against truth, tune and generators

Table 14 collects the bin-by-bin χ^2 of the unfolded fake data against each prediction, for every observable and every beam configuration. All predictions are A_C -smeared, so the comparison is made in the measurement space, and the full covariance is used. The *truth* column is the closure test: it is small throughout, confirming the chain, and is visibly larger for RHC than for FHC, consistent with the poorer RHC angular response noted in Sec. 5.1 of the technical supplement.

An anomaly that turned out to be a bug. Earlier versions of this note reported χ^2 values for NEUT and NuWro in $\cos\theta_\mu$ and θ_μ an order of magnitude above every other observable — 40.6/5 and 38.6/5 in FHC θ_μ — and devoted a paragraph to explaining them as an artefact of A_C suppression acting on sharply peaked generators.

The explanation was wrong and so were the numbers. The predictions were being multiplied by A_C *twice*: once in the extractor, which applies the additional smearing to every entry of the prediction map in place, and again on the plotting path. Verified as $A_C^2 \cdot \text{raw} = \text{plotted} \times \text{width}$ to 1.0000 in every bin. The second application has been removed.

With the correction the anomaly disappears: FHC θ_μ gives 8.96/5 for NEUT and 9.43/5 for NuWro, and $\cos\theta_\mu$ gives 4.90/5 and 4.93/5. No observable now sits an order of magnitude apart from the others, and the earlier argument for preferring θ over $\cos\theta$ on χ^2 grounds falls away with the effect that motivated it.

The bug affected only the model comparisons. The unfolded cross sections are unchanged (FHC p_μ integrates to 0.8275 before and after), and the closure test was never affected, because the fake-data truth is the one curve that was not passed through the second multiplication.

Table 14: χ^2/ndf of the unfolded fake data against each A_C -smeared prediction, for the inclusive selection. “truth” is the fake-data truth (the closure test). [release 2026-09-04, `current_results.tsv`, per-run EXT, re-thrown fake data]

Beam	Observable	truth	uB tune	GENIE	GiBUU	NEUT	NuWro
FHC	p_μ	1.45/7	1.54/7	2.51/7	1.69/7	3.08/7	1.42/7
	p_π	0.11/2	0.10/2	0.37/2	1.25/2	0.74/2	0.57/2
	$\cos\theta_\mu$	0.67/5	1.09/5	1.67/5	1.27/5	3.88/5	4.04/5
	$\cos\theta_\pi$	0.80/4	0.51/4	1.23/4	2.17/4	1.37/4	0.92/4
	$\theta_{\mu\pi}$	0.30/5	0.86/5	1.20/5	1.44/5	3.87/5	3.82/5
	θ_μ	1.52/5	2.14/5	5.55/5	6.07/5	8.46/5	9.10/5
RHC	p_μ	0.64/7	2.51/7	3.44/7	2.81/7	3.19/7	3.31/7
	p_π	0.02/2	0.01/2	0.56/2	0.30/2	0.25/2	0.11/2
	$\cos\theta_\mu$	0.35/5	0.37/5	1.11/5	0.24/5	2.71/5	2.67/5
	$\cos\theta_\pi$	0.43/4	0.34/4	0.99/4	2.69/4	1.81/4	0.69/4
	$\theta_{\mu\pi}$	0.19/5	0.34/5	2.52/5	3.54/5	0.46/5	1.08/5
	θ_μ	0.12/5	0.21/5	0.78/5	0.34/5	5.60/5	5.76/5
comb	p_μ	0.41/7	0.74/7	2.08/7	0.86/7	1.46/7	0.98/7
	p_π	0.01/2	0.01/2	0.45/2	0.92/2	0.47/2	0.30/2
	$\cos\theta_\mu$	0.05/5	0.09/5	0.84/5	0.47/5	3.27/5	2.92/5
	$\cos\theta_\pi$	1.05/4	0.78/4	1.80/4	3.82/4	2.17/4	0.91/4
	$\theta_{\mu\pi}$	0.06/5	0.54/5	1.76/5	2.64/5	2.41/5	2.89/5
	θ_μ	0.56/5	0.83/5	3.16/5	4.71/5	8.20/5	7.87/5

Table 15: The same comparison for the muon angle in both parameterisations. The θ bin edges are exactly arccos of the $\cos \theta$ edges, so the phase space is identical; only the variable differs. Values are post-fix: a single application of A_C , on the width-aware regularisation. The two parameterisations give comparable χ^2 , and neither shows the order-of-magnitude separation reported in earlier drafts, which was the double- A_C bug rather than a property of either variable. [release 2026-09-04, `current.results.tsv`, per-run EXT, re-thrown fake data]

Beam	Variable	truth	uB tune	GENIE	GiBUU	NEUT	NuWro
FHC	$\cos \theta_\mu$	0.67/5	1.09/5	1.67/5	1.27/5	3.88/5	4.04/5
FHC	θ_μ	1.52/5	2.14/5	5.55/5	6.07/5	8.46/5	9.10/5
RHC	$\cos \theta_\mu$	0.35/5	0.37/5	1.11/5	0.24/5	2.71/5	2.67/5
RHC	θ_μ	0.12/5	0.21/5	0.78/5	0.34/5	5.60/5	5.76/5
Comb	$\cos \theta_\mu$	0.05/5	0.09/5	0.84/5	0.47/5	3.27/5	2.92/5
Comb	θ_μ	0.56/5	0.83/5	3.16/5	4.71/5	8.20/5	7.87/5

12 Full-exposure results: FHC, RHC, and combined

12.1 Scope and status of each result

The analysis has been through a number of corrections, several of which changed results and two of which invalidated conclusions (all recorded, with before-and-after values, in the change log). This subsection states plainly what each result currently is, so that no number in this note has to be interpreted by working out which fixes preceded it.

All fifty-one extractions now carry a full systematic covariance including detector variations. The proton-tagged results previously carried none — the detector variation samples had been processed but never referenced by the proton-tagged configuration — and were rebuilt and re-unfolded on 2026-08-31 to correct that. Their detector terms (10.8%–65.5%, median 24.0%) are of the same order as the inclusive ones (16.9%–43.1%).

The proton-tagged results remain secondary scope for the narrower reason that their model comparisons have not been recomputed since, and because the exposure does not support the differential $W_{\pi p}$ shape at any binning.

Result	Status	Basis
<i>Primary scope</i>		
Total flux-averaged cross section (cut-and-count), all three configurations	Final	Not suppressed by A_C ; the most directly interpretable total.
$d\sigma/dp_\mu$	Final	Full covariance including detector variations.
$d\sigma/d\cos\theta_\mu$ and $d\sigma/d\theta_\mu$	Final	Full covariance; discriminates between models after the width-aware regularisation fix.
$d\sigma/d\cos\theta_\pi$	Final	Full covariance. (No θ_π parameterisation is extracted; only the muon angle is released in both forms.)
$d\sigma/d\theta_{\mu\pi}$	Final	Full covariance; the best-conditioned shape observable.
$d\sigma/dp_\pi$, two-bin	Final, coarse	Full covariance, but a two-region partition rather than a spectrum: one near-threshold interval and one broad inclusive interval above 0.205 GeV/c. See below.
<i>Secondary scope — complete covariance, secondary results</i>		
Proton-tagged integrated cross section	Complete	Detector systematics propagated, central value postdates the weighting fix, closure χ^2 recomputed, A_C and covariance released.
Proton-tagged differential results, including transverse kinematic imbalance	Complete	Detector systematics propagated; beam-frame reconstruction corrected; central values post-date the weighting fix; A_C and covariance released.
<i>Withdrawn — do not quote</i>		
Differential $W_{\pi p}$ shape	Withdrawn	No binning satisfies both the migration and the statistics requirement at this exposure; supports an integrated cross section only.
Five-bin $d\sigma/dp_\pi$	Withdrawn	Four of five bins fall below the 68% diagonal criterion.
Transverse kinematic imbalance about detector z	Withdrawn	Measured and predicted quantities were different observables.
Model comparisons predating the width-aware regularisation	Withdrawn	A_C was near rank one; apparent agreement was an artefact.

On the two-bin p_π result. Charged pions interact hadronically before ranging out, so the reconstructed momentum saturates near 0.34 GeV/c: a pion of true momentum 0.95 GeV/c reconstructs at almost the same value as one of 0.55 GeV/c. The two bins have migration diagonals of 91% and 76% (column-normalised M , §11.3) and are sound, but the upper bin integrates over a very wide physical interval and is not a differential point in the usual sense. The true upper bin is open-ended, $p_\pi > 0.205$ GeV/c (`ccpi_ppi_bin_config_2bin.txt` places no upper edge on it); the 1 GeV/c that appears in the data release and in Appendix A is only the finite boundary adopted to assign the bin a width for plotting, not a kinematic limit. The released $d\sigma/dp_\pi$ values are therefore bin-averaged densities over very different intervals (0.030 and 0.795 GeV/c), and the panel draws the two at equal visual width, flagged on the axis, so the plotted area does not represent cross section. The less ambiguous statement of the same result is the pair of bin-integrated partial cross sections, $\sigma_i = (d\sigma/dp_\pi)_i \Delta p_{\pi,i}$, which do not depend on the choice of plotting boundary for the lower bin and depend on it for the upper bin only through the physical content above 1 GeV/c, which the open true bin already includes:

Table 16: Two-bin p_π result as bin-integrated partial cross sections (10^{-38} cm²/Ar), computed from the released `curves_incl.*_ppi2bin.tsv` as density $\times$ physical width; the uncertainty is the data-statistical term of the release (the total fractional uncertainties are those of Tables 17–19). Both bins are A_C -smeared like every other unfolded quantity, so a prediction must be smeared by the released A_C before comparison. [release 2026-09-04, `current_results.tsv`, per-run EXT, re-thrown fake data]

Region	FHC	RHC	Combined
$\sigma(0.175 < p_\pi < 0.205 \text{ GeV}/c)$	0.059 ± 0.041	0.054 ± 0.034	0.062 ± 0.038
$\sigma(p_\pi > 0.205 \text{ GeV}/c)$	0.764 ± 0.293	0.694 ± 0.268	0.713 ± 0.270

The estimator survey establishing that this is a physical information limit rather than an unrefined estimator is in the technical supplement.

On the closure χ^2 values. The closure χ^2 quoted for each extraction compares the unfolded result with the fake-data truth using the *full* systematic covariance, and the resulting p -values are almost all above 0.9 (median 0.95 across all fifty-one extractions). That should not be read as excellent agreement with the model. For a self-closure with correctly-sized uncertainties the p -values would be roughly uniform on $[0, 1]$, and they are not.

The reason is that the measurement and the truth it is being compared against share their systematics: both come from the same generator, the same flux and the same exposure, so the flux, cross-section and exposure terms are common-mode and largely cancel in the difference while still inflating the covariance used to judge it. Recomputing the same comparison with the statistical covariance alone raises χ^2/ndf from 0.02–0.2 to 0.5–0.8, much nearer the expected value:

Extraction	full	prediction-only	statistics-only
inclusive FHC p_μ	0.14/7	0.22/7	1.01/7
inclusive comb p_μ	1.43/7	2.87/7	4.71/7
proton-tagged FHC p_μ	1.01/7	2.20/7	3.02/7
proton-tagged comb $W_{\pi p}$	0.48/6	0.77/6	4.88/6

The closure χ^2 is therefore a check that the extraction does not *disagree* with its input beyond the quoted band, which it passes, and not a measure of how well the band is sized. A test of the latter needs either a covariance restricted to the terms that differ between measurement and reference, or an ensemble of throws. Neither is done here, and the p -values should be read with that limitation in mind.

Ensemble coverage: are the uncertainties the right size? The closure test above uses a single throw. It can show that the extraction reproduces its input, but a single residual has no width, so it cannot say whether the quoted uncertainty is correctly sized. To answer that, thirty-two statistically independent Poisson realisations of the FHC data were each put through the *whole* chain — thrown, universes rebuilt from that throw, unfolded — and the per-bin pull

$$p_i = \frac{\hat{x}_i - (A_C x_{CV})_i}{\sigma_i}$$

formed against the fixed central-value truth smeared by that member’s own A_C . (The per-throw *fake-data* truth is not a valid reference: it follows the Poisson draw, so numerator and reference fluctuate together and the width comes out too small.)

Two observables were run, each with 32 members, chosen to bracket the conditioning: the well-behaved p_μ , and $\cos\theta_\mu$, whose A_C row sums fall as low as 0.050 so that the regularisation removes up to 95% of a bin’s normalisation. If the uncertainties were mis-sized anywhere, the second is where it would show.

Reference covariance	pull mean	pull width	68% / 95% coverage
<i>statistical only</i>			
$d\sigma/dp_\mu$	-0.20	1.04	63.8% / 95.5%
$d\sigma/d\cos\theta_\mu$	-0.18	0.96	73.1% / 94.4%
<i>full</i>			
$d\sigma/dp_\mu$	-0.06	0.32	99.6% / 100%
$d\sigma/d\cos\theta_\mu$	-0.05	0.31	98.8% / 100%
expected	0	1	68% / 95%

Against the statistical covariance both widths are consistent with unity — 1.04 and 0.96, each determined to $\pm 13\%$ with this many members — and the interval coverage is correct. **The statistical uncertainties are correctly sized, and the strongly suppressed observable behaves no differently from the well-conditioned one.** That the severe A_C suppression in $\cos\theta_\mu$ does not distort its uncertainties is the more informative of the two results, since p_μ passing was the expected outcome.

Two further things the ensemble shows. Against the *full* covariance the pull width is 0.31–0.32 for both observables and coverage is essentially 100%: the quoted uncertainty is about three times the throw-to-throw spread, because the ensemble varies only the data statistics while the denominator carries flux, detector and cross-section terms that are identical in every member. That is the same effect as the closure p -values clustered near unity, now measured rather than argued.

Second, the ensemble mean sits 2.5% below the reference (0.7887 against 0.8090), a 2.1σ offset on the mean, and the per-bin pull means are all negative and consistent at -0.10 to -0.25 . This is a real if small negative bias in the extraction, around a fifth of the statistical uncertainty and far inside the 41% systematic band, but it is a bias rather than a fluctuation and should be quoted as such rather than absorbed silently.

The study covers two observables in one configuration (FHC). Extending it to RHC and to the proton-tagged family would test whether the same holds where the statistics are sparser; the machinery is in place and each observable costs about four hours.

On the blinding status. Every “data” point in this note is Poisson-thrown central-value fake data. No beam-on signal region has been unblinded. Statements comparing configurations, or comparing the result to a generator, are therefore properties of the injected sample and the extraction machinery, not measurements.

This is the primary result of the analysis: the $CC1\mu1\pi^\pm$ cross section extracted over the full available exposure in forward horn current (FHC), reverse horn current (RHC), and their combination, all on Poisson-thrown central-value fake data. The FHC measurement uses Runs 1, 2, 4 and 5 (data POT 8.857×10^{20}); RHC uses Runs 1, 2, 3, 4a, 4b, 4c (1.108×10^{21}); the combined measurement uses all fourteen overlay files at the joint POT 1.994×10^{21} . All three close cleanly against the A_C -smeared truth for every observable (Tables 17–19); the differential results are shown in Figs. 22–24.

Table 17: FHC (Runs 1, 2, 4, 5; data POT 8.857×10^{20}): integrated cross section, the two largest systematic terms, the total, and the fake-data closure against A_C -smeared truth. All values are read from the released covariance and results table. The reco-level flux and unfolding-amplification columns of earlier drafts are dropped here and treated in §14.2, since a reco-level flux is not re-derivable from the current extraction logs and mixing passes in one table is how stale numbers survive. [release 2026-09-04, `current.results.tsv`, per-run EXT, re-thrown fake data]

Observable	σ_{int}	Flux	Detector	Total	Closure χ^2/ndf
p_μ	0.838	28.0%	24.7%	40.7%	1.45/7 ($p = 0.98$)
p_π	0.822	32.5%	30.3%	50.2%	0.11/2 ($p = 0.95$)
$\cos \theta_\mu$	0.754	28.0%	28.3%	42.0%	0.67/5 ($p = 0.98$)
$\cos \theta_\pi$	0.726	26.1%	20.9%	36.9%	0.81/4 ($p = 0.94$)
$\theta_{\mu\pi}$	0.819	28.3%	19.6%	36.5%	0.30/5 ($p = 1.00$)
θ_μ	0.960	24.7%	15.6%	31.0%	1.52/5 ($p = 0.91$)

We extend the measurement to the reverse-horn-current beam as both a standalone measurement and a combined $\nu_\mu + \bar{\nu}_\mu$ result. The RHC *flux* is $\bar{\nu}_\mu$ -dominant (54.6% $\bar{\nu}_\mu$, versus FHC's 35%), with authoritative > 60 MeV integrals $\Phi_{\nu_\mu}^{\text{RHC}} = 2.923 \times 10^{-10}$, $\Phi_{\bar{\nu}_\mu}^{\text{RHC}} = 3.523 \times 10^{-10}$, total $6.446 \times 10^{-10} \text{ cm}^{-2}\text{POT}^{-1}$ (the FHC-convention computation reproduces the authoritative FHC values exactly). The *measured event rate*, however, remains ν_μ -dominant (76% ν_μ , 20% $\bar{\nu}_\mu$) because the ν_μ cross section is $\sim 2\text{--}3\times$ larger than the $\bar{\nu}_\mu$ one. The RHC overlays (Runs 1, 2, 3, 4a, 4b, 4c) are all verified as genuine ν_μ overlays and processed, 4.05×10^6 events, combined MC POT 1.546×10^{22} , mutually consistent (2.62×10^{-16} events/POT) with the PPF weights, covering the full RHC exposure (Run 3, the largest period at 5.52×10^{21} MC POT / 1.44×10^6 events, was added last). Native Run-4 RHC detector-variation samples (the standard eight-knob set: light-yield down/Rayleigh, recombination, space charge, and the four wire-modification variations) are now used for the RHC detector systematic, verified as genuine ν_μ overlays with a uniform 3.80×10^{-16} events/POT across all knobs; they replace the Run-1 FHC stand-in used previously. Native RHC dirt is still taken from FHC. The standalone result below uses Poisson-thrown central-value fake data at the RHC data POT 1.108×10^{21} . The generator predictions are re-averaged to the RHC flux composition; the combined FHC+RHC predictions are the fluence-weighted combination of the two horn modes (0.458 FHC / 0.542 RHC, each mode weighted by $\Phi_{\text{tot}} \times \text{POT}$).

The combined FHC+RHC measurement uses the summed 14-file response over the full exposure (5.6×10^6 events, combined data POT $D_{\text{comb}} = 1.994 \times 10^{21} = D_{\text{FHC}} + D_{\text{RHC}}$), with each horn mode normalised to its *own* data exposure. This is a genuine subtlety: the detector-variation covariance is absolute (POT-scaled reco counts), so a single POT scale applied to all files would weight the modes by their Monte-Carlo POT (0.62:1 FHC:RHC) rather than by data exposure (0.80:1). Each mode's numuMC and detVar samples therefore carry a per-mode `summed_pot` (FHC 2.158×10^{22} , RHC 2.782×10^{22} , $= S_{\text{mode}} \times D_{\text{comb}}/D_{\text{mode}}$) so the framework's $D_{\text{comb}}/\text{summed_pot}$ scaling reduces to $D_{\text{FHC}}/S_{\text{FHC}}$ and $D_{\text{RHC}}/S_{\text{RHC}}$; the fake data is Poisson-thrown with the matching per-mode rate.

Contrary to an earlier concern, the joint *flux systematic* is correct as built. All fourteen numuMC files carry the *same* 600 PPF multisim universes, which the universe maker accumulates index-matched across both horn modes in a single chain, so universe u is the same hadron-production draw in FHC and RHC. Per-universe summing therefore propagates one parameter draw through both beam configurations and lets each mode's flux response set the result, capturing the physical (not unit) cross-horn correlation automatically; this is the rigorously correct multisim treatment, superior to any hand-built block covariance. The joint result was in fact blocked by a different, purely technical issue: a framework guard that forbade a second detector-variation file of the same type (needed to carry Run-4 FHC *and* Run-4 RHC as `detVarCV`). That guard now accumulates the per-mode detVar samples exactly as it already did for the alternate-CV samples, matching its own in-code TODO. With both in place the combined closure passes for every observable (Table 19), with an unfolded flux systematic of 25–33% of the same size as the standalone FHC (26–31%) and RHC (24–33%) values (Tables 17–19), as expected for a common flux model propagated through three similarly conditioned responses.

The standalone RHC extraction (native Run-4 RHC detector variations) closes cleanly for every observable (Table 18): the realised A_C -smeared truth is recovered with $\chi^2/\text{ndf} \leq 0.4$ and $p > 0.8$ throughout. Two features stand out. First, the reco-level flux systematic, in the earlier

pass in which it was evaluated (§14.2), was $\sim 15\%$, below the FHC $\sim 17\%$ floor; the $\bar{\nu}_\mu$ -dominant flux carries a smaller PPFX hadron-production uncertainty. Second, the unfolding amplifies it in the same way as for FHC, so the unfolded flux term (24–33%, Table 18) is comparable to the FHC one, again limited by the response conditioning and the data statistics.

Table 18: RHC (Runs 1, 2, 3, 4; data POT 1.1082×10^{21}): integrated cross section, the two largest systematic terms, the total, and the fake-data closure against A_C -smeared truth. All values are read from the released covariance and results table. The reco-level flux and unfolding-amplification columns of earlier drafts are dropped here and treated in §14.2, since a reco-level flux is not re-derivable from the current extraction logs and mixing passes in one table is how stale numbers survive. [release 2026-09-04, `current.results.tsv`, per-run EXT, re-thrown fake data]

Observable	σ_{int}	Flux	Detector	Total	Closure χ^2/ndf
p_μ	0.690	27.6%	21.5%	37.7%	0.64/7 ($p = 1.00$)
p_π	0.748	28.4%	34.7%	48.6%	0.01/2 ($p = 0.99$)
$\cos \theta_\mu$	0.733	25.8%	35.0%	46.3%	0.35/5 ($p = 1.00$)
$\cos \theta_\pi$	0.663	24.7%	26.1%	39.1%	0.43/4 ($p = 0.98$)
$\theta_{\mu\pi}$	0.831	25.2%	16.6%	32.6%	0.18/5 ($p = 1.00$)
θ_μ	0.805	25.6%	22.4%	36.1%	0.12/5 ($p = 1.00$)

Table 19: Combined (data POT 1.9939×10^{21}): integrated cross section, the two largest systematic terms, the total, and the fake-data closure against A_C -smeared truth. All values are read from the released covariance and results table. The reco-level flux and unfolding-amplification columns of earlier drafts are dropped here and treated in §14.2, since a reco-level flux is not re-derivable from the current extraction logs and mixing passes in one table is how stale numbers survive. [release 2026-09-04, `current.results.tsv`, per-run EXT, re-thrown fake data]

Observable	σ_{int}	Flux	Detector	Total	Closure χ^2/ndf
p_μ	0.827	24.2%	21.4%	34.9%	0.41/7 ($p = 1.00$)
p_π	0.775	30.2%	30.8%	47.9%	0.01/2 ($p = 1.00$)
$\cos \theta_\mu$	0.812	25.7%	23.8%	37.3%	0.04/5 ($p = 1.00$)
$\cos \theta_\pi$	0.753	24.6%	20.6%	35.3%	1.05/4 ($p = 0.90$)
$\theta_{\mu\pi}$	0.879	25.7%	17.7%	33.2%	0.06/5 ($p = 1.00$)
θ_μ	0.851	25.2%	20.6%	34.4%	0.56/5 ($p = 0.99$)

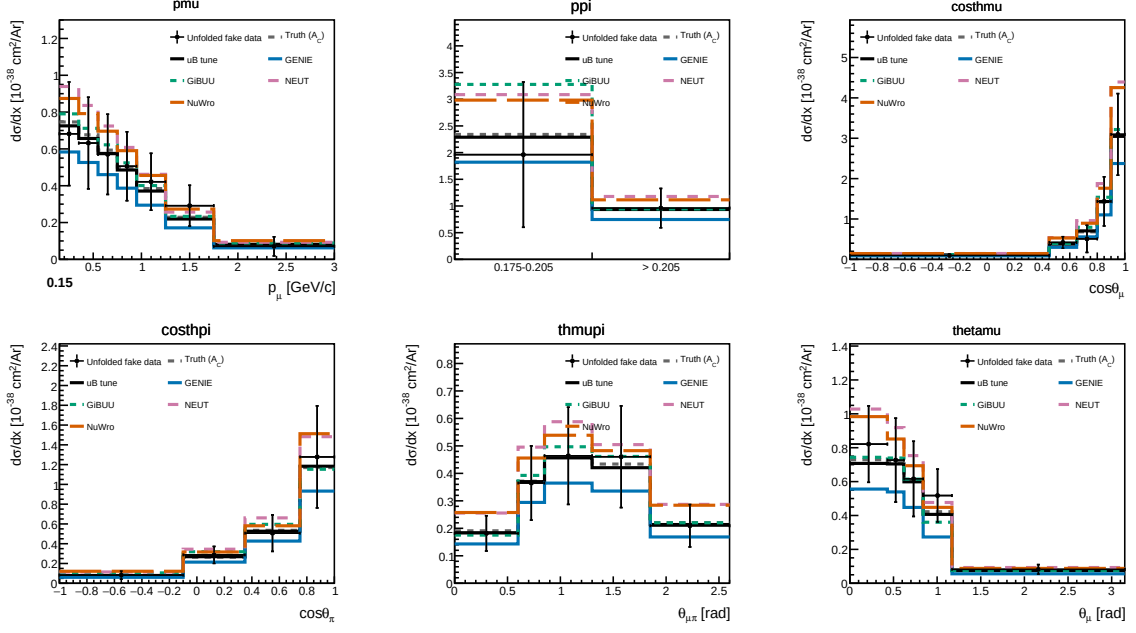


Figure 22: Differential cross sections $d\sigma/dx$ for the six released distributions on the current Poisson-thrown central-value fake data, FHC full exposure: unfolded fake data (black points), A_C -smeared fake-data truth (grey dashed), the MicroBooNE tune (black solid), and the four generator predictions (GENIE, GiBUU, NEUT, NuWro). The unfolded points recover the injected truth for every observable, the same closure summarised in Table 17.

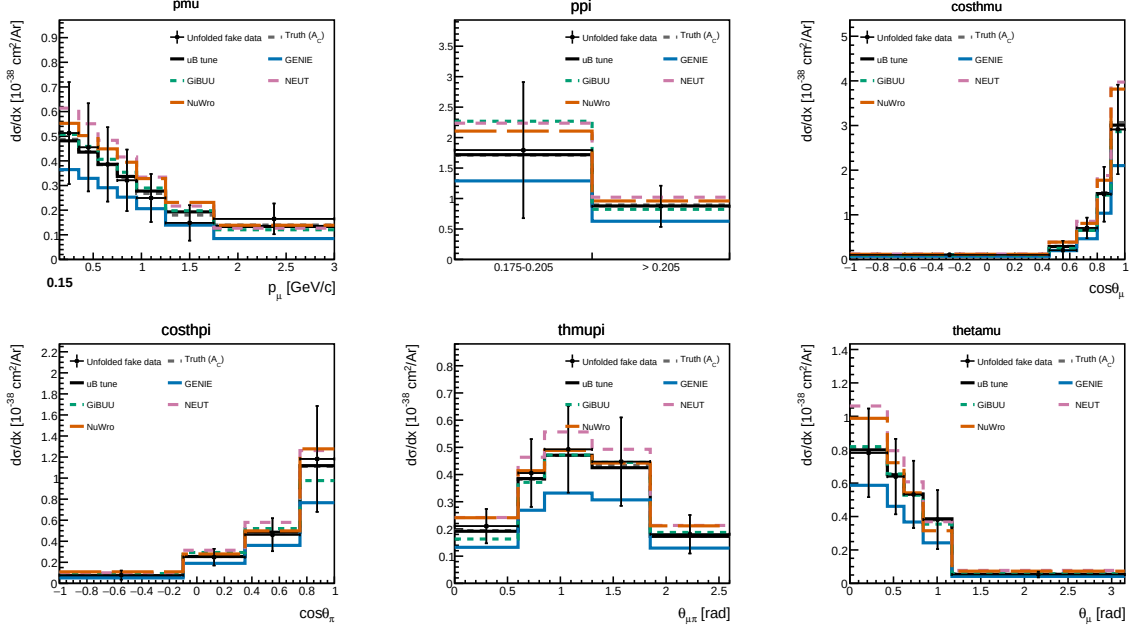


Figure 23: Standalone RHC ($\bar{\nu}_\mu$ -dominant) differential cross sections on the current Poisson-thrown fake data, with the RHC-flux-averaged generator predictions. Same layout as Fig. 22.

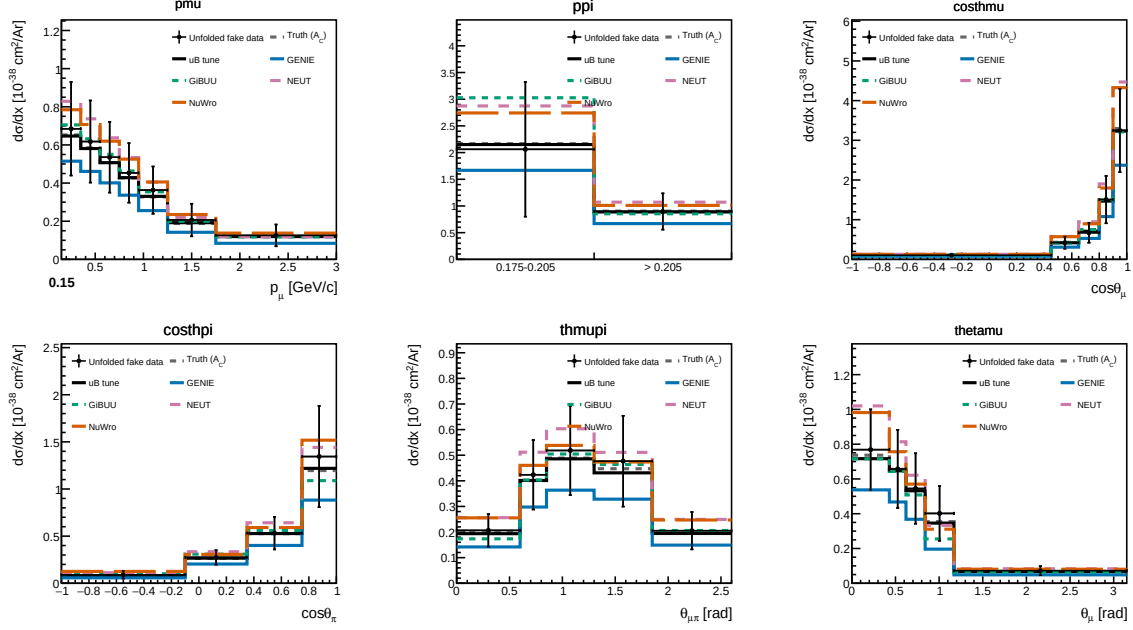


Figure 24: Combined FHC+RHC ($\nu_\mu + \bar{\nu}_\mu$) differential cross sections on the current per-mode-normalised fake data, with the fluence-weighted combined generator predictions.

Table 20: Integrated (unfolded) cross section σ_{int} over the measured phase space, for each observable and configuration (current fake data, coarsened $\cos \theta_\pi / \theta_{\mu\pi}$ binning, per-configuration flux; $10^{-38} \text{ cm}^2/\text{Ar}$). The value differs between observables because the Wiener-SVD additional-smearing A_C is not norm-preserving; the D’Agostini cross-check (Sec. 4.5 of the technical supplement) recovers a flatter, $\sim 20\%$ higher total, confirming the spread is a smearing effect. All entries were recomputed in a single pass with the current code and a forced rebuild of the cached universes. The MC POT normalization scales each ntuple file by its own stored POT ($D_{\text{run}}/\text{POT}_{\text{file}}$): the processing writes the *total* run MC POT into every file of a run, so for the multi-file RHC runs (Run 3 split across 5 ntuple files, Run 4 across 3) the per-file scaling summed over the splits gives the correct total. An intermediate version of this analysis instead divided by the *sum* of the per-file POT values; because those values are already run totals, that over-counted the denominator by the split multiplicity and scaled the RHC/Combined simulation a factor of 5 (Run 3) and 3 (Run 4) too low, which under-subtracted the background and inflated the RHC/Combined cross sections. That change has been reverted and the RHC/Combined columns here are the corrected values, roughly 10–25% below the intermediate ones; FHC (one MC file per run) is unaffected by either convention. The final row gives the alternative muon-angle parameterisation θ_μ (Sec. 5.1 of the technical supplement), complete for all three configurations; no θ_π parameterisation is extracted. (Values are fake-data closures; the signal is $\nu_\mu + \bar{\nu}_\mu$, so the RHC flux-average is not expected to lie far below FHC.) [release 2026-09-04, `current_results.tsv`, per-run EXT, re-thrown fake data]

Observable	FHC	RHC	Combined
p_μ	0.838	0.690	0.827
p_π	0.822	0.748	0.775
$\cos \theta_\mu$	0.754	0.733	0.812
$\cos \theta_\pi$	0.726	0.663	0.753
$\theta_{\mu\pi}$	0.819	0.831	0.879
θ_μ	0.960	0.805	0.851

Uncertainties shown. Every differential cross section and every integrated value quoted in this note carries *both* statistical and systematic uncertainties. The error bars on the unfolded points are the square root of the diagonal of the *total* covariance matrix, propagated through the unfolding: flux (PPFX multisims), cross-section (GENIE multisims and unisims), detector variations [15], hadron re-interaction [25], POT counting and target number, together with the data, MC and EXT statistical terms. For p_μ in FHC the bin-averaged total is 40.7% (Table 32), dominated by flux (28.0%) and detector (24.7%), with data statistics 14.3% and cross-section 10.7%. The cut-and-count total cross sections of §12.3 quote the prediction total of the same breakdown, written there as systematic $\oplus$ data-statistical (FHC: 40.7% $\oplus$ 4.3%).

12.2 How far A_C is from preserving normalisation

Because the Wiener-SVD result estimates $A_C x_{\text{true}}$ rather than x_{true} , and A_C is not norm preserving, summing the differential result over its bins does not give the physical total cross section. This subsection quantifies the effect, which has previously only been described qualitatively.

The row sums of A_C are the relevant measure: a row summing to unity passes the normalisation of that bin through unchanged, and a row summing to 0.2 removes four fifths of it.

Observable	FHC	RHC	combined
<i>Inclusive</i>			
p_μ	0.66–0.91	0.58–0.99	0.68–1.01
p_π (2-bin)	0.35–0.54	0.40–0.51	0.29–0.54
$\cos \theta_\mu$	0.17–0.39	0.05–0.45	0.09–0.43
$\cos \theta_\pi$	0.71–0.82	0.79–0.86	0.80–0.93
$\theta_{\mu\pi}$	0.52–1.03	0.58–1.15	0.63–1.21
θ_μ	0.58–0.78	0.22–1.16	0.53–0.67
<i>Proton-tagged</i>			
p_μ	0.46–1.00	0.55–0.89	0.65–0.96
p_π (2-bin)	0.34–0.84	0.41–0.45	0.47–0.48
$\cos \theta_\mu$	0.61–0.70	−0.08–0.57	0.28–0.46
$\cos \theta_\pi$	0.66–0.77	0.62–0.77	0.65–0.76
$\theta_{\mu\pi}$	0.37–0.87	0.31–0.75	0.35–0.78
$W_{\pi p}$	0.45–1.22	−0.14–1.26	0.32–1.24
W_{had}	0.54–0.81	0.30–0.86	0.18–0.87
δp_T	0.54–0.86	0.45–0.82	0.47–0.87
$\delta \phi_T$	0.65–0.78	0.53–0.72	0.28–0.67
$\delta \alpha_T$	0.59–0.98	0.57–0.93	0.60–0.96
p_n	0.71–0.86	0.61–0.82	0.59–0.86

Three features matter for interpretation.

The suppression is large: inclusive $\cos \theta_\mu$ retains as little as 5% of a bin’s normalisation in RHC. It also differs strongly between observables, which is precisely why the integrated cross section obtained by summing a differential result varies from observable to observable while the underlying physical total cannot. The D’Agostini cross-check, whose integral is flat across observables, is what establishes that this spread is the smearing rather than a physical or acceptance effect.

Two extractions have *negative* row sums, and the pattern identifies their cause. In $W_{\pi p}$ the top row carries negative off-diagonal terms in *all three* configurations (summing to −0.14 FHC, −0.30 combined, −0.61 RHC against diagonals of 0.73, 0.62 and 0.47); only RHC tips negative overall (row sum −0.14), because its diagonal element is the smallest of the three. The same happens in the first $\cos \theta_\mu$ bin of the proton-tagged RHC sample, whose diagonal is 0.079 against off-diagonal terms summing to −0.16 (row sum −0.08).

So this is not an RHC-specific defect but the regularisation pushing back against the least-determined bin of each observable — for $W_{\pi p}$, the same top bin that is only 2% diagonal in the six-bin response and that already led to withdrawing the differential shape. A negative row is not forbidden, since A_C is a regularisation operator rather than a probability matrix, but a prediction smeared through one is being told that raising the truth in that bin *lowers* the measured content, which is a statement about the regularisation and not about the detector. Treat those two rows with corresponding caution.

Numerical data release. All fifty-one extractions are released in `report/data_release/`, indexed by `index_extractions.tsv`: the additional-smearing matrix, the full per-component covariance in cross-section units, and `curves_*.tsv` holding the unfolded data together with every model curve shown here.

A prediction supplied as a per-bin cross section is compared with these results by

$$\left(\frac{d\sigma}{dx}\right)_i^{\text{smeared}} = \frac{1}{w_i} \sum_j (A_C)_{ij} p_j,$$

with p_j the predicted cross section in true bin j and w_i the width of bin i . This reproduces the published generator curves exactly — the worst per-bin deviation over the released matrices is 1.2×10^{-10} — so an external user can verify their implementation against any `curves_*.tsv` column before applying it to a new model.

That check is what exposed the double-smearing bug described above: before the fix the same expression missed the published curves by up to 55%, because the plotted predictions carried A_C twice.

The practical consequence is stated in the results section and repeated here because it is easy to get wrong: quote the physical total from the cut-and-count extraction, which is not subject to A_C , and treat the sum of a Wiener-SVD differential result as a quantity in smeared space rather than as a cross section.

12.3 Total flux-averaged cross section (cut-and-count)

As a check independent of the unfolding, the total flux-averaged cross section is also extracted directly from the selected event counts,

$$\sigma = \frac{N_{\text{sel}} - N_{\text{bkg}}}{\epsilon \Phi N_{\text{Ar}}}, \quad (16)$$

where N_{sel} is the number of selected events, N_{bkg} the predicted background (ν -background + EXT + dirt), $\epsilon = N_{\text{sel}}^{\text{sig}}/N_{\text{gen}}^{\text{sig}}$ the selection efficiency in the measured phase space ($\simeq 17.5\%$ for all configurations), Φ the integrated flux ($\Phi = \text{POT} \times \Phi_{\text{POT}}$, Table 5) and $N_{\text{Ar}} = 8.71 \times 10^{29}$ the number of argon nuclei in the fiducial volume (the dead z region removed), giving the same normalisation $N_{\text{Ar}} \Phi / 10^{38}$ used for the differential result. This uses no matrix inversion; by avoiding the Wiener-SVD additional smearing it recovers the true (un-suppressed) total, consistent with the D’Agostini cross-check (Sec. 4.5 of the technical supplement) and lying above the smearing-suppressed σ_{int} of Table 20, most visibly for RHC. The uncertainty is the quadrature of the systematic breakdown (Table 18 of the technical supplement–Table 20 of the technical supplement, flux-dominated) and the cut-and-count data-statistical term $\sqrt{N_{\text{sel}}}/(N_{\text{sel}} - N_{\text{bkg}})$, which is smaller (3–7%) than the unfolded data-statistical term because no regularisation amplifies it.

Table 21 is the headline total-cross-section table, for the inclusive selection and for the proton-tagged subsample, produced by `macros/total_xsec_counting.C` (`logs/counting_review6.log`, 2026-09-02, with the per-run beam-off treatment). The MC counts carry the framework’s full central-value weight, $\text{tune} \times \text{PPFX} \times \text{normalisation}$ (§6); the PPFX central-value weight averages

0.944 on selected signal, so a count that omits it, as earlier drafts of Table 10 did, sits 3–6% above the prediction the unfolding actually uses. Because the fake data are the central-value prediction, $N_{\text{sel}} - N_{\text{bkg}}$ equals the selected signal exactly and the extracted total is the injected tune, $\sigma = N_{\text{gen}}^{\text{sig}}/(\Phi N_{\text{Ar}})$, with a closure ratio of unity by construction; the table therefore demonstrates the normalisation chain, and the uncertainty column is what the measurement will carry at unblinding. The systematic fraction is taken, conservatively, as the prediction total of the released p_μ extraction of the same configuration (Table 32; `current_results.tsv`, `current_results_1p_new.tsv`), which includes the unfolding amplification of the flux and detector terms that a cut-and-count does not suffer; a reco-level evaluation of the same covariance would be smaller (the reco-level flux term alone is $\sim 17\%$, §14.2) and is left for the unblinded measurement. The RHC/FHC ratio of the inclusive totals is 0.901 (proton-tagged 0.878); these are the ratios of the injected tune in the two phase spaces, and are the numbers to be compared with data once unblinded. Because the extraction is observable-independent, this is also the only proton-tagged total that should be quoted: the integrals of Tables 28–30 are A_C -suppressed and observable-dependent.

Table 21: Headline cut-and-count total flux-averaged cross sections ($10^{-38} \text{ cm}^2/\text{Ar}$) in the measured phase space, on the Poisson-thrown central-value fake data. N_{sel} is the selected (fake-)data count, N_{bkg} the predicted background (ν -background + EXT + dirt), ε the selection efficiency, “truth” the injected central-value total $N_{\text{gen}}^{\text{sig}}/(\Phi N_{\text{Ar}})$; “stat” is the cut-and-count data-statistical term, “syst” the prediction-total fraction of the released p_μ extraction (see text), “total” their quadrature sum. MC counts carry the framework’s central-value weight (tune $\times$ PPFX $\times$ normalisation), the same as Table 10; the two differ by $\lesssim 1\%$ because the macro uses the Run-1 dirt overlay with per-mode scale factors rather than the per-run dirt files and applies the framework’s weight clamp. [release 2026-09-04, `current_results.tsv`, per-run EXT, re-thrown fake data]

Config	N_{sel}	N_{bkg}	ε [%]	σ	stat [%]	syst [%]	total [%]	truth	σ/truth
<i>Inclusive CC $1\mu 1\pi X p$</i>									
FHC	1612	668	17.4	1.031 ± 0.422	4.3	40.7	41.0	1.031	1.000
RHC	1759	744	17.6	0.928 ± 0.352	4.1	37.7	37.9	0.928	1.000
Combined	3371	1412	17.5	0.975 ± 0.342	3.0	34.9	35.0	0.975	1.000
<i>Proton-tagged CC $1\mu 1\pi 1p$ ($p_p > 0.3 \text{ GeV}/c$)</i>									
FHC	723	347	11.4	0.627 ± 0.317	7.1	50.0	50.5	0.627	1.000
RHC	786	395	11.4	0.551 ± 0.297	7.2	53.5	54.0	0.551	1.000
Combined	1509	742	11.4	0.586 ± 0.284	5.1	48.2	48.5	0.586	1.000

Against the standalone flux-averaged generator totals in the inclusive phase space (GENIE 0.82, GiBUU 1.13, NEUT 1.32, NuWro 1.25 for FHC; Table 17 of the technical supplement for all three configurations) the injected tune sits $0.5\text{--}0.6\sigma$ above GENIE and within 0.7σ of the other three; the generator spread itself, a factor 1.6, is larger than any of these differences. For the proton-tagged phase space the FHC generator totals are GENIE 0.50, NuWro 0.65, GiBUU 0.74 and NEUT 0.78, bracketing the injected 0.63.

12.4 Background-control sidebands

The pseudo-data studies in this section are not a closure test of the complete beam-on sideband prediction. The pseudo-data contain only the central-value neutrino-MC component, so the ratios reported here validate the selection logic, the run scaling and the central-value weighting only; the full-stack variant of §12.6 additionally exercises the bookkeeping path. At sideband unblinding, beam-on data will be compared with the full neutrino-MC + EXT + dirt prediction under the frozen procedure of §12.8.

Because the analysis is blind, the background model cannot yet be confronted with beam-on data in the signal region. To exercise it before unblinding, four signal-depleted control regions (“sidebands”) are defined, each anchored by the inversion of one nominal signal requirement, with the requirements that the inversion leaves undefined dropped explicitly (branch-level definitions in Table 1):

- **CC0 π** : the muon selection with no reconstructed charged pion (charged-current quasi-elastic / $2p2h$ -enriched);
- **multi- π** : two or more charged pions (deep-inelastic / resonant multi-pion);
- **π^0** : a reconstructed shower present but the π^0 shower veto failed (CC π^0 / NC π^0);
- **cosmic**: the full signal selection with the muon–pion opening angle inverted ($\theta_{\mu\pi} > 2.6$), enriched in out-of-time cosmic activity.

Each region is evaluated with the same per-run POT scaling and CV event weighting (§6) used for the measurement. Because the current “data” is a Poisson throw of the neutrino Monte Carlo central value only (it contains no EXT or dirt component), the meaningful closure quantity here is the ratio of the pseudo-data to the CV-weighted *neutrino* MC, $N_{\text{data}}/N_{\nu\text{MC}}$; the EXT and dirt yields are shown for scale but enter the background-model test only once real beam-on data are available. Table 22 summarises the four regions for the three beam configurations.

Table 22: Background-control sidebands. “Signal fraction” is the signal contamination of the (CV-weighted neutrino) MC in the region; all four are strongly signal-depleted. $N_{\text{data}}/N_{\nu\text{MC}}$ is the pseudo-data to neutrino-MC ratio: the high-statistics regions (CC0 π , π^0) close to unity across FHC/RHC/combined, validating the per-run weighting and POT-scaling machinery; the low-statistics regions (multi- π , cosmic) scatter within Poisson expectation ($\sqrt{N} \approx 6\text{--}10\%$). The background-model test against the full stack (with EXT and dirt) is performed at control-region unblinding with beam-on data. [release 2026-09-04, `current_results.tsv`, per-run EXT, re-thrown fake data]

Sideband	Signal frac. (of νMC)	$N_{\text{data}}/N_{\nu\text{MC}}$		
		FHC	RHC	Combined
CC0 π	$\sim 10\%$	1.00	0.99	1.00
multi- π	$\sim 23\%$	1.01	1.10	1.06
π^0	$\sim 10\%$	1.00	1.01	1.00
cosmic	$\sim 8\%$	1.09	0.99	1.04

Yields and composition. Table 23 gives the full breakdown for the three configurations. The regions differ enormously in size and in what they are sensitive to, which is what makes them complementary rather than redundant:

- **CC0 π** is by far the largest ($\sim 12\text{k}$ pseudo-data events per beam, $\sim 24\text{k}$ combined). Its neutrino component is dominated by the quasi-elastic and $2p2h$ modes that feed the signal region only through pion *absorption* in the final state. It is therefore the natural handle on the FSI model that converts CC $1\pi^\pm$ into CC0 π and vice versa; the largest single migration path into and out of the signal definition (§7).
- **π^0** is the second largest (6.5–14.7k). It is enriched in CC/NC π^0 events, whose showers are precisely what the shower veto (Cut 8) is designed to remove; it therefore tests the veto efficiency and the π^0 production normalisation together. These are the events that leak into the signal region when a photon is missed.

- **multi- π** is small (127–303 events) but has the highest signal contamination of the four ($\sim 23\%$), because a true multi-pion event with one pion lost is exactly the wrong-topology background the selection fights. Its size is what limits its constraining power at present.
- **cosmic** is the region with by far the largest beam-off (EXT) share of the prediction: 53 EXT against 75 neutrino-MC events for FHC, a ratio of 0.7, against 0.1 in CC0 π (with the per-run beam-off treatment of §6; earlier drafts, with the beam-off over-scaled by an order of magnitude, had it EXT-dominated). It is the region that tests the beam-off normalisation and shape, and the one whose interpretation changes most at unblinding, since the current pseudo-data contain no EXT component at all. Its beam-off purity is only 41% (FHC; 34% RHC), so under the target-purity requirement of §12.8 it is diagnostic: it cannot by itself produce a fitted EXT normalisation correction, only an additional uncertainty.

What the present numbers do and do not show. With Poisson-thrown neutrino-MC pseudo-data, the ratios of Table 22 test the *machinery*, per-run POT scaling, CV weighting, the selection logic in each inverted region, and not the background model. That test passes: the two high-statistics regions sit at $N_{\text{data}}/N_{\nu\text{MC}} = 0.99\text{--}1.01$ in all three configurations, which is a non-trivial check of the per-run normalisation that caused the difficulties described in Sec. B of the technical supplement. The apparent deviations in the small regions are statistical: with 82 cosmic-sideband events for FHC, a $\sqrt{N}$ fluctuation is 11%, so the observed 1.09, and the RHC multi- π 1.10 (on 175 events, 8%), are within about one standard deviation.

Use at unblinding. Once the control regions are opened with beam-on data the same table becomes a genuine background test, and the statistical precision available is set by the yields above: the combined CC0 π and π^0 samples have roughly 1% and 1.5% *raw data statistical* precision. Their effective constraint on the signal-region background is weaker and depends on the correlated transfer uncertainty (§12.7): the transfer factors are small (0.03 and 0.02) but the transfer from a CC0 π event to a selected CC1 π topology is itself model dependent, so the achievable constraint is set by the cross-section and detector terms of that transfer rather than by the sample size. The cosmic region would constrain the EXT normalisation only to $\sim 17\%$ (94 EXT events in ~ 250 combined), weaker than the 3–5% gate-ratio uncertainty already assigned; with the beam-off now at 1.9% of the signal-region sample that is adequate. The multi- π region, at ~ 300 combined events, gives $\sim 6\%$, useful as a cross-check but not competitive with the others. The intended procedure is to unblind the control regions first, use them to constrain or reweight the dominant background components, and only then open the signal region.

Table 23: Sideband yields at full exposure. “Signal” is the signal contamination of the neutrino MC (events satisfying the inclusive signal definition of §7; all yields and the pseudo-data throw apply the safe-weight rule of Eq. (5)), “ ν bkg” the remaining neutrino MC, EXT the per-run beam-off (cosmic) samples scaled by the run gate ratios (§6), and dirt the out-of-cryostat overlay. The pseudo-data column is a Poisson throw of the neutrino MC only, so it should be compared with Signal + ν bkg; EXT and dirt are listed for scale and enter the comparison only with real beam-on data. [release 2026-09-04, `current_results.tsv`, per-run EXT, re-thrown fake data]

Beam	Sideband	Pseudo-data	Signal	ν bkg	EXT	Dirt	Sig. frac.
FHC	CC0 π	12 181	1 124	11 068	1 358	56	9.2%
FHC	multi- π	125	30	94	7.2	0.1	24.0%
FHC	π^0	6 628	662	5 985	778	12	10.0%
FHC	cosmic	82	6.0	69	53	2.2	7.9%
RHC	CC0 π	12 590	1 206	11 452	1 214	43	9.5%
RHC	multi- π	175	35	124	5.9	0.1	22.0%
RHC	π^0	8 176	780	7 300	678	9.1	9.6%
RHC	cosmic	81	6.4	75	42	1.7	7.9%
Comb	CC0 π	24 771	2 329	22 520	2 572	99	9.4%
Comb	multi- π	300	65	218	13	0.2	22.9%
Comb	π^0	14 804	1 441	13 285	1 455	21	9.8%
Comb	cosmic	163	12	145	94	3.9	7.9%

12.5 Region overlaps

The four regions are disjoint from the signal region *by construction*: CC0 π and multi- π require $n_\pi = 0$ and $n_\pi \geq 2$ against the signal region’s $n_\pi = 1$; the π^0 region requires the shower veto to fail; the cosmic region requires the opening-angle cut to fail. This was also checked event by event on the sideband-instrumented files (`macros/sb_overlap.C`): the number of events carrying both the signal-region flag and any control-region flag is exactly zero in the Monte Carlo and in the pseudo-data of every configuration. The regions are *not* mutually exclusive: multi- π does not apply the shower veto, so events with two pion candidates and a failed shower veto belong to both multi- π and π^0 (47 of the 124 FHC multi- π MC events, 68 of 159 RHC); all other pairs are disjoint. Any simultaneous fit of the regions must account for that overlap; the per-region comparisons of §12.8 do not need to.

12.6 Full-stack pseudo-data test

Table 22 compares the neutrino-MC throw with the neutrino MC alone. To exercise the exact path that beam-on data will take, the same macro was run with pseudo-data consisting of the neutrino-MC throw *plus* an independent Poisson throw of the beam-off and dirt expectations (`sideband_compare.C`, mode `fakestack`), compared with the complete prediction. The data/prediction ratios are 0.996, 1.047, 0.992, 1.000 (FHC; CC0 π , multi- π , π^0 , cosmic), 0.992, 1.122, 1.012, 1.080 (RHC) and 0.995, 1.079, 1.006, 1.027 (combined), each within its statistical uncertainty (0.6–10%). This tests bookkeeping, not physics. Figures 25 and 26 show, for every region and both beam modes, the frozen distributions of `configs/sideband_plots.txt` exactly as they will be drawn at unblinding, with the same full-stack pseudo-data in place of the beam-on points.

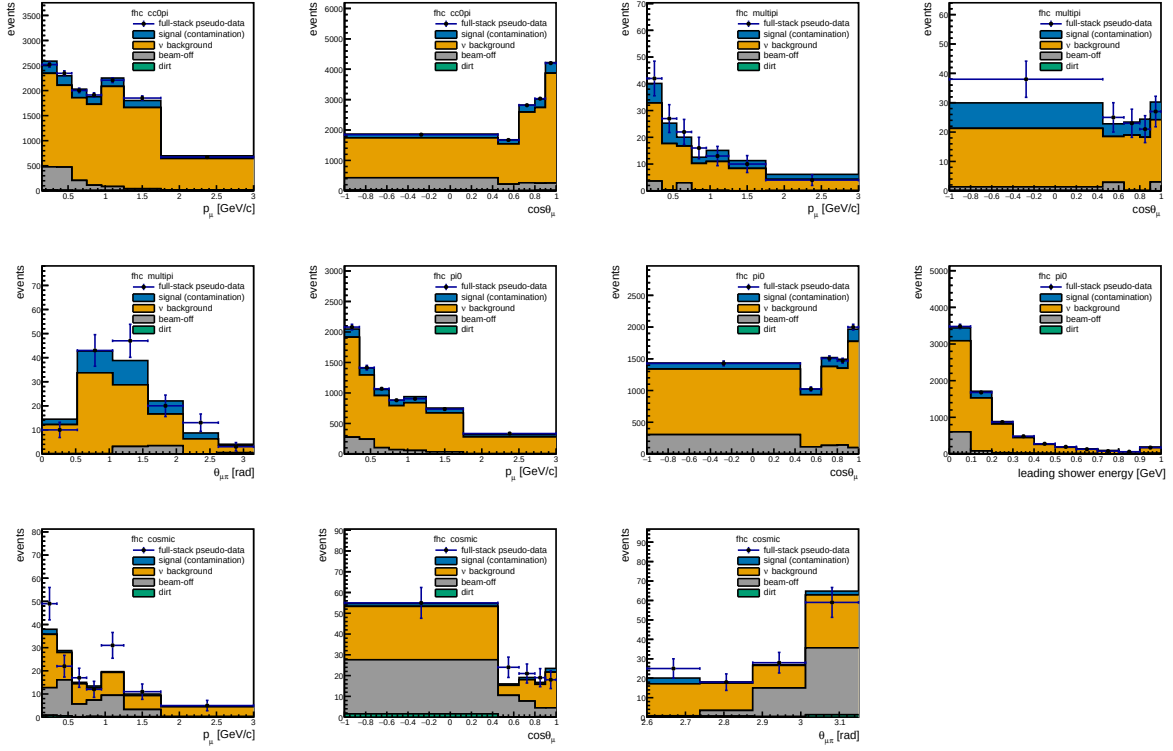


Figure 25: FHC control-region distributions at the frozen binning (configs/sideband.plots.txt): by row, CC0 π , multi- π , π^0 and cosmic; p_μ , $\cos\theta_\mu$ and, where defined, the muon-pion opening angle or the leading shower energy. Stack: signal contamination, neutrino background, beam-off (per-run scaled) and dirt; points: full-stack pseudo-data (neutrino-MC throw plus a Poisson throw of the beam-off and dirt expectations). These are the plots the collaboration will see with beam-on points after opening; overflow is in the last bin.

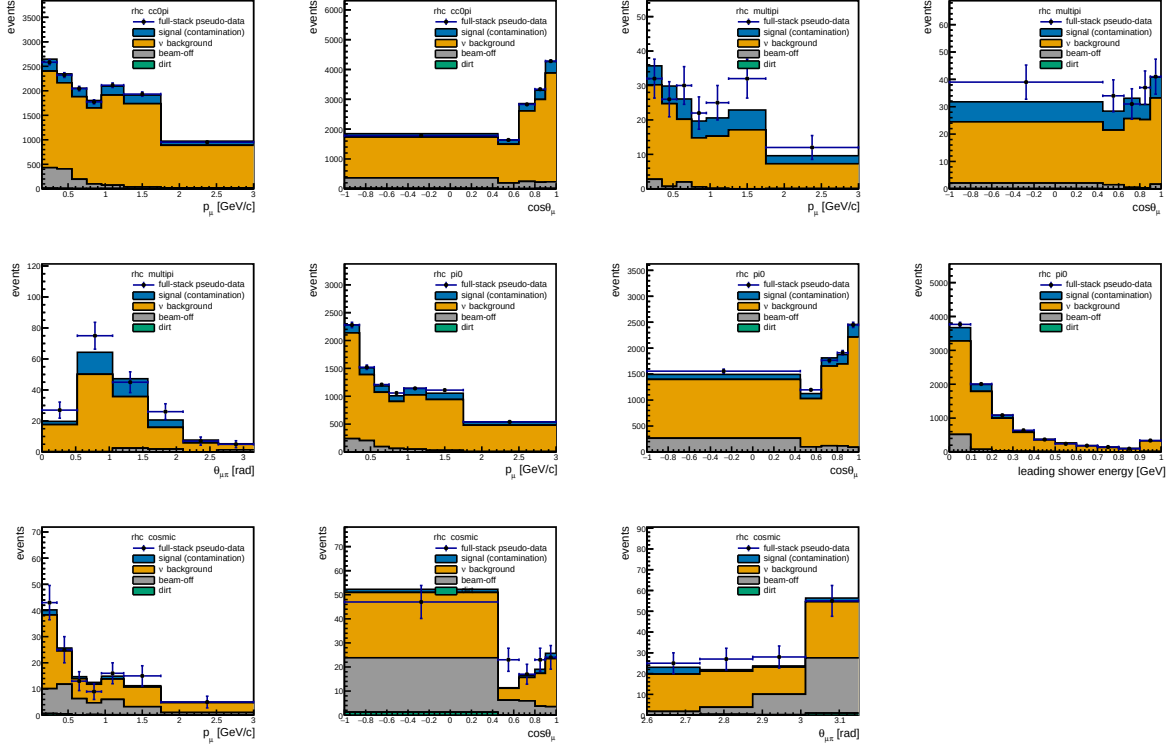


Figure 26: As Fig. 25, for RHC.

12.7 Transfer factors and control-region uncertainties

Whether a control-region discrepancy can be propagated into the signal region depends on the transfer factor $T_{b,c} = N_b^{\text{SR}}/N_b^{\text{CR}_c}$ for each background class b : a region constrains a class only where T is small and the ratio is stable against the model variations. Table 24 gives the composition of each region by background class and the transfer factors (`macros/sb.transfer.C`).

Table 24: Composition of the signal region (SR) and of each control region by background class (CV-weighted neutrino MC, per-run POT-scaled; beam-off and dirt excluded), and the transfer factors $T = N^{\text{SR}}/N^{\text{CR}}$ of each class. Classes are built from the selection’s truth counters: $\nu_\mu\text{CC } 0\pi$ (no charged or neutral pion above threshold), $1\pi^\pm$ outside the measured phase space, any π^0 , $\geq 2\pi^\pm$ without π^0 , out-of-fiducial-volume, and NC plus $\nu_e\text{CC}$. A region constrains a class only where T is small: $\text{CC}0\pi$ constrains the 0π class ($T \simeq 0.03$) and π^0 the π^0 class ($T \simeq 0.02$); the multi- π region contains *fewer* multi-pion events than the signal region ($T \simeq 2.5$) and is a weak constraint; the cosmic region constrains only the beam-off component (not listed: 53 EXT events in the region against 30 in the signal region, $T = 0.57$, FHC). [release 2026-09-04, `current_results.tsv`, per-run EXT, re-thrown fake data]

Class	SR	CC0 π	multi- π	π^0	cosmic	$T_{\text{CC}0\pi}$	T_{multi}	T_{π^0}	T_{cosmic}
<i>FHC</i>									
signal (contamination)	943	1 124	29.7	662	6.0	0.84	32	1.43	158
$\nu_\mu\text{CC } 0\pi$ (proton or other track as pion)	244	8 781	2.6	1 011	23.2	0.03	95	0.24	11
$\nu_\mu\text{CC } 1\pi^\pm$ outside the measured phase space	52.7	629	2.7	361	14.8	0.08	19	0.15	3.57
$\nu_\mu\text{CC}$ with a π^0	58.0	408	20.2	3 070	2.5	0.14	2.87	0.02	23
$\nu_\mu\text{CC } \geq 2\pi^\pm$, no π^0	145	222	57.8	658	3.9	0.65	2.51	0.22	37
out of fiducial volume	47.4	480	3.3	253	18.5	0.10	14	0.19	2.56
NC and $\nu_e\text{CC}$	89.5	549	7.1	632	6.4	0.16	13	0.14	14
total ν -MC	1 581	12 192	124	6 646	75.3				
<i>RHC</i>									
signal (contamination)	1 015	1 206	34.9	780	6.4	0.84	29	1.30	158
$\nu_\mu\text{CC } 0\pi$ (proton or other track as pion)	257	8 981	2.4	1 117	25.7	0.03	108	0.23	9.98
$\nu_\mu\text{CC } 1\pi^\pm$ outside the measured phase space	49.1	678	3.4	384	16.7	0.07	14	0.13	2.93
$\nu_\mu\text{CC}$ with a π^0	75.5	445	29.0	3 756	2.7	0.17	2.61	0.02	28
$\nu_\mu\text{CC } \geq 2\pi^\pm$, no π^0	176	238	71.8	885	4.1	0.74	2.45	0.20	43
out of fiducial volume	54.5	511	4.9	302	17.2	0.11	11	0.18	3.17
NC and $\nu_e\text{CC}$	108	599	12.5	857	8.9	0.18	8.59	0.13	12
total ν -MC	1 735	12 658	159	8 080	81.8				

The neutrino-MC prediction in each region carries the flux, cross-section and hadron re-interaction uncertainties evaluated directly from the multisim weights on the sideband-instrumented files with the same weight conventions as the extraction (`macros/sb_syst.C`), and the detector-variation term from the Run-4 detector-variation samples, which carry the region flags. Table 26 gives the uncertainty on the transfer factor of each region’s target class, computed as a ratio in every universe and variation.

Table 25: Normalisation uncertainty (%) of the neutrino-MC prediction in the signal region and in each control region: flux (PPFX), cross-section (GENIE multisim) and hadron re-interaction multisims evaluated on the sideband-instrumented files with the extraction’s weight conventions (`sb_syst.C`), and the detector term from the Run-4 detector-variation samples, which carry the region flags (`sb_detvar.C`; eight variations against the detector-variation central value, summed in quadrature as in the extraction). “Total” is the quadrature sum. The last column is the correlation, across the flux universes, between the region’s yield and the signal-region yield: the flux term is a common normalisation and cancels in a control-region-to-signal-region transfer (Table 26). The ν -MC yields apply the safe-weight rule of Eq. (5), as do Tables 10 and 23 and the pseudo-data throws since 2026-09-03; the yields agree between the tables. [release 2026-09-04, `current_results.tsv`, per-run EXT, re-thrown fake data]

Beam	Region	ν -MC yield	flux	cross section	re-interaction	detector	total	corr. with SR (flux)
FHC	signal region	1 581	18.0	16.9	2.7	20.7	32.3	1.00
FHC	CC0 π	12 192	16.7	11.9	1.1	21.8	30.0	1.00
FHC	multi- π	124	20.0	20.5	5.5	18.5	34.6	0.99
FHC	π^0	6 646	18.8	15.8	2.4	21.8	32.9	1.00
FHC	cosmic	75	15.9	13.2	1.8	24.3	31.9	0.98
RHC	signal region	1 735	16.2	16.2	2.7	14.2	27.1	1.00
RHC	CC0 π	12 658	15.1	11.8	1.1	8.7	21.1	0.99
RHC	multi- π	159	17.0	19.3	6.1	19.4	32.8	0.99
RHC	π^0	8 080	16.3	15.4	2.8	7.3	23.7	1.00
RHC	cosmic	82	15.2	12.8	2.5	11.9	23.3	0.99

Table 26: Transfer factor $T = N^{\text{SR}}/N^{\text{CR}}$ of each control region’s target class and its fractional uncertainty (%), with T recomputed as a ratio in every universe so that the numerator–denominator covariance is included (`sb_transfer_unc.C`, `sb_detvar.C`). The flux term, 15–20% on either yield alone (Table 25), cancels to 2–3% in the ratio, as the yield correlations anticipated; the cross-section term does not cancel, because it changes the composition of the two regions differently, and the detector term is the largest. The detector entries carry the finite statistics of the detector-variation samples (a few thousand events per region), so those for the smaller classes are upper limits rather than measurements; the multi- π transfer is dominated by the Monte-Carlo statistics of the region itself. The last row uses the total neutrino MC of the CC0 π region as a reference. [release 2026-09-04, `current_results.tsv`, per-run EXT, re-thrown fake data]

Beam	Region	Target class	T	flux	cross section	re-int.	MC stat	detector	total
FHC	CC0 π	$\nu_\mu\text{CC } 0\pi$	0.0278	3.3	11.7	6.7	2.5	27.1	30.5
FHC	multi- π	$\nu_\mu\text{CC } \geq 2\pi^\pm$	2.5094	3.0	4.7	2.6	5.1	18.5	20.2
FHC	π^0	$\nu_\mu\text{CC with } \pi^0$	0.0189	2.6	6.1	1.8	4.4	15.1	17.1
FHC	CC0 π	all neutrino MC	0.1296	1.7	13.2	2.6	1.0	9.7	16.7
RHC	CC0 π	$\nu_\mu\text{CC } 0\pi$	0.0286	3.0	11.5	6.6	2.0	30.6	33.6
RHC	multi- π	$\nu_\mu\text{CC } \geq 2\pi^\pm$	2.4527	1.9	4.0	2.8	3.9	21.3	22.3
RHC	π^0	$\nu_\mu\text{CC with } \pi^0$	0.0201	1.8	6.3	1.8	3.3	11.5	13.7
RHC	CC0 π	all neutrino MC	0.1371	2.0	12.5	2.6	0.8	9.8	16.3

12.8 Frozen procedure for opening the control regions

The following is the analysers’ proposed procedure, to be ratified with the approval; once ratified it is frozen and any departure from it is itself a change requiring approval.

1. **What is examined.** For each of the four regions: the event total, and the reconstructed distributions of p_μ , $\cos\theta_\mu$ and the muon–pion opening angle, and for the π^0 region the

leading shower energy (multiplicities are not stored in the processed trees and cannot be added without a reprocess, which the frozen procedure forbids after inspection), at the binning of the released observables where one exists and otherwise at the binning fixed in `configs/sideband_plots.txt` and shown, with full-stack pseudo-data, in Figs. 25 and 26. FHC and RHC are examined separately; the combined configuration is a cross-check only. Data are compared with the complete prediction, neutrino MC + beam-off + dirt, with the run-period beam-off scaling of §6.

2. **Statistic.** A normalisation comparison, $N_{\text{data}}/N_{\text{pred}}$ with its total uncertainty, and a binned shape χ^2 using the full covariance (flux, cross-section, re-interaction and detector terms, plus MC, beam-off and data statistics). Goodness of fit is evaluated against the *pre-fit* prediction.
3. **Acceptable agreement.** A region passes if the normalisation ratio is within 2σ of unity and the shape p -value exceeds 0.05, in FHC and RHC independently, where σ is the *pre-fit* predictive uncertainty of the difference, $\sigma^2(N_{\text{data}} - N_{\text{pred}}) = N_{\text{data}} + \mathbf{1}^T V_{\text{pred}} \mathbf{1}$, with V_{pred} the summed covariance of the complete prediction (neutrino MC, beam-off and dirt): their finite statistics, flux, cross-section, re-interaction and detector terms, with all correlations among the prediction components; the same V_{pred} enters the shape χ^2 . A single-bin excursion beyond 3σ with a passing total is recorded and examined for a reconstruction pathology but does not by itself fail the region.
4. **Permitted responses and the rule for choosing between them.** For a region that passes: no change (a). For a region that fails: (i) the pre-declared distributions are first examined for a localised reconstruction failure (a single-bin or single-run excursion), which is treated as a reconstruction issue, not a model correction; (ii) if the discrepancy is consistent with a pure normalisation shift and the region's target-class purity exceeds 50%, a one-parameter normalisation fit of that class is performed, and the fitted correction (c) is applied only if it improves the fit by a likelihood-ratio test at $p < 0.01$ and the transfer-factor uncertainty of Table 26 is available and is added to it; (iii) otherwise the central prediction is left unchanged and an additional normalisation uncertainty covering the discrepancy is assigned to the class (b); (iv) if neither (b) nor (c) gives acceptable post-treatment agreement, the background model is revised (d), after which renewed collaboration approval is required before the signal region is opened. The pass/fail test of item 3 is always the pre-fit one; any post-treatment goodness of fit is reported separately and labelled as such. If more than one region fails, no independent corrections are combined without accounting for shared events and correlated systematic variations: in particular, multi- π and π^0 share events (§12.5) and are treated jointly, or multi- π is designated diagnostic only, which its transfer factor already argues for.
5. **Prohibited responses.** Changing any signal-selection cut; changing the binning of any released observable; choosing a generator or tune after inspecting the regions; bin-by-bin reweighting; and any use of the beam-on signal-region sample.
6. **Sign-off.** The analysis conveners, the flux and detector-systematics contacts, and the collaboration review committee, on the basis of a short report containing the comparisons of item 1 and the response chosen from item 4.

12.9 Scope: inclusive regions only

The request of §2 covers the four inclusive regions. The proton-tagged regions of §12.10 apply the proton tag rather than inverting it and therefore do not constrain the $\approx 13\%$ proton mis-tag background that most affects the hadronic observables; the inverted-proton-PID region that would do so has not yet been implemented. The proton-tagged control-region programme is

consequently *not* ready for equivalent approval, and opening the inherited proton-tagged regions, while useful diagnostically, would not validate the proton-tagged background model.

12.10 Control regions for the proton-tagged subsample

The proton-tagged selection inherits its cut logic from `CC1mu1piXp`, and with it the four sideband definitions of §12.4: the reprocessed proton-tagged files already carry `CC1mu1pi1p_sb_cc0pi`, `_sb_multi`, `_sb_pi0` and `_sb_cosmic`. No new definitions or reprocessing are required to exercise the background model of the W/TKI measurement; the same four regions can be evaluated either as inherited (the inclusive phase space) or with a reconstructed tagged proton additionally required, which places them in the same phase space as the measurement itself. Table 27 gives both.

Three things follow from the numbers. First, all four regions are *more* signal-depleted for the proton-tagged signal than they were for the inclusive one: 6.0–7.3% signal contamination in $CC0\pi$ and π^0 , against 9.5–10.0% for the same regions in the inclusive phase space and a 48% purity in the signal region, because requiring a proton does not make a pion-less or π^0 event look like signal. Second, the proton requirement costs about a third of the statistics ($CC0\pi$: 11 887 $\rightarrow$ 8 268 for FHC) but suppresses the beam-off contribution by roughly a factor six (1 358 $\rightarrow$ 225, FHC $CC0\pi$, Table 27): cosmic-induced events rarely contain a reconstructed proton above threshold, so the proton tag is itself a strong cosmic rejector. This is worth noting for the systematic budget; the EXT normalisation matters far less in the proton-tagged measurement than in the inclusive one. Third, the pseudo-data to neutrino-MC ratios remain consistent with unity in every region and configuration (0.956–1.158 with the tag), so the per-run normalisation machinery is validated in the proton-tagged phase space as well.

What these regions cannot do. All four *apply* the proton tag rather than invert it, so none of them constrains the proton *mis-tag* background; the $\sim 13\%$ of tagged protons that are really a pion or another track, which smears $W_{\pi p}$ and the TKI variables directly. Testing that requires a region defined by inverting the proton PID (selecting candidates that pass the momentum and containment requirements but fail the LLR cut), which does need a new branch in `CC1mu1pi1p` and a reprocess. Slicing a measured W/TKI observable is *not* an alternative: because the signal definition is inclusive in those variables, cutting on them splits the signal region rather than depleting it. Explicitly, on the Run-1 FHC overlay the signal region has 53.7% purity and the slices $W_{\pi p} > 1.5$ GeV, $W_{\pi p} < 1.15$ GeV, $p_n > 0.8$ GeV/ c and $\delta p_T > 0.7$ GeV/ c give 50.0%, 48.9%, 47.3% and 52.0% respectively; no depletion at all. Even the kinematically forbidden region $W_{\text{had}} < m_\pi + m_p$ retains 53.4% signal, because the calorimetric mass is systematically under-reconstructed and true signal populates it through resolution.

Table 27: Sideband yields for the proton-tagged subsample at full exposure, evaluated with the inherited definitions both as-is and with a tagged proton additionally required. “Sig.” is the proton-tagged signal contamination of the neutrino MC. $N_{\text{data}}/N_{\nu\text{MC}}$ compares the Poisson-thrown pseudo-data with the CV-weighted neutrino MC; EXT and dirt are listed for scale. [release 2026-09-04, `current_results.tsv`, per-run EXT, re-thrown fake data]

Beam	Region	N_{data}	Signal	ν bkg	EXT	Sig. frac.	$N_{\text{data}}/N_{\nu\text{MC}}$
<i>inherited definitions (inclusive phase space)</i>							
FHC	CC0 π	12 181	705	11 487	1 358	5.8%	0.999
FHC	multi- π	125	30	94	7.2	23.9%	1.012
FHC	π^0	6 628	466	6 181	778	7.0%	0.997
FHC	cosmic	82	3.5	72	53	4.6%	1.089
RHC	CC0 π	12 590	731	11 927	1 214	5.8%	0.995
RHC	multi- π	175	34	125	5.9	21.5%	1.101
RHC	π^0	8 176	524	7 556	678	6.5%	1.012
RHC	cosmic	81	3.5	78	42	4.3%	0.990
Comb	CC0 π	24 771	1 436	23 414	2 572	5.8%	0.997
Comb	multi- π	300	64	219	13	22.6%	1.062
Comb	π^0	14 804	990	13 736	1 455	6.7%	1.005
Comb	cosmic	163	6.9	150	94	4.4%	1.037
<i>with a tagged proton required (measurement phase space)</i>							
FHC	CC0 π	8 386	486	7 886	225	5.8%	1.002
FHC	multi- π	59	9.8	46	0.2	17.7%	1.064
FHC	π^0	4 011	293	3 769	189	7.2%	0.987
FHC	cosmic	39	1.7	24	0.1	6.5%	1.517
RHC	CC0 π	8 621	500	8 127	197	5.8%	0.999
RHC	multi- π	62	14	58	1.3	19.8%	0.852
RHC	π^0	4 902	333	4 534	162	6.8%	1.007
RHC	cosmic	37	2.3	29	0.5	7.5%	1.187
Comb	CC0 π	17 007	985	16 013	422	5.8%	1.000
Comb	multi- π	121	24	104	1.5	18.9%	0.944
Comb	π^0	8 913	626	8 303	351	7.0%	0.998
Comb	cosmic	76	4.0	53	0.6	7.1%	1.336

13 Secondary scope: proton-tagged subsample (hadronic invariant mass and transverse kinematic imbalance)

Scope note. This section is secondary scope, but the reason has changed and the change is substantive.

Detector variations have now been propagated through the proton-tagged selection. All thirty-three proton-tagged extractions were rebuilt on 2026-08-31 with the nine detector variation samples included and re-unfolded from those universes, so every uncertainty quoted here now carries a detector term. Those terms run from 10.8% to 65.5% with a median of 24.0%, against 16.9%–43.1% for the inclusive results: the same order, as expected for the same detector. The central values were regenerated in the same pass and therefore also postdate the central-value weighting fix.

The closure χ^2 values were recomputed in the same pass, so no quantity in this section is carried over from before the fix, and the additional-smearing and covariance matrices are released with the rest (Sec. 12.2).

What remains is a binning limitation that no systematic treatment addresses: the differential $W_{\pi p}$ shape stays withdrawn, because at this exposure no binning satisfies both the migration and the statistics requirement. The proton-tagged sample supports

an integrated cross section, not a resonance-shape measurement. See Sec. 12.1.

On the six-bin $W_{\pi p}$ material below. Six-bin $W_{\pi p}$ distributions and their closure appear in this section. They are *diagnostic*: they show why the shape is not measurable, and they are not a result. The differential $W_{\pi p}$ shape is withdrawn (Sec. 12.1), and closure against A_C -smeared truth cannot restore shape information the response does not contain — the top bin is 2% diagonal in that scheme. The proton-tagged deliverable is the integrated cross section.

Requiring an identified proton in the final state, in addition to the muon and charged pion, reconstructs the full visible hadronic system and opens two classes of observable inaccessible to the inclusive $\text{CC } 1\mu 1\pi X p$ measurement: the *hadronic invariant mass*, which probes the $\Delta(1232)$ and higher resonances directly, and the *transverse kinematic imbalance* (TKI) between the muon and the hadronic system, which is sensitive to Fermi motion and final-state interactions. This subsample is implemented as a dedicated selection, **CC1mu1pi1p**, that inherits the full $\text{CC } 1\mu 1\pi X p$ selection (§9) and additionally requires a leading proton candidate; the most proton-like track (log-likelihood-ratio PID score below the proton cut), contained in the fiducial volume and above the 0.3 GeV/ c tracking threshold. The signal is correspondingly narrowed to the inclusive signal plus at least one true proton above 0.3 GeV/ c . On the Run-1 FHC sample this yields a selection efficiency of 11.5% and a purity of 52% (versus 17.5% and $\sim 59\%$ for the inclusive selection); the proton tag trades signal statistics for a fully reconstructed hadronic final state.

13.1 Extracted cross sections

The proton tag costs efficiency and buys exclusivity: it lowers the selection efficiency from $\approx 17.5\%$ to $\approx 12\%$ and roughly halves the selected signal, while leaving purity at $\approx 47\text{--}48\%$ under the more restrictive exclusive truth definition. About 87% of tagged events contain a true proton above threshold, so the mis-tag rate is $\approx 13\%$. Its clearest benefit is cosmic rejection: requiring a contained proton above 0.3 GeV/ c removes beam-off background by up to a factor of six. The full cut-flow, the per-observable binning study and the response tables are in the technical supplement; only the extracted cross sections are given here. The full-exposure cut-flow of the proton-tagged subsample is the inherited $\text{CC } 1\mu 1\pi X p$ sequence followed by the leading-proton identification (“IdentProton”), which defines the subsample. Table 6 of the technical supplement gives the per-category, per-cut breakdown for the three beam configurations and Fig. 5 of the technical supplement the corresponding stacks; both use the same per-run POT scaling as the inclusive cut-flow (Table 10), and “signal” here is the proton-tagged signal (inclusive signal plus a true proton above 0.3 GeV/ c). The proton-identification cut is the key discriminator: at FHC it rejects $\sim 64\%$ of the neutrino background while retaining $\sim 71\%$ of the signal, lifting the purity from 33% at the inclusive final cut (Nonprotons) to 48% in the proton-tagged sample.

Proton-tag performance. Beyond the yields, the leading-proton identification (log-likelihood-ratio PID score below the proton cut, $p_p \geq 0.3$ GeV/ c , contained) is a reliable tag. Truth-matched on the FHC ν_μ overlay, 87% of the tagged events contain an energetic true proton ($p_p > 0.3$ GeV/ c); a $\sim 13\%$ mis-tag rate, dominated by a charged pion or other track being taken for the proton, and 52% are the full $\mu\pi p$ signal topology. The reconstructed proton momentum, from the proton-hypothesis range kinetic energy, carries a -9% bias (underestimate) and a 21% resolution. In cross-section terms the tag rejects $\sim 64\%$ of the neutrino background for a $\sim 30\%$ signal loss, improving the signal-to-background figure of merit $S/\sqrt{B}$ by $\sim 17\%$; the residual mis-tag and momentum smearing are folded into the response matrix and corrected by the unfolding.

Eleven observables are extracted in the proton-tagged subsample. Ten of them exist as differential extractions and are released as *secondary* results with their covariances; the robust proton-tagged deliverable proposed as a headline is the integrated cut-and-count cross section

(Table 21). The ten are: the six hadronic-mass and transverse-kinematic-imbalance (W/TKI) variables specific to this selection (the differential $W_{\pi p}$ shape being withdrawn and used only through the integrated cross section, Table 21), plus the five muon/pion kinematic observables of the inclusive analysis (p_μ , p_π , $\cos\theta_\mu$, $\cos\theta_\pi$, $\theta_{\mu\pi}$) re-measured on the proton-tagged phase space so that the two selections share a common set of kinematic distributions. The two hadronic-mass variables are the *directly measured* πp invariant mass, $W_{\pi p} = [(E_\pi + E_p)^2 - (\vec{p}_\pi + \vec{p}_p)^2]^{1/2}$, which makes no assumption on the neutrino direction, and the *calorimetric* hadronic mass $W_{\text{had}} = [m_p^2 + 2m_p(E_{\text{cal}} - E_\mu) - Q^2]^{1/2}$ built from the reconstructed calorimetric energy. The four TKI observables; the transverse-momentum imbalance δp_T , its two angles $\delta\alpha_T$ and $\delta\phi_T$ [28], and the STV-inferred nucleon-momentum proxy p_n [29] (an approximation, see below), are computed from the muon versus the summed ($\pi + p$) hadronic system using the framework STV tool. Proton momenta are obtained from the proton-hypothesis range-based kinetic energy. W_{had} is measured in six equal-population bins ($W_{\pi p}$ was binned the same way for the diagnostic study in the supplement); the four TKI variables use the coarsened two-bin scheme adopted after the finer binning was found to sit inside the resolution (Sec. 4.2 of the technical supplement). The per-run POT normalisation (Sec. B of the technical supplement) removes the background-subtraction bias that had previously forced a coarser three-bin binning for the lower-statistics reverse-horn-current sample, while the five inherited kinematic observables use the inclusive-analysis binning (four to seven bins). All bin edges are set from the validated true-signal distributions (Table 7 of the technical supplement).

Reconstruction from input quantities

All six observables are built from the three candidate tracks (muon, charged pion, proton) using only reconstructed quantities stored in the input ntuples: the track direction cosines $\hat{n} = (\text{track_dirx}, \text{track_diry}, \text{track_dirz})$, the muon-hypothesis range and multiple-scattering momenta ($\text{track_range_mom_mu}$, track_mcs_mom_mu) and the proton-hypothesis calorimetric kinetic energy ($\text{track_kinetic_energy_p}$).

Candidate three-momenta. The muon momentum magnitude is range-based when the muon track ends inside the fiducial volume, $p_\mu = \text{track_range_mom_mu}[\mu]$, and multiple-Coulomb-scattering based otherwise, $p_\mu = \text{track_mcs_mom_mu}[\mu]$. The charged pion, being minimum-ionising, is assigned the muon-hypothesis range momentum $p_\pi = \text{track_range_mom_mu}[\pi]$. The proton momentum follows from its calorimetric kinetic energy under the proton hypothesis, $p_p = \sqrt{T_p^2 + 2m_p T_p}$ with $T_p = \text{track_kinetic_energy_p}[p]$. Each three-momentum is $\vec{p}_i = p_i \hat{n}_i$, and the energies are $E_\mu = \sqrt{p_\mu^2 + m_\mu^2}$, $E_\pi = \sqrt{p_\pi^2 + m_\pi^2}$, $E_p = T_p + m_p$ (masses $m_\mu = 105.7$, $m_\pi = 139.6$, $m_p = 938.3$ MeV).

Directly measured mass $W_{\pi p}$. The invariant mass of the pion-proton four-vector sum, using no neutrino-direction assumption:

$$W_{\pi p} = \sqrt{(E_\pi + E_p)^2 - |\vec{p}_\pi + \vec{p}_p|^2}.$$

Transverse kinematic imbalance. The TKI observables treat the muon against the visible hadronic system $\vec{p}_{\text{had}} = \vec{p}_\pi + \vec{p}_p$, $E_{\text{had}} = E_\pi + E_p$. With the components transverse to the beam ($\hat{z}$) written $\vec{p}_T^\mu$ and $\vec{p}_T^{\text{had}}$,

$$\delta p_T = |\vec{p}_T^\mu + \vec{p}_T^{\text{had}}|, \quad \delta\alpha_T = \arccos \frac{-\vec{p}_T^\mu \cdot (\vec{p}_T^\mu + \vec{p}_T^{\text{had}})}{|\vec{p}_T^\mu| \delta p_T}, \quad \delta\phi_T = \arccos \frac{-\vec{p}_T^\mu \cdot \vec{p}_T^{\text{had}}}{|\vec{p}_T^\mu| |\vec{p}_T^{\text{had}}|},$$

both angles in degrees folded onto $[0, 180]$. For a stationary nucleon with no final-state interactions $\delta p_T \rightarrow 0$; Fermi motion broadens it and FSI drive $\delta\alpha_T \rightarrow 180^\circ$.

Calorimetric mass W_{had} and struck-nucleon momentum p_n . The calorimetric available energy is $E_{\text{cal}} = E_\mu + (E_{\text{had}} - m_p) + E_b$ with the argon removal energy $E_b = 24.78$ MeV, giving the

momentum transfer $Q^2 = -[(0, 0, E_{\text{cal}}, E_{\text{cal}}) - p^\mu]^2$ and

$$W_{\text{had}} = \sqrt{m_p^2 + 2m_p(E_{\text{cal}} - E_\mu) - Q^2}$$

(returned as 0 where reconstruction fluctuations make the argument negative). The nucleon-momentum proxy p_n combines the transverse imbalance with a longitudinal component p_L from energy-momentum balance, $p_n = \sqrt{\delta p_T^2 + p_L^2}$, using the MINERvA prescription [45] as implemented in STVTools (option 1):

$$p_L = \frac{R}{2} - \frac{M_{A'}^2 + \delta p_T^2}{2R}, \quad R = M_A + p_z^\mu + p_z^{\text{had}} - E_\mu - E_{\text{had}},$$

with the target mass $M_A = 22m_n + 18m_p - 0.34381$ GeV (argon-40), the residual mass $M_{A'} = M_A - m_n + E_b$ and $E_b = 24.78$ MeV; z is the beam axis (§5.2). The tool was written for a single-proton hadronic system: it receives the summed three-momentum $\vec{p}_\pi + \vec{p}_p$ and energy $E_{\text{had}} = E_\pi + E_p$ in the proton slot and forms the hadronic kinetic energy as $E_{\text{had}} - m_p$, which is why m_p appears in E_{cal} and W_{had} above. δp_T , $\delta\alpha_T$ and $\delta\phi_T$ depend only on the transverse momenta and are exact for the $\mu + (\pi p)$ system; p_L , p_n and W_{had} inherit the single-nucleon residual-mass and removal-energy assumptions of a quasi-elastic-oriented prescription and are therefore *proxies*, not the initial-state nucleon momentum. The generator predictions apply the identical formulas to the generator final states (Sec. 3.1 of the technical supplement), so the comparison is like for like; a truth-level closure of this prescription against a pion-production-specific definition, and a quantification of the bias from the residual-mass assumption, have not been performed and are listed in Sec. 16.

The pipeline is validated end-to-end on FHC with a fake-data closure: the Poisson-thrown central-value pseudo-data are unfolded with the same Wiener-SVD procedure (§11) and compared to the A_C -smeared truth. Table 28 shows the result. The bin-by-bin χ^2/ndf is well below unity for every observable, confirming the unfolded pseudo-data are statistically consistent with truth and that the response, background subtraction and normalisation are self-consistent. The integrated-cross-section ratios (0.94–1.19) reflect the same non-norm-preserving Wiener-SVD additional smearing A_C seen in the inclusive measurement (Sec. 4.5 of the technical supplement), largest for $\delta\alpha_T$ and $\theta_{\mu\pi}$. The systematic treatment here includes the flux, cross-section and reinteraction and detector-variation terms. The detector term was added on 2026-08-31, when the detector-variation samples were referenced by this selection for the first time and all thirty-three proton-tagged universes were rebuilt; before that these results carried none.

The reverse-horn-current measurement is complete at the same binning (Table 29). All ten released RHC observables close positive, with σ_{int} clustered in $0.38\text{--}0.49 \times 10^{-38}$ cm²/Ar, below the FHC band (0.43–0.56), as expected for the $\bar{\nu}_\mu$ -enriched RHC flux; and $\chi^2/\text{ndf} \lesssim 1$ for almost every observable.

Reaching this point required resolving a normalisation subtlety that is worth recording, because it produced spurious *negative* cross sections for this subsample at an intermediate stage. The processed ntuples use two different POT conventions: the files behind the inclusive selection store the *total* run MC POT duplicated in every file of a multi-file run, whereas the proton-tagged files store each file's *own* POT. Normalising by the individual file POT is therefore correct for the first set but scales every split of a multi-file run to the full run exposure in the second, inflating the simulated background by the split multiplicity ($\sim 3\times$ for RHC) and driving the background-subtracted spectrum negative. Normalising by the summed per-file POT has the mirror problem. The analysis now normalises by the sum of the *distinct* per-file POT values, which reduces to the correct run total under either convention; FHC, with one MC file per run, is unaffected by the choice.

The combined-sample measurement is complete on the same binning, is included in the results tables and the data release, and covers all ten released combined proton-tagged observables.

Table 28: FHC fake-data closure for the ten released proton-tagged observables. σ_{int} is the integrated (unfolded) cross section over the measured phase space ($10^{-38} \text{ cm}^2/\text{Ar}$); “unf./truth” is the ratio of integrals; χ^2 is the bin-by-bin comparison of unfolded pseudo-data to A_C -smeared truth (ndf = number of bins). The upper block is the five released W/TKI observables; W_{had} is at six bins, while the four TKI variables use the coarsened two-bin scheme adopted after the finer binning was found to sit inside the resolution; the six-bin $W_{\pi p}$ closure rows are diagnostic only and are given in the technical supplement (Sec. 4.2 of the technical supplement); the lower block is the five kinematic observables inherited from the inclusive analysis. The small χ^2/ndf validates the chain; the integral ratios are the A_C smearing effect, not a bias. [release 2026-09-04, `current_results.tsv`, per-run EXT, re-thrown fake data]

Observable	σ_{int}	unf./truth	χ^2/ndf
W_{had}	0.389	1.00	0.88/6
δp_T	0.461	0.98	0.01/2
$\delta \alpha_T$	0.485	0.96	0.02/2
$\delta \phi_T$	0.474	0.99	0.11/2
p_n	0.490	0.98	0.06/2
p_μ	0.496	1.04	1.89/7
p_π	0.498	0.98	0.01/2
$\cos \theta_\mu$	0.445	1.08	0.98/5
$\cos \theta_\pi$	0.394	0.99	0.30/4
$\theta_{\mu\pi}$	0.469	1.02	0.50/5

Figure 27 shows the five released FHC W/TKI differential cross sections (W_{had} and the four TKI observables), overlaid with all four generator predictions; the withdrawn six-bin $W_{\pi p}$ distribution is deliberately not shown here, and its diagnostic closure and generator curves are in the technical supplement (Sec. 4.2 of the technical supplement). The generator W/TKI predictions were regenerated at this binning (GENIE and GiBUU natively, NuWro and NEUT from their event vectors inside the SL7 container). All curves, data, tune and generators, are A_C -smeared, so the comparison is made consistently in the measurement space (Sec. 3.1 of the technical supplement). The predictions are built at truth level from each generator’s event-level final state, GENIE `gst` events, GiBUU `FinalEvents`, and the NuWro and NEUT event vectors reprocessed in their respective SL7-container environments, by applying the inclusive signal plus a leading true proton above 0.3 GeV/ c and evaluating the six observables with the same formulas as the selection (§13.1), then combining the ν_μ and $\bar{\nu}_\mu$ samples over the FHC flux to a per-argon cross section. The four generators span a factor of ~ 1.6 in the integrated proton-tagged cross section, GENIE lowest (0.50), then NuWro (0.65), GiBUU (0.74) and NEUT highest ($0.78 \times 10^{-38} \text{ cm}^2/\text{Ar}$), bracketing the injected tune (proton-tagged cut-and-count total, Table 21). In the released two-bin TKI scheme the differences between generators are almost entirely normalisation: their shapes, and that of the tune, agree with the unfolded points within the uncertainties in every panel, so the present exposure does not discriminate between the generators’ treatments of Fermi motion and final-state interactions. (An earlier version of this figure was drawn from the withdrawn fine-binned extraction of 2026-08-18, which predates the beam-axis correction: its “data” were transverse quantities about detector z while the generator curves were in the beam frame, which produced apparent shape disagreements of a factor of several. Those sidecars are quarantined.) As throughout, the unfolded fake data sit above the A_C -smeared truth by the Wiener-SVD normalisation factor.

Table 29: RHC fake-data closure for the ten released proton-tagged observables, binning and columns as in Table 28. All observables close positive with the convention-agnostic MC POT normalisation described above; σ_{int} sits just below the FHC band, consistent with the $\bar{\nu}_\mu$ -enriched RHC flux. [release 2026-09-04, `current.results.tsv`, per-run EXT, re-thrown fake data]

Observable	σ_{int}	unf./truth	χ^2/ndf
W_{had}	0.314	0.96	0.69/6
δp_T	0.418	1.10	0.52/2
$\delta \alpha_T$	0.438	1.08	0.19/2
$\delta \phi_T$	0.389	1.02	0.10/2
p_n	0.391	1.00	1.06/2
p_μ	0.380	1.09	3.80/7
p_π	0.429	1.09	1.21/2
$\cos \theta_\mu$	0.392	1.11	0.84/5
$\cos \theta_\pi$	0.365	1.10	6.01/4
$\theta_{\mu\pi}$	0.367	0.96	0.14/5

Table 30: Combined FHC+RHC fake-data closure for the ten released proton-tagged observables at binning and columns as in Table 28. All ten close positive and every χ^2 is comfortably below its number of bins. This completes the proton-tagged measurement in all three beam configurations. Note that the combined result is an independent extraction, not a fluence-weighted average of the FHC and RHC ones, so it is not required to lie between them; see the discussion below. [release 2026-09-04, `current.results.tsv`, per-run EXT, re-thrown fake data]

Observable	σ_{int}	unf./truth	χ^2/ndf
W_{had}	0.345	0.99	0.85/6
δp_T	0.540	1.09	0.17/2
$\delta \alpha_T$	0.459	1.01	0.02/2
$\delta \phi_T$	0.503	1.09	0.24/2
p_n	0.449	1.02	0.85/2
p_μ	0.437	1.03	1.53/7
p_π	0.462	1.03	0.40/2
$\cos \theta_\mu$	0.410	1.06	1.52/5
$\cos \theta_\pi$	0.382	1.02	3.40/4
$\theta_{\mu\pi}$	0.445	1.04	0.26/5

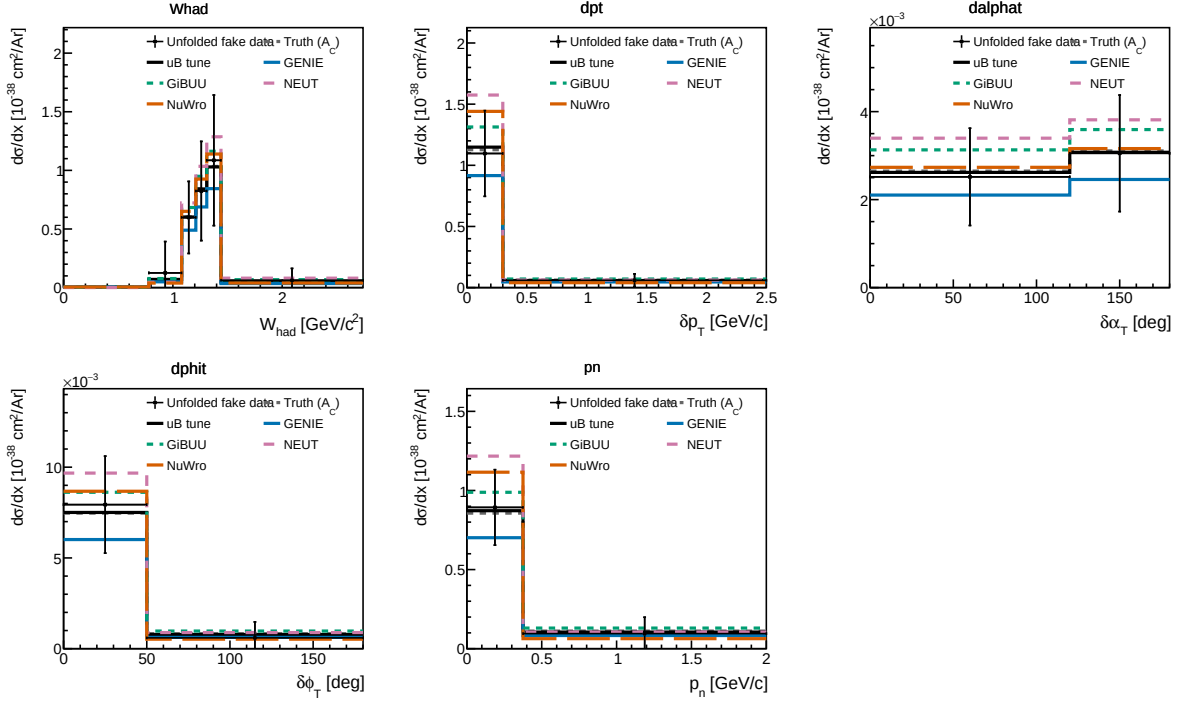


Figure 27: Proton-tagged (**CC1mu1pi1p**) released differential cross sections for FHC: the calorimetric hadronic mass W_{had} and the four transverse-kinematic-imbalance observables δp_T , $\delta \alpha_T$, $\delta \phi_T$ and p_n (the withdrawn $W_{\pi p}$ shape is not shown; see the technical supplement). Unfolded fake data (black points) are compared to the A_C -smeared fake-data truth (grey dashed), the MicroBooNE tune (black), and the GENIE (blue), GiBUU (green), NEUT (purple) and NuWro (orange) predictions built for the same proton-tagged phase space. $\delta \alpha_T$ rises toward 180° , the signature of final-state interactions.

With all three configurations in hand, the combined σ_{int} lies between the FHC and RHC values for all ten released observables. Earlier drafts reported four exceptions, at the few-per-cent level; they were artefacts of the pre-rebuild proton-tagged extractions and do not survive the detector-variation rebuild and the correction of the duplicated A_C .

That the combined result now brackets cleanly is reassuring but should not be over-interpreted: the combined measurement is *not* required to lie between the two single-mode ones. It is an independent extraction with its own response matrix, its own regularisation and its own A_C , not a fluence-weighted average, so excursions at the few-per-cent level would have been unremarkable against a $\sim 30\%$ systematic uncertainty.

The spread across observables within a configuration, 0.41 to 0.52 for the combined set, is likewise not a physics spread but the residual of the additional smearing discussed in the next section: each observable's A_C rescales the integral by a different factor, so the ten numbers are not ten independent measurements of the same quantity.

14 Systematic Uncertainties

Full systematic covariance propagation is complete for all fifty-one extractions. Every result carries flux, cross-section-model, hadron-reinteraction, detector-variation, POT, target-count and statistical terms, propagated through the unfolding by the error propagation matrix and released per component (Sec. 12.2). The proton-tagged detector term was the last to be added, on 2026-08-31; before that those thirty-three results carried none. Table 31 lists the sources.

Table 31: Systematic sources and their bin-averaged fractional contribution, computed from the released per-component covariance rather than estimated. Ranges span observables and configurations; the two families are given separately because the proton-tagged sample is both statistically sparser and more detector-sensitive. [release 2026-09-04, `current_results.tsv`, per-run EXT, re-thrown fake data]

Source	Branch / method	Inclusive (%)	Proton-tagged (%)
Flux (PPFX multisims)	<code>weightsFlux</code>	24.2–32.5	18.3–74.6
Detector response	Dedicated samples	15.6–35.0	10.5–108.5
Cross-section model	<code>weightsGenie</code>	8.5–16.9	7.3–53.8
Hadron re-interaction	<code>weightsReint</code>	3.5–11.9	3.6–24.9
POT counting	Beam toroids	2.7–3.8	2.2–8.7
Target count	FV geometry	1.4–1.9	1.1–4.4
MC statistics	universe spread	1.9–6.3	2.3–20.0
EXT statistics	beam-off sample	0.6–3.1	0.6–4.4
Data statistics	thrown fake data	6.3–18.3	7.4–62.1
Prediction total	quadrature sum	31.0–50.2	23.2–133.6
Total	incl. data stats	32.4–53.9	25.1–147.7

Flux and detector response are the two largest terms in both families; which of the two leads depends on the observable and configuration (Table 32; Table 19 of the technical supplement–Table 20 of the technical supplement). Every entry is propagated into the released covariance; no source declared in the systematics configuration is left unpropagated.

14.1 Systematic breakdown by source

The framework builds a covariance matrix for every systematic. The bin-averaged fractional uncertainty from each source group for the released FHC extractions is given below; the RHC and combined tables, generated from the same release dumps, are Table 19 of the technical supplement and Table 20 of the technical supplement. (An earlier draft showed here an illustrative single-configuration table from a pre-rebuild pass, with detector terms of $\sim 10\%$; those values are superseded by the ones below, which are the same numbers as Table 17.)

Table 32: FHC (Runs 1, 2, 4, 5): bin-averaged fractional uncertainty (%) by source group for the six released extractions, read from the release systematics dumps (`logs/systdump`, 2026-08-31 pass). “Prediction total” excludes the data statistics; the flux row is the PPFX multisim term. [release 2026-09-04, `current_results.tsv`, per-run EXT, re-thrown fake data]

Source	p_μ	p_π	$\cos\theta_\mu$	$\cos\theta_\pi$	$\theta_{\mu\pi}$	θ_μ
Prediction total	40.7	50.2	42.0	36.9	36.5	31.0
Cross section (GENIE)	10.7	16.9	10.3	11.5	10.5	8.5
Flux (PPFX)	28.0	32.5	28.0	26.1	28.3	24.7
Detector	24.7	30.3	28.3	20.9	19.6	15.6
Reinteraction	4.4	11.9	3.6	4.9	4.0	3.5
MC stat	4.9	6.3	5.3	4.1	2.7	2.9
EXT stat	0.9	3.1	2.3	1.1	1.2	1.4
Data stat	14.3	18.3	15.4	12.0	8.2	8.6
POT + targets	3.4	4.2	3.6	3.3	3.5	3.0
Total (incl. data stat)	43.5	53.9	44.9	39.0	37.5	32.4

The NuMI flux uncertainty has two parts: the PPFX hadron-production term (the dominant systematic) and a beamline-geometry term. The latter is built from the 12 geometry-variation weights (Horn_2kA, Horn1/2_x/y_3mm, Beam_spot_*, Beam_shift_*, Horns_*_water, Target_z_7mm) injected into the numuMC overlay via `AddBeamlineGeometryWeights`, summed in quadrature as two-sided unisim variations. It was evaluated at 1.6–3.3% across the observables in the earlier pass and is sub-dominant; it is *not* part of the released covariance, because the release systematics configuration (`ccpi_systcalc_numi.conf`) omits the beamline block that the earlier pass evaluated with `ccpi_systcalc_numi_beamline.conf`. Added in quadrature to the $\sim 28\%$ PPFX term it would raise the flux row by $\lesssim 0.2\%$ absolute, so the omission is immaterial to the budget, but the term is listed here as absent rather than propagated. (Building it required a framework fix: the covariance code dereferenced a null histogram when a systematic, here the beamline weights, present in the numuMC but not the dirt overlay, was absent from some MC sample; it now falls back to that sample’s central value, i.e. unit weight.) The per-bin breakdowns are shown in Fig. 28.

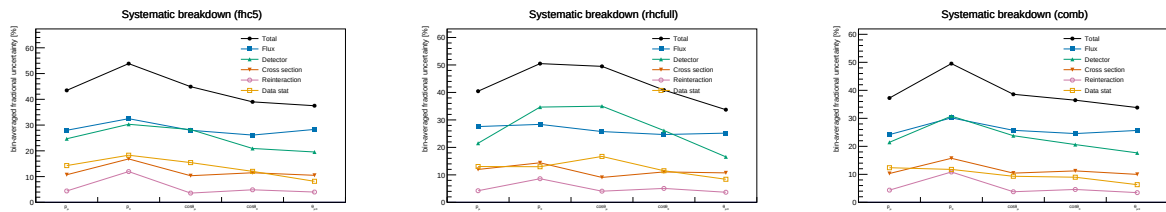


Figure 28: Bin-averaged fractional systematic uncertainty by source vs. observable, for FHC (left), RHC (centre) and combined (right); the figure form of Table 18 of the technical supplement–Table 20 of the technical supplement. Colours use the Okabe-Ito colourblind-safe palette with distinct markers per source: total (black), flux (blue), detector (green), cross section (vermillion), reinteraction (purple), data stat (orange).

14.2 Anatomy of the flux term: genuine flux vs. unfolding amplification

The current unfolded PPFX term is 24–34% (Tables 17–19), larger than the $\sim 17\%$ normalisation uncertainty usually quoted for the NuMI flux. An earlier pass, retained below because it is the only one in which the flux term was evaluated at reconstructed level as well as after unfolding (the numbers are therefore those of that pass, 17–26% unfolded, and not the current ones), separates the two. It is therefore worth asking how much of it is the flux model itself and how much is statistical amplification introduced by the unfolding. To separate the two we evaluate the flux covariance at two stages, on the reconstructed spectrum (before unfolding) and on the unfolded result, and, at each stage, both integrated (the fully correlated, normalisation-like number, from the sum of all covariance elements) and bin-averaged (the mean of the diagonal fractional errors, i.e. the number quoted in Table 32 and Fig. 28).

Table 33: Flux (PPFX) fractional uncertainty before and after unfolding. The reconstructed-level value is $\sim 17\%$ and flat across observables; the genuine flux-normalisation uncertainty. The unfolded value is that same $\sim 17\%$ amplified by the unfolding operator, which magnifies variations in the poorly constrained modes of the response; the amplification is largest for the observables with the weakest response conditioning and lowest effective statistics. **These values are from an earlier, single-configuration extraction pass** (pre-detector-variation-rebuild FHC binning, five observables); the reco-level flux term is not re-derivable from the current extraction logs, so the table is kept as an illustration of the mechanism only. The current unfolded flux terms are those of Tables 17–19.

	p_μ	p_π	$\cos \theta_\mu$	$\cos \theta_\pi$	$\theta_{\mu\pi}$
Flux, reco (integrated / norm)	16.9	17.0	16.7	16.7	16.7
Flux, reco (bin-averaged)	17.7	16.4	16.6	17.0	17.0
Flux, unfolded (bin-averaged)	19.8	22.4	19.8	25.5	17.5
Amplification factor	1.12	1.37	1.20	1.49	1.03
Excess from unfolding (abs.)	+2.1	+6.0	+3.2	+8.5	+0.5
Data+MC stats (unfolded)	7.9	6.3	8.4	12.2	8.8

Three points follow. First, the *genuine* flux uncertainty is $\sim 17\%$ and identical for all observables, as expected for a normalisation term. Second, it is not itself contaminated by Monte Carlo statistics: at reco level the integrated (correlated) and bin-averaged (diagonal) numbers agree to better than a percent, whereas statistical noise leaking into the covariance would inflate the diagonal well above the correlated norm. This is expected; the ~ 600 PPFX universes reweight the *same* events, so there is no independent per-universe statistical scatter. Third, the elevated unfolded values are the $\sim 17\%$ flux *amplified* by the unfolding operator, $A_C M V_{\text{flux}} M^\top A_C^\top$ with M the error-propagation matrix. This is a property of the inverse problem: with a fixed response and a fixed unfolding operator, a flux variation that projects onto a poorly constrained mode of the response is magnified even if the response carried no statistical uncertainty at all. Finite overlay statistics enter only indirectly, by worsening the conditioning of that response, and are not fluctuated inside the PPFX universes. Empirically the amplification is largest where the conditioning is weakest and the effective statistics lowest ($\cos \theta_\pi$: stat. 12.2%, amplification $1.49\times$, $+8.5\%$) and negligible where the response is well conditioned ($\theta_{\mu\pi}$: $1.03\times$, $+0.5\%$). The lever for reducing the flux term is therefore the response conditioning (binning, regularisation strength, and secondarily overlay statistics), not the flux model; $\cos \theta_\pi$, the noisiest observable across the entire budget, is where it would matter most.

15 Analysis Pipeline and Automated Guards

Every released number is produced by the XSECANALYZER chain in `xsec_analyzer/`, in four stages that are re-runnable from the raw PeLEE ntuples:

1. `ProcessNTuples` applies the selection(s) to each PeLEE ntuple and writes the processed `stv_tree` files (`run_process_ntuples.sh`; beam-off samples `run_process_ext_perrun.sh`).
2. `univmake` builds, per observable and configuration, the true/reco histograms of every systematic universe from the files listed in the configuration's `file.properties` (`slurm/slurm_extfix.sbatch` for a full rebuild on the cluster; the last one, 2026-09-02, rebuilt all 51 in one array job).
3. `UnfolderNuMI` sums the universes with the per-run exposure scaling, builds the covariance, unfolds, and writes the closure sidecar, the systematics dump and the matrix tables (`regen_fdfix.sh` runs all 51; `slurm/unfold_incl_local.sh` and `unfold_1p_local.sh` harvest the covariances).
4. `export_curves.C` and `export_matrices.C` write the data release; the figure macros draw from the closure sidecars; `total_xsec_counting.C`, `cutflow_yields.C` and `sideband_compare.C` produce the count-level tables with the same weights and scalings.

Five automated guards protect against the failure modes this analysis has actually suffered (each is listed with the incident that motivated it in the change log):

- `norm_manifest.py` reads the authoritative configuration files and asserts that exposure, flux, target count and beam-off scaling agree across all configurations, and that every declared systematic is backed by a file.

- The systematics/file-list coupling assertion in `UnfolderNuMI` refuses a configuration whose declared detector variations are not present in its file list.
- `check_staleness.py` refuses to ship any released artefact or result figure older than the extraction it depicts.
- `report/check_tables.py` parses every result table in this note and fails if a value disagrees with `current_results.tsv`, if a table does not sit under its own label, or if a row’s column count differs from its header (added after a caption-length-dependent patcher rewrote the wrong tables in a draft of this note).
- The blinding guard in `SystematicsCalculator`: a beam-on file without MC truth is real data and is refused unless `XSEC_UNBLIND=1` is set; the same detection selects the beam-off treatment (added to pseudo-data, subtracted from real data), so no manual switch exists to forget.

The clean-cache reproducibility test (all 51 extractions rebuilt from the processed ntuples with every cached universe deleted, agreeing bin by bin) is recorded in the change log and was last repeated on 2026-09-02.

16 Limitations

This section states what limits the measurement as it now stands. It is deliberately about the present, not the past: the corrections that got the analysis here, with their before-and-after values, are in the change log, and the studies behind the scope decisions are in the technical supplement. Earlier drafts carried two retrospective sections in this position, and a reader looking for the limitations found a catalogue of fixed bugs instead.

The analysis is blind, and that is the dominant limitation. No beam-on signal region has been opened. Every “data” point is Poisson-thrown from the central-value Monte Carlo, so everything demonstrated here concerns the extraction machinery rather than nature. In particular, no statement in this note tests the background model against data. At $\approx 59\%$ purity, with proton-to-pion misidentification the largest neutrino background, the measurement depends on that model substantially, and fake-data closure cannot probe it: the same simulation supplies both the pseudo-data and the background estimate. This is the single largest untested assumption in the analysis.

Pion momentum is not recoverable, and that sets the scope. Charged pions interact hadronically before ranging out, so the reconstructed momentum saturates near $0.34 \text{ GeV}/c$ and carries little information above roughly $0.4 \text{ GeV}/c$. Every alternative estimator was tried — golden-pion tagging, multiple Coulomb scattering, calorimetry, multivariate regression, daughter-energy recovery, the proton-tagged selection — and a truth-level golden selection performs no better than the trained classifier, which places the limit in the physics rather than the estimator. This is why p_π is reported in two coarse bins and why the differential $W_{\pi p}$ shape is withdrawn: at this exposure no binning satisfies both the migration and the statistics requirement.

The extraction carries a small negative bias. The ensemble of thirty-two throws puts the mean 2.5% below the reference, 2.1σ on the mean, with per-bin pull means consistently negative in both observables tested. It is about a fifth of the statistical uncertainty and far inside the systematic band, but it is a bias rather than a fluctuation and is not currently corrected for or assigned an uncertainty.

Coverage is demonstrated for two observables, in one configuration. The pull and coverage study covers FHC p_μ and $\cos\theta_\mu$. Both give widths consistent with unity, and the second was chosen because its A_C suppression is the most severe in the analysis, but RHC and the proton-tagged family — where the statistics are sparser — have not been tested.

The proton-tagged results are secondary, and their model comparisons are the weakest part. Their systematics are complete and their central values current, but the exposure does not support a differential $W_{\pi p}$ shape, and the sample has no dedicated control region for the $\approx 13\%$ proton mis-tag: every existing sideband applies the proton requirement rather than inverting it, so nothing directly constrains a wrong-track proton assignment, which is exactly what distorts the hadronic observables.

The TKI prescription is borrowed from the single-proton case. δp_T , $\delta\alpha_T$ and $\delta\phi_T$ are exact for the $\mu + (\pi p)$ system, but p_n and W_{had} use the single-nucleon residual-mass and removal-energy assumptions of STVTools with the summed hadronic system in the proton slot (§13.1). Data and generators are treated identically, so the comparison is consistent, but no truth-level closure of this prescription against a pion-production-specific definition has been made and the bias from the residual-mass assumption is not quantified; p_n should be read as a proxy.

A_C is not norm preserving, so two different totals exist. The Wiener-SVD differential result estimates $A_C x_{\text{true}}$, and the row sums fall as low as 0.050. Summing it over bins therefore does not give a physical total cross section; the physical total is the cut-and-count value, quoted separately. Two matrices have negative row sums in their least-determined bin.

Actions required before unblinding. The prerequisites, split by the decision they gate, are the checklist of §2. In brief: the four inclusive control regions are to be opened first, under the frozen procedure of §12.8 once the conveners have ratified it; the signal region only after any control-region discrepancy has been treated under that procedure, the flux constants have been confirmed as PPFX-corrected central values (§5.4), the real beam-on gate counts have replaced the fake-data ones (the Run-1 beam-off sample stands in for Run 2 permanently, §6), and a final clean-cache reproduction has been made; and any proton-tagged result only after the inverted-proton-PID control region exists (§12.9). Nothing has to be switched for the beam-off treatment: the framework detects fake data by the presence of MC truth and selects the treatment itself, and refuses a real beam-on file unless XSEC_UNBLIND=1 is set.

The five automated guards that protect against regression — the normalisation manifest, the systematics/file-list coupling assertion, the staleness check, the table-consistency check and the blinding guard — are described in Sec. 15.

17 Summary

We have developed a measurement of $\nu_\mu + \bar{\nu}_\mu$ charged-current single charged-pion production on argon using the full MicroBooNE NuMI exposure: forward horn current (8.857×10^{20} POT), reverse horn current (1.1082×10^{21} POT), and their combination (1.9939×10^{21} POT).

The analysis is blind. No beam-on signal region has been opened. Every “data” point in this note is Poisson-thrown from the central-value Monte Carlo, so the results demonstrate that the extraction works; they are not measurements, and no statement here about a generator is a statement about nature.

Measured phase space and selection. $|\text{PDG}_\nu| = 14$, charged current, true vertex in the fiducial volume, exactly one muon and one charged pion, no neutral pions or kaons, any number of protons, with $p_\mu > 0.15 \text{ GeV}/c$, $p_\pi > 0.175 \text{ GeV}/c$ and $\theta_{\mu\pi} < 2.6 \text{ rad}$. The selection reaches

$\approx 17.5\%$ efficiency at $\approx 59\%$ purity. Angles and transverse observables are defined about the per-event neutrino direction, not the detector z axis.

What is measured. Fifty-one extractions: eighteen inclusive and thirty-three proton-tagged, across the three configurations. The primary scope is the total flux-averaged cross section, $d\sigma/dp_\mu$, the muon and pion angular cross sections, $d\sigma/d\theta_{\mu\pi}$, and a deliberately coarse two-bin $d\sigma/dp_\pi$. Inclusive total uncertainties run 33.8–59.3%, with the detector term 16.9–42.8% and flux the largest single contribution.

The proton-tagged subsample is secondary scope, with total uncertainties 24.3–77.2%. Its systematics are complete, but the exposure does not support a differential $W_{\pi p}$ shape at any binning, so that result is withdrawn and only the integrated proton-tagged cross section is offered.

What limits the measurement. Charged pions interact hadronically before ranging out, so the reconstructed momentum saturates near $0.34 \text{ GeV}/c$. A survey of alternatives — golden-pion tagging, multiple Coulomb scattering, calorimetry, multivariate regression, daughter-energy recovery, and the proton-tagged selection — recovers none of it, and a truth-level golden selection performs no better than the classifier, which places the limit in the physics rather than the estimator. This is what reduces p_π to two bins and $W_{\pi p}$ to an integrated quantity, and it is the single most consequential limitation of the analysis.

Validation. The extraction reproduces its own input: the mean closure ratio over the eighteen inclusive extractions is 1.018 with second-derivative regularisation (the identity-regularised variant was not re-extracted after the fake-data-truth correction). The algebraic identity $UR = A_C$ holds to 4.4×10^{-6} , which tests the unfolding construction rather than only its behaviour on one sample. D’Agostini iterative unfolding, whose integral is flat across observables where the Wiener-SVD spread is large, establishes that the observable-to-observable spread of the Wiener-SVD integrals is the additional smearing rather than physics or acceptance.

The closure χ^2 values should not be read as agreement: their p -values cluster near unity because the measurement and the truth it is compared against share their systematics, which cancel in the residual while still inflating the covariance used to judge it.

Corrections that changed results. Nine, each recorded in the change log with before-and-after values. The largest were an obsolete flux (a factor ~ 3.48), a double application of the central-value weight to the fake data (which moved every cross section by 3.8–12.9%), a regularisation operator that assumed a uniform grid and drove A_C towards rank one, an angular convention defined about detector z rather than the beam, and — most recently — a double application of A_C to every model prediction, which inflated the generator χ^2 and produced an apparent anomaly in the muon-angle comparisons that does not exist.

That last one was found by asking whether an external user could reproduce a published generator curve from the released matrix. Internal closure could not have found it: the data and the truth were both correct, and only the comparison was wrong.

Reproducibility from raw inputs. All fifty-one extractions were rebuilt from the processed ntuples with every cached universe deleted — new universe files, new unfolding, new covariance — into a separate tree so that a failure could not damage the release it was verifying. The rebuilt spectra agree with the released ones *bin by bin*, with no bin differing by more than 10^{-9} in any of the fifty-one. The release is therefore regenerable from the raw inputs and the recorded configuration, not merely from a preserved intermediate.

This is worth separating from the closure and ensemble tests. Those ask whether the extraction is right; this asks whether what is published is what the code currently produces.

Given that this analysis has twice shipped a downstream product that lagged a fix, that is a distinct question and it now has an answer.

Before unblinding. Control regions must be opened in data first, with the permitted response to any discrepancy fixed beforehand; a clean rebuild from empty caches must reproduce the release; and the configuration-to-systematics coupling needs a check that every declared source has its expected universes, since it was exactly that interface which allowed the proton-tagged detector uncertainty to vanish silently. An ensemble of throws, rather than the single one used here, would turn the closure caveat above into a quantitative coverage statement.

A Observable Binning

Generated directly from the released `curves_*.tsv` files, so these are by construction the bin edges of the published results rather than a transcription. The binnings are identical across the three beam configurations. Earlier drafts of this appendix listed a finer exploratory binning that predates the resolution-driven optimisation; that binning is not used for any released result.

Table 34: Bin edges of every released observable, read from the data release.

Observable	Bins	Edges
<i>Inclusive</i>		
p_μ (GeV/c)	7	0.15, 0.35, 0.55, 0.75, 0.95, 1.25, 1.75, 3
p_π (GeV/c)	2	0.175, 0.205, 1
$\cos \theta_\mu$	5	-1, 0.45, 0.65, 0.8, 0.9, 1
$\cos \theta_\pi$	4	-1, -0.1, 0.35, 0.75, 1
$\theta_{\mu\pi}$ (rad)	5	0, 0.6, 0.85, 1.3, 1.85, 2.6
θ_μ (rad)	5	0, 0.43, 0.62, 0.84, 1.17, 3.15
<i>Proton-tagged</i>		
p_μ (GeV/c)	7	0.15, 0.35, 0.55, 0.75, 0.95, 1.25, 1.75, 3
p_π (GeV/c)	2	0.175, 0.205, 1
$\cos \theta_\mu$	5	-1, 0.45, 0.65, 0.8, 0.9, 1
$\cos \theta_\pi$	4	-1, -0.1, 0.35, 0.75, 1
$\theta_{\mu\pi}$ (rad)	5	0, 0.6, 0.85, 1.3, 1.85, 2.6
$W_{\pi p}$ (GeV/c ²)	6	1.08, 1.19, 1.23, 1.27, 1.34, 1.47, 2.9
W_{had} (GeV/c ²)	6	0, 0.78, 1.08, 1.21, 1.31, 1.44, 2.74
δp_T (GeV/c)	2	0, 0.3, 2.5
$\delta \phi_T$ (deg)	2	0, 50, 180
$\delta \alpha_T$ (deg)	2	0, 120, 180
p_n (GeV/c)	2	0, 0.375, 2

The top p_π bin is defined at truth level as $p_\pi > 0.205$ GeV/c with no upper edge (`ccpi_ppi_bin_config_2bin.txt`); the 1 GeV/c listed above is the finite boundary used in the numerical release to assign that bin a width, not a kinematic limit of the sample. The bin is an inclusive high-momentum region, not a differential point, for the reasons set out in the technical supplement; the bin-integrated partial cross sections are given in Table 16.

B Selection Cut Parameter Summary

Table 35: All selection threshold constants from `selection/ccpi_selection.C`.

Parameter	Value	Ntuple variable
Topological score	> 0.67	<code>topological_score</code>
Muon track score	> 0.80	<code>trk_score_v</code>
Muon vertex distance	≤ 4.0 cm	
Muon track length	> 10.0 cm	<code>trk_len_v</code>
Muon LLR PID	> 0.20	<code>trk_llr_pid_score_v</code>
Muon MIP BDT	≥ -0.10	TMVA BDT
Pion track length	> 20.0 cm	<code>trk_len_v</code>
Pion vertex distance	< 4.0 cm	
Pion LLR PID	> 0.10	<code>trk_llr_pid_score_v</code>
Pion MIP BDT	> -0.10	TMVA BDT
Pion pion-BDT	> -0.10	TMVA BDT
Pion Bragg score	≥ 0.08	<code>trk_bragg_pion_v</code>
Max. opening angle	< 2.6 rad	
Shower Y-plane hits	< 50 (if 1 shower)	
Shower vertex distance	> 10 cm (if 1 shower)	
Non-proton multiplicity	< 4	LLR PID > 0.1
Primary track multiplicity	< 5	
Cosmic BDT	> 0.0 (disabled)	<code>bdt_cosmic</code>
Software trigger	$= 1$	<code>swtrig</code>

References

- [1] MicroBooNE Collaboration, “Measurement of the flux-averaged differential ν_e CC1 π cross section in the NuMI beam,” Internal Note v3.2 (2025).
- [2] K. Mistry, “First Measurement of the Flux-Averaged Differential Charged-Current Electron-Neutrino and Antineutrino Cross Section on Argon with the MicroBooNE Detector,” PhD thesis (2021).
- [3] NOvA Collaboration, M. A. Acero *et al.*, “New constraints on oscillation parameters from ν_e appearance and ν_μ disappearance in the NOvA experiment,” *Phys. Rev. D* **98**, 032012 (2018).
- [4] T2K Collaboration, K. Abe *et al.*, “Constraint on the matter–antimatter symmetry-violating phase in neutrino oscillations,” *Nature* **580**, 339–344 (2020).
- [5] DUNE Collaboration, B. Abi *et al.*, “Prospects for Beyond the Standard Model Physics Searches at the Deep Underground Neutrino Experiment,” *Eur. Phys. J. ST* **231**, 3159–3260 (2022).
- [6] J. S. Marshall and M. A. Thomson, “The Pandora Software Development Kit for Pattern Recognition,” *Eur. Phys. J. C* **75**, 439 (2015).
- [7] W. Tang, X. Li, X. Qian, H. Wei, and C. Zhang, “Data Unfolding with Wiener-SVD Method,” *JINST* **12**, P10002 (2017), arXiv:1705.03568 [physics.data-an].
- [8] L. Aliaga *et al.* (MINERvA Collaboration), “Neutrino Flux Predictions for the NuMI Beam,” *Phys. Rev. D* **94**, 092005 (2016).

- [9] C. Andreopoulos *et al.*, “The GENIE Neutrino Monte Carlo Generator,” *Nucl. Instrum. Methods A* **614**, 87–104 (2010).
- [10] MicroBooNE Collaboration, P. Abratenko *et al.*, “Measurement of space charge effects in the MicroBooNE LArTPC using cosmic muons,” *JINST* **15**, P12037 (2020).
- [11] MicroBooNE Collaboration, C. Adams *et al.*, “A Method to Determine the Electric Field of Liquid Argon Time Projection Chambers Using a UV Laser System and Its Application in MicroBooNE,” *JINST* **15**, P07010 (2020).
- [12] MicroBooNE Collaboration, R. Acciarri *et al.*, “The Pandora multi-algorithm approach to automated pattern recognition of cosmic-ray muon and neutrino events in the MicroBooNE detector,” *Eur. Phys. J. C* **78**, 82 (2018).
- [13] MicroBooNE Collaboration, P. Abratenko *et al.*, “Calorimetric classification of track-like signatures in liquid argon TPCs using MicroBooNE data,” *JHEP* **12**, 153 (2021).
- [14] MicroBooNE Collaboration, P. Abratenko *et al.*, “Determination of muon momentum in the MicroBooNE LArTPC using an improved model of multiple Coulomb scattering,” *JINST* **12**, P10010 (2017).
- [15] MicroBooNE Collaboration, P. Abratenko *et al.*, “Novel approach for evaluating detector-related uncertainties in a LArTPC using MicroBooNE data,” *Eur. Phys. J. C* **82**, 454 (2022).
- [16] MicroBooNE Collaboration, P. Abratenko *et al.*, “New CC0 π GENIE model tune for MicroBooNE,” *Phys. Rev. D* **105**, 072001 (2022), arXiv:2110.14028 [hep-ex].
- [17] MicroBooNE Collaboration, P. Abratenko *et al.*, “First double-differential measurement of kinematic imbalance in neutrino interactions with the MicroBooNE detector,” *Phys. Rev. Lett.* **131**, 101802 (2023), arXiv:2301.03700 [hep-ex].
- [18] S. Gardiner *et al.*, “XSecAnalyzer: a cross-section analysis framework for MicroBooNE,” https://github.com/uboone/xsec_analyzer (accessed 2026).
- [19] P. Adamson *et al.*, “The NuMI Neutrino Beam,” *Nucl. Instrum. Methods A* **806**, 279–306 (2016).
- [20] L. Alvarez-Ruso *et al.* (GENIE Collaboration), “Recent highlights from GENIE v3,” *Eur. Phys. J. ST* **230**, 4449–4467 (2021).
- [21] T. Golan, J. T. Sobczyk, and J. Żmuda, “NuWro: the Wrocław Monte Carlo generator of neutrino interactions,” *Nucl. Phys. B Proc. Suppl.* **229–232**, 499 (2012).
- [22] O. Buss *et al.*, “Transport-theoretical description of nuclear reactions,” *Phys. Rept.* **512**, 1–124 (2012); U. Mosel, “Neutrino event generators: foundation, status and future,” *J. Phys. G* **46**, 113001 (2019).
- [23] Y. Hayato and L. Pickering, “The NEUT neutrino interaction simulation program library,” *Eur. Phys. J. ST* **230**, 4469–4481 (2021).
- [24] S. Agostinelli *et al.* (GEANT4 Collaboration), “GEANT4: a simulation toolkit,” *Nucl. Instrum. Methods A* **506**, 250–303 (2003).
- [25] J. Calcutt, C. Thorpe, K. Mahn, and L. Fields, “Geant4Reweight: a framework for evaluating and propagating hadronic interaction uncertainties in Geant4,” *JINST* **16**, P08042 (2021).

- [26] A. Hoecker *et al.*, “TMVA: Toolkit for Multivariate Data Analysis,” PoS **ACAT**, 040 (2007), arXiv:physics/0703039.
- [27] G. D’Agostini, “A multidimensional unfolding method based on Bayes’ theorem,” *Nucl. Instrum. Methods A* **362**, 487–498 (1995).
- [28] X.-G. Lu, L. Pickering, S. Dolan, G. Barr, D. Coplowe, Y. Uchida, D. Wark, M. O. Wascko, A. Weber, and T. Yuan, “Measurement of nuclear effects in neutrino interactions with minimal dependence on neutrino energy,” *Phys. Rev. C* **94**, 015503 (2016).
- [29] A. P. Furmanski and J. T. Sobczyk, “Neutrino energy reconstruction from one-muon and one-proton events,” *Phys. Rev. C* **95**, 065501 (2017).
- [30] D. Rein and L. M. Sehgal, “Coherent π^0 production in neutrino reactions,” *Nucl. Phys. B* **223**, 29–44 (1983).
- [31] C. Berger and L. M. Sehgal, “Partially conserved axial vector current and coherent pion production by low energy neutrinos,” *Phys. Rev. D* **79**, 053003 (2009).
- [32] MicroBooNE Collaboration, P. Abratenko *et al.*, “First Measurement of Energy-Dependent Inclusive Muon-Neutrino Charged-Current Cross Sections on Argon with the MicroBooNE Detector,” *Phys. Rev. Lett.* **128**, 151801 (2022), arXiv:2110.14023 [hep-ex].
- [33] MicroBooNE Collaboration, R. Acciarri *et al.*, “Design and Construction of the MicroBooNE Detector,” *JINST* **12**, P02017 (2017).
- [34] MicroBooNE Collaboration, C. Adams *et al.*, “Design and construction of the MicroBooNE Cosmic Ray Tagger system,” *JINST* **14**, P04004 (2019).
- [35] D. Rein and L. M. Sehgal, “Neutrino Excitation of Baryon Resonances and Single Pion Production,” *Ann. Phys.* **133**, 79–153 (1981).
- [36] K. S. Kuzmin, V. V. Lyubushkin, and V. A. Naumov, “Lepton polarization in neutrino nucleon interactions,” *Mod. Phys. Lett. A* **19**, 2815–2829 (2004).
- [37] K. M. Graczyk and J. T. Sobczyk, “Form factors in the quark resonance model,” *Phys. Rev. D* **77**, 053001 (2008) [Erratum: *Phys. Rev. D* **79**, 079903 (2009)].
- [38] C. Berger and L. Sehgal, “Lepton mass effects in single pion production by neutrinos,” *Phys. Rev. D* **76**, 113004 (2007).
- [39] J. A. Nowak (MiniBooNE Collaboration), “Four-momentum transfer discrepancy in the charged current π^+ production in the MiniBooNE: Data vs. theory,” *AIP Conf. Proc.* **1189**, 243–248 (2009).
- [40] Y. Hayato, “A neutrino interaction simulation program library NEUT,” *Acta Phys. Polon. B* **40**, 2477–2489 (2009).
- [41] G. M. Radecky *et al.* (ANL Collaboration), “Study of single pion production by weak charged currents in low-energy νd interactions,” *Phys. Rev. D* **25**, 1161–1173 (1982).
- [42] T. Kitagaki *et al.* (BNL Collaboration), “Charged-current exclusive pion production in neutrino–deuterium interactions,” *Phys. Rev. D* **34**, 2554–2565 (1986).
- [43] A. Rodriguez *et al.* (K2K Collaboration), “Measurement of single charged pion production in the charged-current interactions of neutrinos in a 1.3 GeV wide band beam,” *Phys. Rev. D* **78**, 032003 (2008).

- [44] A. A. Aguilar-Arevalo *et al.* (MiniBooNE Collaboration), “Measurement of ν_μ -induced charged-current single positive pion production cross sections on mineral oil at $E_\nu \in 0.5\text{--}2.0$ GeV,” *Phys. Rev. D* **83**, 052007 (2011).
- [45] X.-G. Lu *et al.* (MINERvA Collaboration), “Measurement of final-state correlations in neutrino muon-proton mesonless production on hydrocarbon at $\langle E_\nu \rangle = 3$ GeV,” *Phys. Rev. Lett.* **121**, 022504 (2018).
- [46] B. Eberly *et al.* (MINERvA Collaboration), “Charged pion production in ν_μ interactions on hydrocarbon at $\langle E_\nu \rangle = 4.0$ GeV,” *Phys. Rev. D* **92**, 092008 (2015).
- [47] C. L. McGivern *et al.* (MINERvA Collaboration), “Cross sections for ν_μ and $\bar{\nu}_\mu$ induced pion production on hydrocarbon in the few-GeV region using MINERvA,” *Phys. Rev. D* **94**, 052005 (2016).
- [48] MINERvA Collaboration, J. Wolcott *et al.*, “Evidence for Quasielastic Charge Current Antineutrino Scattering Off Protons in Hydrocarbon at $\langle E_{\bar{\nu}} \rangle = 3.5$ GeV,” *Phys. Rev. Lett.* **116**, 081802 (2016).
- [49] K. Abe *et al.* (T2K Collaboration), “Measurement of the muon neutrino charged-current single π^+ production on hydrocarbon using the T2K off-axis near detector ND280,” *Phys. Rev. D* **101**, 012007 (2020), arXiv:1909.03936 [hep-ex].
- [50] R. Acciarri *et al.* (ArgoNEUT Collaboration), “First measurement of the cross section for ν_μ and $\bar{\nu}_\mu$ induced single charged pion production on argon using ArgoNEUT,” *Phys. Rev. D* **98**, 052002 (2018), arXiv:1804.10294 [hep-ex].
- [51] MicroBooNE Collaboration, P. Abratenko *et al.*, “First measurement of ν_e and $\bar{\nu}_e$ charged-current single charged-pion production differential cross sections on argon using the MicroBooNE detector,” *Phys. Rev. Lett.* **135**, 061802 (2025), arXiv:2503.23384 [hep-ex].
- [52] MicroBooNE Collaboration, P. Abratenko *et al.*, “Measurement of single charged pion production in charged-current ν_μ -Ar interactions with the MicroBooNE detector,” *Phys. Rev. D* **113**, 032007 (2026), arXiv:2509.03628 [hep-ex].